

| BRK (00) |                             | Cycles: 8                                                                                                                | Size: 2 * |
|----------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------|
| Implied  |                             |                                                                                                                          |           |
| Cycle    | R/W                         | Desc.                                                                                                                    |           |
| -X.2     | Read <b>PC:PB</b>           | ** Store as OpCode.                                                                                                      |           |
| 1.1      |                             | Inc. <b>PC</b> .                                                                                                         |           |
| 1.2      | Read <b>PC:PB</b>           | Store as Operand.                                                                                                        |           |
| 2.1      |                             | Inc. <b>PC</b> .                                                                                                         |           |
| 2.2      | *** Write to ( <b>S</b> +0) | Push <b>PB</b> onto stack.                                                                                               |           |
| 3.1      |                             |                                                                                                                          |           |
| 3.2      | *** Write to ( <b>S</b> -1) | Push <b>PC</b> .H onto stack.                                                                                            |           |
| 4.1      |                             |                                                                                                                          |           |
| 4.2      | *** Write to ( <b>S</b> -2) | Push <b>PC</b> .L onto stack.                                                                                            |           |
| 5.1      |                             | Use Interrupt flags to decide which vector to use.<br>RESET = \$FFFC.<br>NMI = \$FFEA.<br>IRQ = \$FFEE.<br>BRK = \$FFE6. |           |
| 5.2      | *** Write to ( <b>S</b> -3) | Push <b>P</b> onto stack with <b>B</b> flag if software BRK.                                                             |           |
| 6.1      |                             | Dec. <b>S</b> by 4.                                                                                                      |           |
| 6.2      | Read Vector                 | Store to Address.L.                                                                                                      |           |
| 7.1      |                             | Set <b>I</b> , unset <b>D</b> .                                                                                          |           |
| 7.2      | Read Vector + 1             | Store to Address.H.                                                                                                      |           |
| 8.1      |                             | Enable writes to <b>PC</b> (disabled by NMI/IRQ). Copy Address to <b>PC</b> . Set <b>PB</b> to 0.                        |           |
| 8.2      | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                         |           |
| +X.1     |                             |                                                                                                                          |           |

\* Concerning the BRK instruction, you should also note that although its second byte is basically a “don’t care” byte – that is, it can have any value - the BRK (and COP instruction as well) is a two-byte instruction, the second byte sometimes is used as a signature byte to determine the nature of the BRK being executed. When an RTI instruction is executed, control always returns to the second byte past the BRK opcode. — assembly-programming-manual-for-w65c816.pdf

\*\* If NMI, IRQ, or RESET, OpCode is forced to 0 and writes to PC are disabled.

\*\*\* If handling a RESET, the writes turn into reads. The databus is populated but ignored.

|                                      |                             |                                                                                                                                                                                                                               |         |
|--------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ORA (01)                             |                             | Cycles: 6                                                                                                                                                                                                                     | Size: 2 |
| Direct Indexed Indirect (d,x) (Read) |                             |                                                                                                                                                                                                                               |         |
| Cycle                                | R/W                         | Desc.                                                                                                                                                                                                                         |         |
| -X.2                                 | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1                                  | Read <b>PC:PB</b>           | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2                                  |                             | Store as Pointer.                                                                                                                                                                                                             |         |
| 2.1                                  |                             | Inc. <b>PC</b> . Set Pointer to Pointer + <b>X</b> + <b>D</b> . Address = Pointer.                                                                                                                                            |         |
| 2.2                                  |                             | If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                      |         |
| 3.1                                  |                             | Address.H = Pointer.H (fixes high byte of address).                                                                                                                                                                           |         |
| 3.2                                  |                             |                                                                                                                                                                                                                               |         |
| 4.1                                  | Read Address                | Store as Pointer.                                                                                                                                                                                                             |         |
| 4.2                                  |                             |                                                                                                                                                                                                                               |         |
| 5.1                                  | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                           |         |
| 5.2                                  |                             |                                                                                                                                                                                                                               |         |
| 6.1                                  |                             |                                                                                                                                                                                                                               |         |
| 6.2                                  | Read Pointer: <b>DB</b>     | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 7.1                                  | Read Pointer + 1: <b>DB</b> | Store as Operand.H.                                                                                                                                                                                                           |         |
| 7.2                                  |                             |                                                                                                                                                                                                                               |         |
| 8.1                                  | Read <b>PC:PB</b>           | Perform <b>A</b>  = Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 8.2                                  |                             | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1                                 |                             |                                                                                                                                                                                                                               |         |

|          |                         |                                                 |
|----------|-------------------------|-------------------------------------------------|
| COP (02) | Cycles: 8               | Size: 2                                         |
| Implied  |                         |                                                 |
| Cycle    | R/W                     | Desc.                                           |
| -X.2     | Read <b>PC:PB</b>       | Store as OpCode.                                |
| 1.1      |                         | Inc. <b>PC</b> .                                |
| 1.2      | Read <b>PC:PB</b>       | Store as Operand.                               |
| 2.1      |                         | Inc. <b>PC</b> .                                |
| 2.2      | Write to ( <b>S</b> +0) | Push <b>PB</b> onto stack.                      |
| 3.1      |                         | Inc. <b>PC</b> .                                |
| 3.2      | Write to ( <b>S</b> -1) | Push <b>PC</b> .H onto stack.                   |
| 4.1      |                         |                                                 |
| 4.2      | Write to ( <b>S</b> -2) | Push <b>PC</b> .L onto stack.                   |
| 5.1      |                         | Set Vector to \$FFE4.                           |
| 5.2      | Write to ( <b>S</b> -3) | Push <b>P</b> onto stack.                       |
| 6.1      |                         | Dec. <b>S</b> by 4.                             |
| 6.2      | Read Vector             | Store to Address.L.                             |
| 7.1      |                         | Set <b>I</b> , unset <b>D</b> .                 |
| 7.2      | Read Vector + 1         | Store to Address.H.                             |
| 8.1      |                         | Copy Address to <b>PC</b> . Set <b>PB</b> to 0. |
| 8.2      | Read <b>PC:PB</b>       | Store as OpCode.                                |
| +X.1     |                         |                                                 |

|                       |                   |                                                                                                                                                                                                                                 |         |
|-----------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ORA (03)              |                   | Cycles: 4                                                                                                                                                                                                                       | Size: 2 |
| Stack Relative (Read) |                   |                                                                                                                                                                                                                                 |         |
| Cycle                 | R/W               | Desc.                                                                                                                                                                                                                           |         |
| -X.2                  | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                |         |
| 1.1                   | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 1.2                   |                   | Store as Pointer.                                                                                                                                                                                                               |         |
| 2.1                   |                   | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                                                                                            |         |
| 2.2                   |                   |                                                                                                                                                                                                                                 |         |
| 3.1                   |                   |                                                                                                                                                                                                                                 |         |
| 3.2                   | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |         |
| 4.1                   | Read Address + 1  |                                                                                                                                                                                                                                 |         |
| 4.2                   |                   | Store as Operand.H.                                                                                                                                                                                                             |         |
| 5.1                   | Read <b>PC:PB</b> | Perform <b>A</b>  = Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 5.2                   |                   | Store as OpCode.                                                                                                                                                                                                                |         |
| +X.1                  |                   |                                                                                                                                                                                                                                 |         |

| TSB (04)                   | Cycles: 5            | Size: 2                                                                                                                                                           |
|----------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct (Read/Modify/Write) |                      |                                                                                                                                                                   |
| Cycle                      | R/W                  | Desc.                                                                                                                                                             |
| -X.2                       | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                  |
| 1.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                  |
| 1.2                        | Read <b>PC:PB</b>    | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                        |
| 2.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                  |
| 2.2                        |                      |                                                                                                                                                                   |
| 3.1                        |                      | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                 |
| 3.2                        | Read Address         | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                   |
| 4.1                        |                      |                                                                                                                                                                   |
| 4.2                        | Read Address + 1     | Store as Operand.H.                                                                                                                                               |
| 5.1                        |                      | If <b>M</b> flag is set, set <b>Z</b> based off ( <b>A.L</b> & Operand.L), otherwise set <b>Z</b> based off ( <b>A</b> & Operand). Perform Operand   = <b>A</b> . |
| 5.2                        |                      | If <b>M</b> flag is set, skip the next cycle.                                                                                                                     |
| 6.1                        |                      |                                                                                                                                                                   |
| 6.2                        | Write to Address + 1 | Write Operand.H.                                                                                                                                                  |
| 7.1                        |                      |                                                                                                                                                                   |
| 7.2                        | Write to Address     | Write Operand.L.                                                                                                                                                  |
| 8.1                        |                      |                                                                                                                                                                   |
| 8.2                        | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                  |
| +X.1                       |                      |                                                                                                                                                                   |

|               |                   |                                                                                                                                                                                                                               |         |
|---------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ORA (05)      |                   | Cycles: 3                                                                                                                                                                                                                     | Size: 2 |
| Direct (Read) |                   |                                                                                                                                                                                                                               |         |
| Cycle         | R/W               | Desc.                                                                                                                                                                                                                         |         |
| -X.2          | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2           | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                    |         |
| 2.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 2.2           |                   |                                                                                                                                                                                                                               |         |
| 3.1           |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                             |         |
| 3.2           | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 4.1           |                   |                                                                                                                                                                                                                               |         |
| 4.2           | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                           |         |
| 5.1           |                   | Perform <b>A</b>  = Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 5.2           | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1          |                   |                                                                                                                                                                                                                               |         |

| ASL (06)                   | Cycles: 5            | Size: 2                                                                                                                                                                                                                                                                                                                              |
|----------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct (Read/Modify/Write) |                      |                                                                                                                                                                                                                                                                                                                                      |
| Cycle                      | R/W                  | Desc.                                                                                                                                                                                                                                                                                                                                |
| -X.2                       | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                     |
| 1.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                     |
| 1.2                        | Read <b>PC:PB</b>    | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                           |
| 2.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                     |
| 2.2                        |                      |                                                                                                                                                                                                                                                                                                                                      |
| 3.1                        |                      | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                    |
| 3.2                        | Read Address         | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                      |
| 4.1                        |                      |                                                                                                                                                                                                                                                                                                                                      |
| 4.2                        | Read Address + 1     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                  |
| 5.1                        |                      | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$80), perform Operand.L <= 1, and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.H & \$80), perform Operand <= 1, and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |
| 5.2                        |                      | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                        |
| 6.1                        |                      |                                                                                                                                                                                                                                                                                                                                      |
| 6.2                        | Write to Address + 1 | Write Operand.H.                                                                                                                                                                                                                                                                                                                     |
| 7.1                        |                      |                                                                                                                                                                                                                                                                                                                                      |
| 7.2                        | Write to Address     | Write Operand.L.                                                                                                                                                                                                                                                                                                                     |
| 8.1                        |                      |                                                                                                                                                                                                                                                                                                                                      |
| 8.2                        | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                     |
| +X.1                       |                      |                                                                                                                                                                                                                                                                                                                                      |

|                             |                                    |                                                                                                                                                                                                                               |         |
|-----------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ORA (07)                    |                                    | Cycles: 6                                                                                                                                                                                                                     | Size: 2 |
| Direct Indirect Long (Read) |                                    |                                                                                                                                                                                                                               |         |
| Cycle                       | R/W                                | Desc.                                                                                                                                                                                                                         |         |
| -X.2                        | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1                         |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2                         | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                    |         |
| 2.1                         |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 2.2                         |                                    |                                                                                                                                                                                                                               |         |
| 3.1                         |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                             |         |
| 3.2                         | Read Address                       | Store as Pointer.                                                                                                                                                                                                             |         |
| 4.1                         |                                    |                                                                                                                                                                                                                               |         |
| 4.2                         | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                           |         |
| 5.1                         |                                    |                                                                                                                                                                                                                               |         |
| 5.2                         | Read Address + 2                   | Store as Bank.                                                                                                                                                                                                                |         |
| 6.1                         |                                    |                                                                                                                                                                                                                               |         |
| 6.2                         | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 7.1                         |                                    |                                                                                                                                                                                                                               |         |
| 7.2                         | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                           |         |
| 8.1                         |                                    | Perform <b>A</b>  = Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 8.2                         | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1                        |                                    |                                                                                                                                                                                                                               |         |



|          |                         |                             |
|----------|-------------------------|-----------------------------|
| PHP (08) | Cycles: 3               | Size: 1                     |
| Implied  |                         |                             |
| Cycle    | R/W                     | Desc.                       |
| -X.2     | Read <b>PC:PB</b>       | Store as OpCode.            |
| 1.1      |                         | Inc. <b>PC</b> .            |
| 1.2      |                         |                             |
| 2.1      |                         | Set Operand.L to <b>P</b> . |
| 2.2      | Write to ( <b>S</b> +0) | Push Operand.L onto stack.  |
| 3.1      |                         | Dec. <b>S</b> by 1.         |
| 3.2      | Read <b>PC:PB</b>       | Store as OpCode.            |
| +X.1     |                         |                             |

|           |                   |                                                                                                                                                                                                                                                |
|-----------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ORA (09)  | Cycles: 2         | Size: 3                                                                                                                                                                                                                                        |
| Immediate |                   |                                                                                                                                                                                                                                                |
| Cycle     | R/W               | Desc.                                                                                                                                                                                                                                          |
| -X.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                               |
| 1.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                               |
| 1.2       | Read <b>PC:PB</b> | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                |
| 2.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                               |
| 2.2       | Read <b>PC:PB</b> | Store as Operand.H.                                                                                                                                                                                                                            |
| 3.1       |                   | Inc. <b>PC</b> . Perform <b>A</b>  = Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |
| 3.2       | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                               |
| +X.1      |                   |                                                                                                                                                                                                                                                |

|          |                   |                                                                                                                                                                                     |
|----------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASL (0A) | Cycles: 2         | Size: 1                                                                                                                                                                             |
| Implied  |                   |                                                                                                                                                                                     |
| Cycle    | R/W               | Desc.                                                                                                                                                                               |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                    |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                    |
| 1.2      |                   | Discard.                                                                                                                                                                            |
| 2.1      |                   | If <b>M</b> flag is set, set <b>C</b> based off ( <b>A.L</b> & \$80), perform <b>A.L</b> <<= 1, and set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , |
| 2.2      |                   | otherwise set <b>C</b> based off ( <b>A.H</b> & \$80), perform <b>A</b> <<= 1, and set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> .                    |
| +X.1     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                    |

|          |                         |                              |
|----------|-------------------------|------------------------------|
| PHD (0B) | Cycles: 4               | Size: 1                      |
| Implied  |                         |                              |
| Cycle    | R/W                     | Desc.                        |
| -X.2     | Read <b>PC:PB</b>       | Store as OpCode.             |
| 1.1      |                         | Inc. <b>PC</b> .             |
| 1.2      |                         |                              |
| 2.1      |                         |                              |
| 2.2      | Write to ( <b>S</b> +0) | Push <b>D</b> .H onto stack. |
| 3.1      |                         |                              |
| 3.2      | Write to ( <b>S</b> -1) | Push <b>D</b> .L onto stack. |
| 4.1      |                         |                              |
| 4.2      | Read <b>PC:PB</b>       | Store as OpCode.             |
| +X.1     |                         |                              |

| TSB (0C)                     | Cycles: 6                                    | Size: 3                                                                                                                                                            |
|------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute (Read/Modify/Write) |                                              |                                                                                                                                                                    |
| Cycle                        | R/W                                          | Desc.                                                                                                                                                              |
| -X.2                         | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                   |
| 1.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                   |
| 1.2                          | Read <b>PC:PB</b>                            | Store as Address.                                                                                                                                                  |
| 2.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                   |
| 2.2                          | Read <b>PC:PB</b>                            | Store as Address.L.                                                                                                                                                |
| 3.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                   |
| 3.2                          | Read Address: <b>DB</b>                      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                    |
| 4.1                          |                                              |                                                                                                                                                                    |
| 4.2                          | Read Address + 1: <b>DB</b> (With Carry)     | Store as Operand.H.                                                                                                                                                |
| 5.1                          |                                              | If <b>M</b> flag is set, set <b>Z</b> based off ( <b>A</b> .L & Operand.L), otherwise set <b>Z</b> based off ( <b>A</b> & Operand). Perform Operand   = <b>A</b> . |
| 5.2                          |                                              | If <b>M</b> flag is set, skip the next cycle.                                                                                                                      |
| 6.1                          |                                              |                                                                                                                                                                    |
| 6.2                          | Write to Address + 1: <b>DB</b> (With Carry) | Write Operand.H.                                                                                                                                                   |
| 7.1                          |                                              |                                                                                                                                                                    |
| 7.2                          | Write to Address: <b>DB</b>                  | Write Operand.L.                                                                                                                                                   |
| 8.1                          |                                              |                                                                                                                                                                    |
| 8.2                          | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                   |
| +X.1                         |                                              |                                                                                                                                                                    |

|                 |                                          |                                                                                                                                                                                                                          |
|-----------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ORA (0D)        | Cycles: 4                                | Size: 3                                                                                                                                                                                                                  |
| Absolute (Read) |                                          |                                                                                                                                                                                                                          |
| Cycle           | R/W                                      | Desc.                                                                                                                                                                                                                    |
| -X.2            | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                         |
| 1.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                         |
| 1.2             | Read <b>PC:PB</b>                        | Store as Address.                                                                                                                                                                                                        |
| 2.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                         |
| 2.2             | Read <b>PC:PB</b>                        | Store as Address.L.                                                                                                                                                                                                      |
| 3.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                         |
| 3.2             | Read Address: <b>DB</b>                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                          |
| 4.1             |                                          |                                                                                                                                                                                                                          |
| 4.2             | Read Address + 1: <b>DB</b> (With Carry) | Store as Operand.H.                                                                                                                                                                                                      |
| 5.1             |                                          | Perform <b>A</b>  = Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set N based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |
| 5.2             | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                         |
| +X.1            |                                          |                                                                                                                                                                                                                          |

| ASL (0E)                     | Cycles: 6                                    | Size: 3                                                                                                                                                                                                                                                                                                                                |
|------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute (Read/Modify/Write) |                                              |                                                                                                                                                                                                                                                                                                                                        |
| Cycle                        | R/W                                          | Desc.                                                                                                                                                                                                                                                                                                                                  |
| -X.2                         | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| 1.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 1.2                          | Read <b>PC:PB</b>                            | Store as Address.                                                                                                                                                                                                                                                                                                                      |
| 2.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 2.2                          | Read <b>PC:PB</b>                            | Store as Address.L.                                                                                                                                                                                                                                                                                                                    |
| 3.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 3.2                          | Read Address: <b>DB</b>                      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                        |
| 4.1                          |                                              |                                                                                                                                                                                                                                                                                                                                        |
| 4.2                          | Read Address + 1: <b>DB</b> (With Carry)     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                    |
| 5.1                          |                                              | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$80), perform Operand.L <<= 1, and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.H & \$80), perform Operand <<= 1, and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |
| 5.2                          |                                              | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                          |
| 6.1                          |                                              |                                                                                                                                                                                                                                                                                                                                        |
| 6.2                          | Write to Address + 1: <b>DB</b> (With Carry) | Write Operand.H.                                                                                                                                                                                                                                                                                                                       |
| 7.1                          |                                              |                                                                                                                                                                                                                                                                                                                                        |
| 7.2                          | Write to Address: <b>DB</b>                  | Write Operand.L.                                                                                                                                                                                                                                                                                                                       |
| 8.1                          |                                              |                                                                                                                                                                                                                                                                                                                                        |
| 8.2                          | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| +X.1                         |                                              |                                                                                                                                                                                                                                                                                                                                        |

|                      |                                    |                                                                                                                                                                                                                                 |
|----------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ORA (0F)             | Cycles: 5                          | Size: 4                                                                                                                                                                                                                         |
| Absolute Long (Read) |                                    |                                                                                                                                                                                                                                 |
| Cycle                | R/W                                | Desc.                                                                                                                                                                                                                           |
| -X.2                 | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |
| 1.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 1.2                  | Read <b>PC:PB</b>                  | Store as Address.                                                                                                                                                                                                               |
| 2.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 2.2                  | Read <b>PC:PB</b>                  | Store as Address.L.                                                                                                                                                                                                             |
| 3.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 3.2                  | Read <b>PC:PB</b>                  | Stores as Bank.                                                                                                                                                                                                                 |
| 4.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 4.2                  | Read Address:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |
| 5.1                  |                                    |                                                                                                                                                                                                                                 |
| 5.2                  | Read Address + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                             |
| 6.1                  |                                    | Perform <b>A</b>  = Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |
| 6.2                  | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |
| +X.1                 |                                    |                                                                                                                                                                                                                                 |



|          |                   |                                                                                                 |
|----------|-------------------|-------------------------------------------------------------------------------------------------|
| BPL (10) | Cycles: 2         | Size: 2                                                                                         |
| Relative |                   |                                                                                                 |
| Cycle    | R/W               | Desc.                                                                                           |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                |
| 1.1      |                   | Inc. <b>PC</b> . Decide to branch if <b>N</b> is 0.                                             |
| 1.2      | Read <b>PC:PB</b> | Store as Operand. Address = i16(i8(Operand.L)) + <b>PC</b> . If not branching, poll interrupts. |
| 2.1      |                   | Inc. <b>PC</b> . If not branching, end (next half-cycle is 3.2).                                |
| 2.2      |                   | Poll interrupts.                                                                                |
| 3.1      |                   | Set <b>PC</b> to Address.                                                                       |
| 3.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                |
| +X.1     |                   |                                                                                                 |

|                                      |                                    |                                                                                                                                                                                                                               |         |
|--------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ORA (11)                             |                                    | Cycles: 5                                                                                                                                                                                                                     | Size: 2 |
| Direct Indirect Indexed (d),y (Read) |                                    |                                                                                                                                                                                                                               |         |
| Cycle                                | R/W                                | Desc.                                                                                                                                                                                                                         |         |
| -X.2                                 | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1                                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2                                  | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                    |         |
| 2.1                                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 2.2                                  |                                    |                                                                                                                                                                                                                               |         |
| 3.1                                  |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                             |         |
| 3.2                                  | Read Address                       | Store as Pointer.                                                                                                                                                                                                             |         |
| 4.1                                  |                                    |                                                                                                                                                                                                                               |         |
| 4.2                                  | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                           |         |
| 5.1                                  |                                    | Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                            |         |
| 5.2                                  |                                    |                                                                                                                                                                                                                               |         |
| 6.1                                  |                                    |                                                                                                                                                                                                                               |         |
| 6.2                                  | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 7.1                                  |                                    |                                                                                                                                                                                                                               |         |
| 7.2                                  | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                           |         |
| 8.1                                  |                                    | Perform <b>A</b>  = Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 8.2                                  | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1                                 |                                    |                                                                                                                                                                                                                               |         |

|                        |                             |                                                                                                                                                                                                                               |
|------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ORA (12)               | Cycles: 5                   | Size: 2                                                                                                                                                                                                                       |
| Direct Indirect (Read) |                             |                                                                                                                                                                                                                               |
| Cycle                  | R/W                         | Desc.                                                                                                                                                                                                                         |
| -X.2                   | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                              |
| 1.1                    |                             | Inc. <b>PC</b> .                                                                                                                                                                                                              |
| 1.2                    | Read <b>PC:PB</b>           | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                    |
| 2.1                    |                             | Inc. <b>PC</b> .                                                                                                                                                                                                              |
| 2.2                    |                             |                                                                                                                                                                                                                               |
| 3.1                    |                             | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                             |
| 3.2                    | Read Address                | Store as Pointer.                                                                                                                                                                                                             |
| 4.1                    |                             |                                                                                                                                                                                                                               |
| 4.2                    | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                           |
| 5.1                    |                             |                                                                                                                                                                                                                               |
| 5.2                    | Read <b>PC:PB</b>           | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |
| 6.1                    |                             |                                                                                                                                                                                                                               |
| 6.2                    | Read Pointer + 1: <b>DB</b> | Store as Operand.H.                                                                                                                                                                                                           |
| 7.1                    |                             | Perform <b>A</b>  = Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |
| 7.2                    | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                              |
| +X.1                   |                             |                                                                                                                                                                                                                               |

|                                        |                                    |                                                                                                                                                                                                                                 |         |
|----------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ORA (13)                               |                                    | Cycles: 7                                                                                                                                                                                                                       | Size: 2 |
| Stack Relative Indirect Indexed (Read) |                                    |                                                                                                                                                                                                                                 |         |
| Cycle                                  | R/W                                | Desc.                                                                                                                                                                                                                           |         |
| -X.2                                   | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| 1.1                                    | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 1.2                                    |                                    | Store as Pointer.                                                                                                                                                                                                               |         |
| 2.1                                    |                                    | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                                                                                            |         |
| 2.2                                    |                                    |                                                                                                                                                                                                                                 |         |
| 3.1                                    |                                    |                                                                                                                                                                                                                                 |         |
| 3.2                                    | Read Address                       | Store as Pointer.                                                                                                                                                                                                               |         |
| 4.1                                    | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                             |         |
| 4.2                                    |                                    | Perform Orig = Pointer.                                                                                                                                                                                                         |         |
| 5.1                                    |                                    | Tmp = ui32(Pointer + <b>Y</b> ).                                                                                                                                                                                                |         |
|                                        |                                    | Pointer = ui16(Tmp).                                                                                                                                                                                                            |         |
|                                        |                                    | Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                                 |         |
| 5.2                                    | Read Pointer:Bank                  |                                                                                                                                                                                                                                 |         |
| 6.1                                    |                                    |                                                                                                                                                                                                                                 |         |
| 6.2                                    |                                    | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |         |
| 7.1                                    |                                    |                                                                                                                                                                                                                                 |         |
| 7.2                                    |                                    | Store as Operand.H.                                                                                                                                                                                                             |         |
| 8.1                                    | Read Pointer + 1:Bank (With Carry) | Perform <b>A</b>  = Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 8.2                                    |                                    | Store as OpCode.                                                                                                                                                                                                                |         |
| +X.1                                   |                                    |                                                                                                                                                                                                                                 |         |

| TRB (14)                   |                      | Cycles: 5                                                                                                                                                          | Size: 2 |
|----------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Direct (Read/Modify/Write) |                      |                                                                                                                                                                    |         |
| Cycle                      | R/W                  | Desc.                                                                                                                                                              |         |
| -X.2                       | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                   |         |
| 1.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                   |         |
| 1.2                        | Read <b>PC:PB</b>    | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                         |         |
| 2.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                   |         |
| 2.2                        |                      |                                                                                                                                                                    |         |
| 3.1                        |                      | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                  |         |
| 3.2                        | Read Address         | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                    |         |
| 4.1                        |                      |                                                                                                                                                                    |         |
| 4.2                        | Read Address + 1     | Store as Operand.H.                                                                                                                                                |         |
| 5.1                        |                      | If <b>M</b> flag is set, set <b>Z</b> based off ( <b>A.L</b> & Operand.L), otherwise set <b>Z</b> based off ( <b>A</b> & Operand). Perform Operand &= ~ <b>A</b> . |         |
| 5.2                        |                      | If <b>M</b> flag is set, skip the next cycle.                                                                                                                      |         |
| 6.1                        |                      |                                                                                                                                                                    |         |
| 6.2                        | Write to Address + 1 | Write Operand.H.                                                                                                                                                   |         |
| 7.1                        |                      |                                                                                                                                                                    |         |
| 7.2                        | Write to Address     | Write Operand.L.                                                                                                                                                   |         |
| 8.1                        |                      |                                                                                                                                                                    |         |
| 8.2                        | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                   |         |
| +X.1                       |                      |                                                                                                                                                                    |         |

|                  |                   |                                                                                                                                                                                                                               |         |
|------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ORA (15)         |                   | Cycles: 4                                                                                                                                                                                                                     | Size: 2 |
| Direct, X (Read) |                   |                                                                                                                                                                                                                               |         |
| Cycle            | R/W               | Desc.                                                                                                                                                                                                                         |         |
| -X.2             | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1              | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2              |                   | Store as Operand.                                                                                                                                                                                                             |         |
| 2.1              |                   | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 2.2              |                   |                                                                                                                                                                                                                               |         |
| 3.1              |                   | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                             |         |
| 3.2              |                   |                                                                                                                                                                                                                               |         |
| 4.1              |                   |                                                                                                                                                                                                                               |         |
| 4.2              | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 5.1              | Read Address + 1  |                                                                                                                                                                                                                               |         |
| 5.2              |                   | Store as Operand.H.                                                                                                                                                                                                           |         |
| 6.1              | Read <b>PC:PB</b> | Perform <b>A</b>  = Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 6.2              |                   | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1             |                   |                                                                                                                                                                                                                               |         |

| ASL (16)                      | Cycles: 6            | Size: 2                                                                                                                                                                                                                                                                                                                              |
|-------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct, X (Read/Modify/Write) |                      |                                                                                                                                                                                                                                                                                                                                      |
| Cycle                         | R/W                  | Desc.                                                                                                                                                                                                                                                                                                                                |
| -X.2                          | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                     |
| 1.1                           | Read <b>PC:PB</b>    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                     |
| 1.2                           |                      | Store as Operand.                                                                                                                                                                                                                                                                                                                    |
| 2.1                           |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                     |
| 2.2                           |                      |                                                                                                                                                                                                                                                                                                                                      |
| 3.1                           |                      | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                                                                                                                                    |
| 3.2                           |                      |                                                                                                                                                                                                                                                                                                                                      |
| 4.1                           |                      |                                                                                                                                                                                                                                                                                                                                      |
| 4.2                           | Read Address         | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                      |
| 5.1                           |                      |                                                                                                                                                                                                                                                                                                                                      |
| 5.2                           | Read Address + 1     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                  |
| 6.1                           |                      | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$80), perform Operand.L <= 1, and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.H & \$80), perform Operand <= 1, and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |
| 6.2                           |                      | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                        |
| 7.1                           |                      |                                                                                                                                                                                                                                                                                                                                      |
| 7.2                           | Write to Address + 1 | Write Operand.H.                                                                                                                                                                                                                                                                                                                     |
| 8.1                           |                      |                                                                                                                                                                                                                                                                                                                                      |
| 8.2                           | Write to Address     | Write Operand.L.                                                                                                                                                                                                                                                                                                                     |
| 9.1                           |                      |                                                                                                                                                                                                                                                                                                                                      |
| 9.2                           | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                     |
| +X.1                          |                      |                                                                                                                                                                                                                                                                                                                                      |

|                                           |                                    |                                                                                                                                                                                                                               |         |
|-------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ORA (17)                                  |                                    | Cycles: 6                                                                                                                                                                                                                     | Size: 2 |
| Direct Indirect Indexed Long [d],y (Read) |                                    |                                                                                                                                                                                                                               |         |
| Cycle                                     | R/W                                | Desc.                                                                                                                                                                                                                         |         |
| -X.2                                      | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1                                       |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2                                       | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                    |         |
| 2.1                                       |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 2.2                                       |                                    |                                                                                                                                                                                                                               |         |
| 3.1                                       |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                             |         |
| 3.2                                       | Read Address                       | Store as Pointer.                                                                                                                                                                                                             |         |
| 4.1                                       |                                    |                                                                                                                                                                                                                               |         |
| 4.2                                       | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                           |         |
| 5.1                                       |                                    |                                                                                                                                                                                                                               |         |
| 5.2                                       | Read Address + 2                   | Store as Bank.                                                                                                                                                                                                                |         |
| 6.1                                       |                                    | Perform $Tmp = ui32(Pointer + Y)$ .                                                                                                                                                                                           |         |
|                                           |                                    | Pointer = $ui16(Tmp)$ .                                                                                                                                                                                                       |         |
|                                           |                                    | Bank += $(Tmp \gg 16)$ .                                                                                                                                                                                                      |         |
| 6.2                                       | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 7.1                                       |                                    |                                                                                                                                                                                                                               |         |
| 7.2                                       | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                           |         |
| 8.1                                       |                                    | Perform $A \mid = Operand$ . If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 8.2                                       | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1                                      |                                    |                                                                                                                                                                                                                               |         |



|          |                   |                             |
|----------|-------------------|-----------------------------|
| CLC (18) | Cycles: 2         | Size: 1                     |
| Implied  |                   |                             |
| Cycle    | R/W               | Desc.                       |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.            |
| 1.1      |                   | Inc. <b>PC</b> .            |
| 1.2      | Read <b>PC:PB</b> | Discard.                    |
| 2.1      |                   | Set the <b>C</b> flag to 0. |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.            |
| +X.1     |                   |                             |

|                    |                                    |                                                                                                                                                                                                                                 |         |
|--------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ORA (19)           |                                    | Cycles: 4                                                                                                                                                                                                                       | Size: 3 |
| Absolute, Y (Read) |                                    |                                                                                                                                                                                                                                 |         |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                           |         |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| 1.1                | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 1.2                |                                    | Store as Pointer.                                                                                                                                                                                                               |         |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 2.2                |                                    | Store as Pointer.H.                                                                                                                                                                                                             |         |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).             |         |
| 3.2                |                                    |                                                                                                                                                                                                                                 |         |
| 4.1                |                                    |                                                                                                                                                                                                                                 |         |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |         |
| 5.1                | Read Pointer + 1:Bank (With Carry) |                                                                                                                                                                                                                                 |         |
| 5.2                |                                    | Store as Operand.H.                                                                                                                                                                                                             |         |
| 6.1                | Read <b>PC:PB</b>                  | Perform <b>A</b>  = Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 6.2                |                                    | Store as OpCode.                                                                                                                                                                                                                |         |
| +X.1               |                                    |                                                                                                                                                                                                                                 |         |

|          |                   |                                                                                                                                                                                                                                                            |
|----------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INC (1A) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                    |
| Implied  |                   |                                                                                                                                                                                                                                                            |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                      |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                           |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                           |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                   |
| 2.1      |                   | If <b>M</b> flag is set, perform <b>A.L</b> += 1, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise perform <b>A</b> += 1, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                           |
| +X.1     |                   |                                                                                                                                                                                                                                                            |

|          |                   |                            |
|----------|-------------------|----------------------------|
| TCS (1B) | Cycles: 2         | Size: 1                    |
| Implied  |                   |                            |
| Cycle    | R/W               | Desc.                      |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.           |
| 1.1      |                   | Inc. <b>PC</b> .           |
| 1.2      | Read <b>PC:PB</b> | Discard.                   |
| 2.1      |                   | Set <b>S</b> to <b>A</b> . |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.           |
| +X.1     |                   |                            |

|                              |                                              |                                                                                                                                                                     |
|------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TRB (1C)                     | Cycles: 6                                    | Size: 3                                                                                                                                                             |
| Absolute (Read/Modify/Write) |                                              |                                                                                                                                                                     |
| Cycle                        | R/W                                          | Desc.                                                                                                                                                               |
| -X.2                         | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                    |
| 1.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                    |
| 1.2                          | Read <b>PC:PB</b>                            | Store as Address.                                                                                                                                                   |
| 2.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                    |
| 2.2                          | Read <b>PC:PB</b>                            | Store as Address.L.                                                                                                                                                 |
| 3.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                    |
| 3.2                          | Read Address: <b>DB</b>                      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                     |
| 4.1                          |                                              |                                                                                                                                                                     |
| 4.2                          | Read Address + 1: <b>DB</b> (With Carry)     | Store as Operand.H.                                                                                                                                                 |
| 5.1                          |                                              | If <b>M</b> flag is set, set <b>Z</b> based off ( <b>A</b> .L & Operand.L), otherwise set <b>Z</b> based off ( <b>A</b> & Operand). Perform Operand &= ~ <b>A</b> . |
| 5.2                          |                                              | If <b>M</b> flag is set, skip the next cycle.                                                                                                                       |
| 6.1                          |                                              |                                                                                                                                                                     |
| 6.2                          | Write to Address + 1: <b>DB</b> (With Carry) | Write Operand.H.                                                                                                                                                    |
| 7.1                          |                                              |                                                                                                                                                                     |
| 7.2                          | Write to Address: <b>DB</b>                  | Write Operand.L.                                                                                                                                                    |
| 8.1                          |                                              |                                                                                                                                                                     |
| 8.2                          | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                    |
| +X.1                         |                                              |                                                                                                                                                                     |

|                    |                                    |                                                                                                                                                                                                                                 |         |
|--------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ORA (1D)           |                                    | Cycles: 4                                                                                                                                                                                                                       | Size: 3 |
| Absolute, X (Read) |                                    |                                                                                                                                                                                                                                 |         |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                           |         |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| 1.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 1.2                | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                               |         |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 2.2                | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                             |         |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>X</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).             |         |
| 3.2                |                                    |                                                                                                                                                                                                                                 |         |
| 4.1                |                                    |                                                                                                                                                                                                                                 |         |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |         |
| 5.1                |                                    |                                                                                                                                                                                                                                 |         |
| 5.2                | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                             |         |
| 6.1                |                                    | Perform <b>A</b>  = Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 6.2                | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| +X.1               |                                    |                                                                                                                                                                                                                                 |         |

| ASL (1E)                       | Cycles: 7                              | Size: 3                                                                                                                                                                                                                                                                                                                              |
|--------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute, X (Read/Modiy/Write) |                                        |                                                                                                                                                                                                                                                                                                                                      |
| Cycle                          | R/W                                    | Desc.                                                                                                                                                                                                                                                                                                                                |
| -X.2                           | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                                                                                                                                                                                     |
| 1.1                            |                                        | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                     |
| 1.2                            | Read <b>PC:PB</b>                      | Store as Pointer.                                                                                                                                                                                                                                                                                                                    |
| 2.1                            |                                        | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                     |
| 2.2                            | Read <b>PC:PB</b>                      | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                  |
| 3.1                            |                                        | Inc. <b>PC</b> .<br>Perform Tmp = ui32(Pointer + <b>X</b> ).<br>Pointer = ui16(Tmp).<br>Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                                              |
| 3.2                            |                                        |                                                                                                                                                                                                                                                                                                                                      |
| 4.1                            |                                        |                                                                                                                                                                                                                                                                                                                                      |
| 4.2                            | Read Pointer:Bank                      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                      |
| 5.1                            |                                        |                                                                                                                                                                                                                                                                                                                                      |
| 5.2                            | Read Pointer + 1:Bank (With Carry)     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                  |
| 6.1                            |                                        | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$80), perform Operand.L <= 1, and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.H & \$80), perform Operand <= 1, and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |
| 6.2                            |                                        | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                        |
| 7.1                            |                                        |                                                                                                                                                                                                                                                                                                                                      |
| 7.2                            | Write to Pointer + 1:Bank (With Carry) | Write Operand.H.                                                                                                                                                                                                                                                                                                                     |
| 8.1                            |                                        |                                                                                                                                                                                                                                                                                                                                      |
| 8.2                            | Write to Pointer:Bank                  | Write Operand.L.                                                                                                                                                                                                                                                                                                                     |
| 9.1                            |                                        |                                                                                                                                                                                                                                                                                                                                      |
| 9.2                            | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                                                                                                                                                                                     |
| +X.1                           |                                        |                                                                                                                                                                                                                                                                                                                                      |

|                         |                                    |                                                                                                                                                                                                                                 |         |
|-------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ORA (1F)                |                                    | Cycles: 5                                                                                                                                                                                                                       | Size: 4 |
| Absolute Long, X (Read) |                                    |                                                                                                                                                                                                                                 |         |
| Cycle                   | R/W                                | Desc.                                                                                                                                                                                                                           |         |
| -X.2                    | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| 1.1                     |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 1.2                     | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                               |         |
| 2.1                     |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 2.2                     | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                             |         |
| 3.1                     |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 3.2                     | Read <b>PC:PB</b>                  | Stores as Bank.                                                                                                                                                                                                                 |         |
| 4.1                     |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
|                         |                                    | Perform Tmp = ui32(Pointer + <b>X</b> ).                                                                                                                                                                                        |         |
|                         |                                    | Pointer = ui16(Tmp).                                                                                                                                                                                                            |         |
|                         |                                    | Bank += (Tmp >> 16).                                                                                                                                                                                                            |         |
| 4.2                     | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |         |
| 5.1                     |                                    |                                                                                                                                                                                                                                 |         |
| 5.2                     | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                             |         |
| 6.1                     |                                    | Perform <b>A</b>  = Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 6.2                     | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| +X.1                    |                                    |                                                                                                                                                                                                                                 |         |



|          |                         |                                                  |
|----------|-------------------------|--------------------------------------------------|
| JSR (20) | Cycles: 6               | Size: 3                                          |
| Absolute |                         |                                                  |
| Cycle    | R/W                     | Desc.                                            |
| -X.2     | Read <b>PC:PB</b>       | Store as OpCode.                                 |
| 1.1      |                         | Inc. <b>PC</b> .                                 |
| 1.2      | Read <b>PC:PB</b>       | Store as Address.                                |
| 2.1      |                         | Inc. <b>PC</b> .                                 |
| 2.2      | Read <b>PC:PB</b>       | Store as Address.L.                              |
| 3.1      |                         | Inc. <b>PC</b> .                                 |
| 3.2      | Read <b>PC:PB</b>       | Discard.                                         |
| 4.1      |                         | Dec. <b>PC</b> .                                 |
| 4.2      | Write to ( <b>S</b> +0) | Push <b>PC</b> .H onto stack.                    |
| 5.1      |                         |                                                  |
| 5.2      | Write to ( <b>S</b> -1) | Push <b>PC</b> .L onto stack.                    |
| 6.1      |                         | Dec. <b>S</b> by 2. Perform <b>PC</b> = Address. |
| 6.2      | Read <b>PC:PB</b>       | Store as OpCode.                                 |
| +X.1     |                         |                                                  |

|                                      |                             |                                                                                                                                                                                                                               |         |
|--------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AND (21)                             |                             | Cycles: 6                                                                                                                                                                                                                     | Size: 2 |
| Direct Indexed Indirect (d,x) (Read) |                             |                                                                                                                                                                                                                               |         |
| Cycle                                | R/W                         | Desc.                                                                                                                                                                                                                         |         |
| -X.2                                 | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1                                  |                             | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2                                  | Read <b>PC:PB</b>           | Store as Pointer.                                                                                                                                                                                                             |         |
| 2.1                                  |                             | Inc. <b>PC</b> . Set Pointer to Pointer + <b>X</b> + <b>D</b> . Address = Pointer.                                                                                                                                            |         |
| 2.2                                  |                             | If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                      |         |
| 3.1                                  |                             | Address.H = Pointer.H (fixes high byte of address).                                                                                                                                                                           |         |
| 3.2                                  |                             |                                                                                                                                                                                                                               |         |
| 4.1                                  |                             |                                                                                                                                                                                                                               |         |
| 4.2                                  | Read Address                | Store as Pointer.                                                                                                                                                                                                             |         |
| 5.1                                  |                             |                                                                                                                                                                                                                               |         |
| 5.2                                  | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                           |         |
| 6.1                                  |                             |                                                                                                                                                                                                                               |         |
| 6.2                                  | Read Pointer: <b>DB</b>     | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 7.1                                  |                             |                                                                                                                                                                                                                               |         |
| 7.2                                  | Read Pointer + 1: <b>DB</b> | Store as Operand.H.                                                                                                                                                                                                           |         |
| 8.1                                  |                             | Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 8.2                                  | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1                                 |                             |                                                                                                                                                                                                                               |         |

|               |                         |                                                                                          |
|---------------|-------------------------|------------------------------------------------------------------------------------------|
| JSL (22)      | Cycles: 8               | Size: 4                                                                                  |
| Absolute Long |                         |                                                                                          |
| Cycle         | R/W                     | Desc.                                                                                    |
| -X.2          | Read <b>PC:PB</b>       | Store as OpCode.                                                                         |
| 1.1           |                         | Inc. <b>PC</b> .                                                                         |
| 1.2           | Read <b>PC:PB</b>       | Store as Address.                                                                        |
| 2.1           |                         | Inc. <b>PC</b> .                                                                         |
| 2.2           | Read <b>PC:PB</b>       | Store as Address.L.                                                                      |
| 3.1           |                         | Inc. <b>PC</b> .                                                                         |
| 3.2           | Write to ( <b>S</b> +0) | Push <b>PB</b> onto stack.                                                               |
| 4.1           |                         |                                                                                          |
| 4.2           | Read ( <b>S</b> +0)     | Discard.                                                                                 |
| 5.1           |                         |                                                                                          |
| 5.2           | Read <b>PC:PB</b>       | Stores as Bank.                                                                          |
| 6.1           |                         |                                                                                          |
| 6.2           | Write to ( <b>S</b> -1) | Push <b>PC</b> .H onto stack.                                                            |
| 7.1           |                         |                                                                                          |
| 7.2           | Write to ( <b>S</b> -2) | Push <b>PC</b> .L onto stack.                                                            |
| 8.1           |                         | Inc <b>PC</b> . Dec. <b>S</b> by 3. Copy Address to <b>PC</b> . Copy Bank to <b>PB</b> . |
| 8.2           | Read <b>PC:PB</b>       | Store as OpCode.                                                                         |
| +X.1          |                         |                                                                                          |

|                       |                   |                                                                                                                                                                                                                                 |         |
|-----------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AND (23)              |                   | Cycles: 4                                                                                                                                                                                                                       | Size: 2 |
| Stack Relative (Read) |                   |                                                                                                                                                                                                                                 |         |
| Cycle                 | R/W               | Desc.                                                                                                                                                                                                                           |         |
| -X.2                  | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                |         |
| 1.1                   | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 1.2                   |                   | Store as Pointer.                                                                                                                                                                                                               |         |
| 2.1                   |                   | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                                                                                            |         |
| 2.2                   |                   |                                                                                                                                                                                                                                 |         |
| 3.1                   |                   |                                                                                                                                                                                                                                 |         |
| 3.2                   | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |         |
| 4.1                   | Read Address + 1  |                                                                                                                                                                                                                                 |         |
| 4.2                   |                   | Store as Operand.H.                                                                                                                                                                                                             |         |
| 5.1                   | Read <b>PC:PB</b> | Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 5.2                   |                   | Store as OpCode.                                                                                                                                                                                                                |         |
| +X.1                  |                   |                                                                                                                                                                                                                                 |         |

|               |                   |                                                                                                                                                                                                                                                                                                                    |
|---------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BIT (24)      | Cycles: 3         | Size: 2                                                                                                                                                                                                                                                                                                            |
| Direct (Read) |                   |                                                                                                                                                                                                                                                                                                                    |
| Cycle         | R/W               | Desc.                                                                                                                                                                                                                                                                                                              |
| -X.2          | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                   |
| 1.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                   |
| 1.2           | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                         |
| 2.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                   |
| 2.2           |                   |                                                                                                                                                                                                                                                                                                                    |
| 3.1           |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                  |
| 3.2           | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                    |
| 4.1           |                   |                                                                                                                                                                                                                                                                                                                    |
| 4.2           | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                                                                                |
| 5.1           |                   | If <b>M</b> flag is set, set <b>Z</b> based off ( <b>A.L</b> & Operand.L), set <b>N</b> based off (Operand.L & \$80), and set <b>V</b> based off (Operand.L & \$40), otherwise set <b>Z</b> based off ( <b>A</b> & Operand), set <b>N</b> based off (Operand.H & \$80), set <b>V</b> based off (Operand.H & \$40). |
| 5.2           | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                   |
| +X.1          |                   |                                                                                                                                                                                                                                                                                                                    |

|               |                   |                                                                                                                                                                                                                               |         |
|---------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AND (25)      |                   | Cycles: 3                                                                                                                                                                                                                     | Size: 2 |
| Direct (Read) |                   |                                                                                                                                                                                                                               |         |
| Cycle         | R/W               | Desc.                                                                                                                                                                                                                         |         |
| -X.2          | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2           | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                    |         |
| 2.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 2.2           |                   |                                                                                                                                                                                                                               |         |
| 3.1           |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                             |         |
| 3.2           | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 4.1           |                   |                                                                                                                                                                                                                               |         |
| 4.2           | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                           |         |
| 5.1           |                   | Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 5.2           | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1          |                   |                                                                                                                                                                                                                               |         |

|                            |                      |                                                                                                                                                                                                                                                                                                                                                                                        |         |
|----------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ROL (26)                   |                      | Cycles: 5                                                                                                                                                                                                                                                                                                                                                                              | Size: 2 |
| Direct (Read/Modify/Write) |                      |                                                                                                                                                                                                                                                                                                                                                                                        |         |
| Cycle                      | R/W                  | Desc.                                                                                                                                                                                                                                                                                                                                                                                  |         |
| -X.2                       | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                       |         |
| 1.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                       |         |
| 1.2                        | Read <b>PC:PB</b>    | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                             |         |
| 2.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                       |         |
| 2.2                        |                      |                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 3.1                        |                      | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                      |         |
| 3.2                        | Read Address         | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                        |         |
| 4.1                        |                      |                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 4.2                        | Read Address + 1     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                    |         |
| 5.1                        |                      | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$80), perform Operand.L = (Operand.L << 1)   <b>C</b> , and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.H & \$80), perform Operand = (Operand << 1)   <b>C</b> , and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |         |
| 5.2                        |                      | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                          |         |
| 6.1                        |                      |                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 6.2                        | Write to Address + 1 | Write Operand.H.                                                                                                                                                                                                                                                                                                                                                                       |         |
| 7.1                        |                      |                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 7.2                        | Write to Address     | Write Operand.L.                                                                                                                                                                                                                                                                                                                                                                       |         |
| 8.1                        |                      |                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 8.2                        | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                       |         |
| +X.1                       |                      |                                                                                                                                                                                                                                                                                                                                                                                        |         |

|                             |                                    |                                                                                                                                                                                                                               |         |
|-----------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AND (27)                    |                                    | Cycles: 6                                                                                                                                                                                                                     | Size: 2 |
| Direct Indirect Long (Read) |                                    |                                                                                                                                                                                                                               |         |
| Cycle                       | R/W                                | Desc.                                                                                                                                                                                                                         |         |
| -X.2                        | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1                         |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2                         | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                    |         |
| 2.1                         |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 2.2                         |                                    |                                                                                                                                                                                                                               |         |
| 3.1                         |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                             |         |
| 3.2                         | Read Address                       | Store as Pointer.                                                                                                                                                                                                             |         |
| 4.1                         |                                    |                                                                                                                                                                                                                               |         |
| 4.2                         | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                           |         |
| 5.1                         |                                    |                                                                                                                                                                                                                               |         |
| 5.2                         | Read Address + 2                   | Store as Bank.                                                                                                                                                                                                                |         |
| 6.1                         |                                    |                                                                                                                                                                                                                               |         |
| 6.2                         | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 7.1                         |                                    |                                                                                                                                                                                                                               |         |
| 7.2                         | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                           |         |
| 8.1                         |                                    | Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 8.2                         | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1                        |                                    |                                                                                                                                                                                                                               |         |



|          |                   |                                                                                   |
|----------|-------------------|-----------------------------------------------------------------------------------|
| PLP (28) | Cycles: 4         | Size: 1                                                                           |
| Implied  |                   |                                                                                   |
| Cycle    | R/W               | Desc.                                                                             |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                  |
| 1.1      |                   | Inc. <b>PC</b> .                                                                  |
| 1.2      |                   |                                                                                   |
| 2.1      |                   |                                                                                   |
| 2.2      |                   |                                                                                   |
| 3.1      |                   | Inc. <b>S</b> by 1.                                                               |
| 3.2      | Read <b>S</b>     | Store as Operand.L.                                                               |
| 4.1      | Read <b>PC:PB</b> | <b>P</b> = Operand.L, set <b>X.H</b> and <b>Y.H</b> to 0 if <b>X</b> flag is set. |
| 4.2      |                   | Store as OpCode.                                                                  |
| +X.1     |                   |                                                                                   |

|           |                   |                                                                                                                                                                                                                                                |
|-----------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AND (29)  | Cycles: 3         | Size: 2                                                                                                                                                                                                                                        |
| Immediate |                   |                                                                                                                                                                                                                                                |
| Cycle     | R/W               | Desc.                                                                                                                                                                                                                                          |
| -X.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                               |
| 1.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                               |
| 1.2       | Read <b>PC:PB</b> | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                |
| 2.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                               |
| 2.2       | Read <b>PC:PB</b> | Store as Operand.H.                                                                                                                                                                                                                            |
| 3.1       |                   | Inc. <b>PC</b> . Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |
| 3.2       | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                               |
| +X.1      |                   |                                                                                                                                                                                                                                                |

|          |                   |                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROL (2A) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                                                                                                                                                                  |
| Implied  |                   |                                                                                                                                                                                                                                                                                                                                                                                                          |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                    |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                         |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                         |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2.1      |                   | If <b>M</b> flag is set, set <b>C</b> based off ( <b>A.L</b> & \$80), perform <b>A.L</b> = ( <b>A.L</b> << 1)   <b>C</b> , and set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>C</b> based off ( <b>A.H</b> & \$80), perform <b>A</b> = ( <b>A</b> << 1)   <b>C</b> , and set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                         |
| +X.1     |                   |                                                                                                                                                                                                                                                                                                                                                                                                          |

|          |                     |                                                                                                                              |
|----------|---------------------|------------------------------------------------------------------------------------------------------------------------------|
| PLD (2B) | Cycles: 5           | Size: 1                                                                                                                      |
| Implied  |                     |                                                                                                                              |
| Cycle    | R/W                 | Desc.                                                                                                                        |
| -X.2     | Read <b>PC:PB</b>   | Store as OpCode.                                                                                                             |
| 1.1      | Read ( <b>S</b> +1) | Inc. <b>PC</b> .                                                                                                             |
| 1.2      |                     |                                                                                                                              |
| 2.1      |                     |                                                                                                                              |
| 2.2      |                     |                                                                                                                              |
| 3.1      |                     |                                                                                                                              |
| 3.2      |                     | Store as Operand.L.                                                                                                          |
| 4.1      |                     |                                                                                                                              |
| 4.2      |                     | Store as Operand.H.                                                                                                          |
| 5.1      |                     | Inc. <b>S</b> by 2. Set <b>D</b> to Operand. Set <b>N</b> based off ( <b>D</b> .H & \$80). Set <b>Z</b> based off <b>D</b> . |
| 5.2      |                     | Store as OpCode.                                                                                                             |
| +X.1     |                     |                                                                                                                              |

|                 |                                          |                                                                                                                                                                                                                                                                                                                         |
|-----------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BIT (2C)        | Cycles: 4                                | Size: 3                                                                                                                                                                                                                                                                                                                 |
| Absolute (Read) |                                          |                                                                                                                                                                                                                                                                                                                         |
| Cycle           | R/W                                      | Desc.                                                                                                                                                                                                                                                                                                                   |
| -X.2            | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                                                                        |
| 1.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                        |
| 1.2             | Read <b>PC:PB</b>                        | Store as Address.                                                                                                                                                                                                                                                                                                       |
| 2.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                        |
| 2.2             | Read <b>PC:PB</b>                        | Store as Address.L.                                                                                                                                                                                                                                                                                                     |
| 3.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                        |
| 3.2             | Read Address: <b>DB</b>                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                         |
| 4.1             |                                          |                                                                                                                                                                                                                                                                                                                         |
| 4.2             | Read Address + 1: <b>DB</b> (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                     |
| 5.1             |                                          | If <b>M</b> flag is set, set <b>Z</b> based off ( <b>A</b> .L & Operand.L), set <b>N</b> based off (Operand.L & \$80), and set <b>V</b> based off (Operand.L & \$40), otherwise set <b>Z</b> based off ( <b>A</b> & Operand), set <b>N</b> based off (Operand.H & \$80), and set <b>V</b> based off (Operand.H & \$40). |
| 5.2             | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                                                                        |
| +X.1            |                                          |                                                                                                                                                                                                                                                                                                                         |

|                 |                                          |                                                                                                                                                                                                                                 |
|-----------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AND (2D)        | Cycles: 4                                | Size: 3                                                                                                                                                                                                                         |
| Absolute (Read) |                                          |                                                                                                                                                                                                                                 |
| Cycle           | R/W                                      | Desc.                                                                                                                                                                                                                           |
| -X.2            | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                |
| 1.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 1.2             | Read <b>PC:PB</b>                        | Store as Address.                                                                                                                                                                                                               |
| 2.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 2.2             | Read <b>PC:PB</b>                        | Store as Address.L.                                                                                                                                                                                                             |
| 3.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 3.2             | Read Address: <b>DB</b>                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |
| 4.1             |                                          |                                                                                                                                                                                                                                 |
| 4.2             | Read Address + 1: <b>DB</b> (With Carry) | Store as Operand.H.                                                                                                                                                                                                             |
| 5.1             |                                          | Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |
| 5.2             | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                |
| +X.1            |                                          |                                                                                                                                                                                                                                 |

| ROL (2E)                     | Cycles: 6                                    | Size: 3                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute (Read/Modify/Write) |                                              |                                                                                                                                                                                                                                                                                                                                                                                        |
| Cycle                        | R/W                                          | Desc.                                                                                                                                                                                                                                                                                                                                                                                  |
| -X.2                         | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                       |
| 1.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                       |
| 1.2                          | Read <b>PC:PB</b>                            | Store as Address.                                                                                                                                                                                                                                                                                                                                                                      |
| 2.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                       |
| 2.2                          | Read <b>PC:PB</b>                            | Store as Address.L.                                                                                                                                                                                                                                                                                                                                                                    |
| 3.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                       |
| 3.2                          | Read Address: <b>DB</b>                      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                        |
| 4.1                          |                                              |                                                                                                                                                                                                                                                                                                                                                                                        |
| 4.2                          | Read Address + 1: <b>DB</b> (With Carry)     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                    |
| 5.1                          |                                              | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$80), perform Operand.L = (Operand.L << 1)   <b>C</b> , and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.H & \$80), perform Operand = (Operand << 1)   <b>C</b> , and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |
| 5.2                          |                                              | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                          |
| 6.1                          |                                              |                                                                                                                                                                                                                                                                                                                                                                                        |
| 6.2                          | Write to Address + 1: <b>DB</b> (With Carry) | Write Operand.H.                                                                                                                                                                                                                                                                                                                                                                       |
| 7.1                          |                                              |                                                                                                                                                                                                                                                                                                                                                                                        |
| 7.2                          | Write to Address: <b>DB</b>                  | Write Operand.L.                                                                                                                                                                                                                                                                                                                                                                       |
| 8.1                          |                                              |                                                                                                                                                                                                                                                                                                                                                                                        |
| 8.2                          | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                       |
| +X.1                         |                                              |                                                                                                                                                                                                                                                                                                                                                                                        |

|                      |                                    |                                                                                                                                                                                                                                 |
|----------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AND (2F)             | Cycles: 5                          | Size: 4                                                                                                                                                                                                                         |
| Absolute Long (Read) |                                    |                                                                                                                                                                                                                                 |
| Cycle                | R/W                                | Desc.                                                                                                                                                                                                                           |
| -X.2                 | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |
| 1.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 1.2                  | Read <b>PC:PB</b>                  | Store as Address.                                                                                                                                                                                                               |
| 2.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 2.2                  | Read <b>PC:PB</b>                  | Store as Address.L.                                                                                                                                                                                                             |
| 3.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 3.2                  | Read <b>PC:PB</b>                  | Stores as Bank.                                                                                                                                                                                                                 |
| 4.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 4.2                  | Read Address:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |
| 5.1                  |                                    |                                                                                                                                                                                                                                 |
| 5.2                  | Read Address + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                             |
| 6.1                  |                                    | Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |
| 6.2                  | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |
| +X.1                 |                                    |                                                                                                                                                                                                                                 |



|          |                   |                                                                                                 |
|----------|-------------------|-------------------------------------------------------------------------------------------------|
| BMI (30) | Cycles: 2         | Size: 2                                                                                         |
| Relative |                   |                                                                                                 |
| Cycle    | R/W               | Desc.                                                                                           |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                |
| 1.1      |                   | Inc. <b>PC</b> . Decide to branch if <b>N</b> is 1.                                             |
| 1.2      | Read <b>PC:PB</b> | Store as Operand. Address = i16(i8(Operand.L)) + <b>PC</b> . If not branching, poll interrupts. |
| 2.1      |                   | Inc. <b>PC</b> . If not branching, end (next half-cycle is 3.2).                                |
| 2.2      |                   | Poll interrupts.                                                                                |
| 3.1      |                   | Set <b>PC</b> to Address.                                                                       |
| 3.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                |
| +X.1     |                   |                                                                                                 |

|                                      |                                    |                                                                                                                                                                                                                               |         |
|--------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AND (31)                             |                                    | Cycles: 5                                                                                                                                                                                                                     | Size: 2 |
| Direct Indirect Indexed (d),y (Read) |                                    |                                                                                                                                                                                                                               |         |
| Cycle                                | R/W                                | Desc.                                                                                                                                                                                                                         |         |
| -X.2                                 | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1                                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2                                  | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                    |         |
| 2.1                                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 2.2                                  |                                    |                                                                                                                                                                                                                               |         |
| 3.1                                  |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                             |         |
| 3.2                                  | Read Address                       | Store as Pointer.                                                                                                                                                                                                             |         |
| 4.1                                  |                                    |                                                                                                                                                                                                                               |         |
| 4.2                                  | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                           |         |
| 5.1                                  |                                    | Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                            |         |
| 5.2                                  |                                    |                                                                                                                                                                                                                               |         |
| 6.1                                  |                                    |                                                                                                                                                                                                                               |         |
| 6.2                                  | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 7.1                                  |                                    |                                                                                                                                                                                                                               |         |
| 7.2                                  | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                           |         |
| 8.1                                  |                                    | Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 8.2                                  | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1                                 |                                    |                                                                                                                                                                                                                               |         |

|                        |                             |                                                                                                                                                                                                                               |
|------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AND (32)               | Cycles: 5                   | Size: 2                                                                                                                                                                                                                       |
| Direct Indirect (Read) |                             |                                                                                                                                                                                                                               |
| Cycle                  | R/W                         | Desc.                                                                                                                                                                                                                         |
| -X.2                   | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                              |
| 1.1                    |                             | Inc. <b>PC</b> .                                                                                                                                                                                                              |
| 1.2                    | Read <b>PC:PB</b>           | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                    |
| 2.1                    |                             | Inc. <b>PC</b> .                                                                                                                                                                                                              |
| 2.2                    |                             |                                                                                                                                                                                                                               |
| 3.1                    |                             | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                             |
| 3.2                    | Read Address                | Store as Pointer.                                                                                                                                                                                                             |
| 4.1                    |                             |                                                                                                                                                                                                                               |
| 4.2                    | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                           |
| 5.1                    |                             |                                                                                                                                                                                                                               |
| 5.2                    | Read Pointer: <b>DB</b>     | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |
| 6.1                    |                             |                                                                                                                                                                                                                               |
| 6.2                    | Read Pointer + 1: <b>DB</b> | Store as Operand.H.                                                                                                                                                                                                           |
| 7.1                    |                             | Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |
| 7.2                    | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                              |
| +X.1                   |                             |                                                                                                                                                                                                                               |

|                                        |                                    |                                                                                                                                                                                                                                 |         |
|----------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AND (33)                               |                                    | Cycles: 7                                                                                                                                                                                                                       | Size: 2 |
| Stack Relative Indirect Indexed (Read) |                                    |                                                                                                                                                                                                                                 |         |
| Cycle                                  | R/W                                | Desc.                                                                                                                                                                                                                           |         |
| -X.2                                   | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| 1.1                                    | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 1.2                                    |                                    | Store as Pointer.                                                                                                                                                                                                               |         |
| 2.1                                    |                                    | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                                                                                            |         |
| 2.2                                    |                                    |                                                                                                                                                                                                                                 |         |
| 3.1                                    |                                    |                                                                                                                                                                                                                                 |         |
| 3.2                                    | Read Address                       | Store as Pointer.                                                                                                                                                                                                               |         |
| 4.1                                    | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                             |         |
| 4.2                                    |                                    | Perform Tmp = ui32(Pointer + <b>Y</b> ).                                                                                                                                                                                        |         |
| 5.1                                    |                                    | Pointer = ui16(Tmp).                                                                                                                                                                                                            |         |
|                                        |                                    | Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                                 |         |
| 5.2                                    |                                    |                                                                                                                                                                                                                                 |         |
| 6.1                                    | Read Pointer:Bank                  |                                                                                                                                                                                                                                 |         |
| 6.2                                    |                                    | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |         |
| 7.1                                    | Read Pointer + 1:Bank (With Carry) |                                                                                                                                                                                                                                 |         |
| 7.2                                    |                                    | Store as Operand.H.                                                                                                                                                                                                             |         |
| 8.1                                    | Read <b>PC:PB</b>                  | Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 8.2                                    |                                    | Store as OpCode.                                                                                                                                                                                                                |         |
| +X.1                                   |                                    |                                                                                                                                                                                                                                 |         |

|                  |                   |                                                                                                                                                                                                                                                                                                                        |
|------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BIT (34)         | Cycles: 4         | Size: 2                                                                                                                                                                                                                                                                                                                |
| Direct, X (Read) |                   |                                                                                                                                                                                                                                                                                                                        |
| Cycle            | R/W               | Desc.                                                                                                                                                                                                                                                                                                                  |
| -X.2             | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                       |
| 1.1              | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                       |
| 1.2              |                   | Store as Operand.                                                                                                                                                                                                                                                                                                      |
| 2.1              |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                       |
| 2.2              |                   |                                                                                                                                                                                                                                                                                                                        |
| 3.1              |                   | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                                                                                                                      |
| 3.2              |                   |                                                                                                                                                                                                                                                                                                                        |
| 4.1              |                   |                                                                                                                                                                                                                                                                                                                        |
| 4.2              | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                        |
| 5.1              | Read Address + 1  |                                                                                                                                                                                                                                                                                                                        |
| 5.2              |                   | Store as Operand.H.                                                                                                                                                                                                                                                                                                    |
| 6.1              | Read <b>PC:PB</b> | If <b>M</b> flag is set, set <b>Z</b> based off ( <b>A.L</b> & Operand.L), set <b>N</b> based off (Operand.L & \$80), and set <b>V</b> based off (Operand.L & \$40), otherwise set <b>Z</b> based off ( <b>A</b> & Operand), set <b>N</b> based off (Operand.H & \$80), and set <b>V</b> based off (Operand.H & \$40). |
| 6.2              |                   | Store as OpCode.                                                                                                                                                                                                                                                                                                       |
| +X.1             |                   |                                                                                                                                                                                                                                                                                                                        |

|                  |                   |                                                                                                                                                                                                                               |         |
|------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AND (35)         |                   | Cycles: 4                                                                                                                                                                                                                     | Size: 2 |
| Direct, X (Read) |                   |                                                                                                                                                                                                                               |         |
| Cycle            | R/W               | Desc.                                                                                                                                                                                                                         |         |
| -X.2             | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1              | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2              |                   | Store as Operand.                                                                                                                                                                                                             |         |
| 2.1              |                   | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 2.2              |                   |                                                                                                                                                                                                                               |         |
| 3.1              |                   | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                             |         |
| 3.2              |                   |                                                                                                                                                                                                                               |         |
| 4.1              |                   |                                                                                                                                                                                                                               |         |
| 4.2              | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 5.1              | Read Address + 1  |                                                                                                                                                                                                                               |         |
| 5.2              |                   | Store as Operand.H.                                                                                                                                                                                                           |         |
| 6.1              | Read <b>PC:PB</b> | Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 6.2              |                   | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1             |                   |                                                                                                                                                                                                                               |         |

|                                                   |                      |                                                                                                                                                                                                                                                                                                                                                                                        |         |
|---------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ROL (36)                                          |                      | Cycles: 6                                                                                                                                                                                                                                                                                                                                                                              | Size: 2 |
| Direct Indexed Indirect (d,x) (Read/Modify/Write) |                      |                                                                                                                                                                                                                                                                                                                                                                                        |         |
| Cycle                                             | R/W                  | Desc.                                                                                                                                                                                                                                                                                                                                                                                  |         |
| -X.2                                              | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                       |         |
| 1.1                                               | Read <b>PC:PB</b>    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                       |         |
| 1.2                                               |                      | Store as Operand.                                                                                                                                                                                                                                                                                                                                                                      |         |
| 2.1                                               |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                       |         |
| 2.2                                               |                      |                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 3.1                                               |                      | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                                                                                                                                                                                      |         |
| 3.2                                               |                      |                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 4.1                                               |                      |                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 4.2                                               | Read Address         | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                        |         |
| 5.1                                               | Read Address + 1     |                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 5.2                                               |                      | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                    |         |
| 6.1                                               |                      | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$80), perform Operand.L = (Operand.L << 1)   <b>C</b> , and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.H & \$80), perform Operand = (Operand << 1)   <b>C</b> , and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |         |
| 6.2                                               |                      | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                          |         |
| 7.1                                               | Write to Address + 1 |                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 7.2                                               |                      | Write Operand.H.                                                                                                                                                                                                                                                                                                                                                                       |         |
| 8.1                                               | Write to Address     |                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 8.2                                               |                      | Write Operand.L.                                                                                                                                                                                                                                                                                                                                                                       |         |
| 9.1                                               | Read <b>PC:PB</b>    |                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 9.2                                               |                      | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                       |         |
| +X.1                                              |                      |                                                                                                                                                                                                                                                                                                                                                                                        |         |

|                                           |                                    |                                                                                                                                                                                                                               |         |
|-------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AND (37)                                  |                                    | Cycles: 6                                                                                                                                                                                                                     | Size: 2 |
| Direct Indirect Indexed Long [d],y (Read) |                                    |                                                                                                                                                                                                                               |         |
| Cycle                                     | R/W                                | Desc.                                                                                                                                                                                                                         |         |
| -X.2                                      | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1                                       |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2                                       | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                    |         |
| 2.1                                       |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 2.2                                       |                                    |                                                                                                                                                                                                                               |         |
| 3.1                                       |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                             |         |
| 3.2                                       | Read Address                       | Store as Pointer.                                                                                                                                                                                                             |         |
| 4.1                                       |                                    |                                                                                                                                                                                                                               |         |
| 4.2                                       | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                           |         |
| 5.1                                       |                                    |                                                                                                                                                                                                                               |         |
| 5.2                                       | Read Address + 2                   | Store as Bank.                                                                                                                                                                                                                |         |
| 6.1                                       |                                    | Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{Y})$ .                                                                                                                                                               |         |
|                                           |                                    | Pointer = $\text{ui16}(\text{Tmp})$ .                                                                                                                                                                                         |         |
|                                           |                                    | Bank += (Tmp >> 16).                                                                                                                                                                                                          |         |
| 6.2                                       | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 7.1                                       |                                    |                                                                                                                                                                                                                               |         |
| 7.2                                       | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                           |         |
| 8.1                                       |                                    | Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 8.2                                       | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1                                      |                                    |                                                                                                                                                                                                                               |         |



|          |                   |                             |
|----------|-------------------|-----------------------------|
| SEC (38) | Cycles: 2         | Size: 1                     |
| Implied  |                   |                             |
| Cycle    | R/W               | Desc.                       |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.            |
| 1.1      |                   | Inc. <b>PC</b> .            |
| 1.2      | Read <b>PC:PB</b> | Discard.                    |
| 2.1      |                   | Set the <b>C</b> flag to 1. |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.            |
| +X.1     |                   |                             |

|                    |                                    |                                                                                                                                                                                                                                 |         |
|--------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AND (39)           |                                    | Cycles: 4                                                                                                                                                                                                                       | Size: 3 |
| Absolute, Y (Read) |                                    |                                                                                                                                                                                                                                 |         |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                           |         |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| 1.1                | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 1.2                |                                    | Store as Pointer.                                                                                                                                                                                                               |         |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 2.2                |                                    | Store as Pointer.H.                                                                                                                                                                                                             |         |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).             |         |
| 3.2                |                                    |                                                                                                                                                                                                                                 |         |
| 4.1                |                                    |                                                                                                                                                                                                                                 |         |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |         |
| 5.1                | Read Pointer + 1:Bank (With Carry) |                                                                                                                                                                                                                                 |         |
| 5.2                |                                    | Store as Operand.H.                                                                                                                                                                                                             |         |
| 6.1                | Read <b>PC:PB</b>                  | Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 6.2                |                                    | Store as OpCode.                                                                                                                                                                                                                |         |
| +X.1               |                                    |                                                                                                                                                                                                                                 |         |

|          |                   |                                                                                                                                                                                                                                                            |
|----------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEC (3A) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                    |
| Implied  |                   |                                                                                                                                                                                                                                                            |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                      |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                           |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                           |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                   |
| 2.1      |                   | If <b>M</b> flag is set, perform <b>A.L</b> -= 1, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise perform <b>A</b> -= 1, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                           |
| +X.1     |                   |                                                                                                                                                                                                                                                            |

|          |                   |                                                                                                           |
|----------|-------------------|-----------------------------------------------------------------------------------------------------------|
| TSC (3B) | Cycles: 2         | Size: 1                                                                                                   |
| Implied  |                   |                                                                                                           |
| Cycle    | R/W               | Desc.                                                                                                     |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                          |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                          |
| 1.2      |                   | Discard.                                                                                                  |
| 2.1      |                   | Set <b>A</b> to <b>S</b> . Set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |
| 2.2      |                   | Store as OpCode.                                                                                          |
| +X.1     |                   |                                                                                                           |

|                    |                                    |                                                                                                                                                                                                                                                                                                                         |
|--------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BIT (3C)           | Cycles: 4                          | Size: 3                                                                                                                                                                                                                                                                                                                 |
| Absolute, X (Read) |                                    |                                                                                                                                                                                                                                                                                                                         |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                   |
| -X.2               | Read PC:PB                         | Store as OpCode.                                                                                                                                                                                                                                                                                                        |
| 1.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                        |
| 1.2                | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                                                                       |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                        |
| 2.2                | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                                                                                                                     |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>X</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                                                                                     |
| 3.2                |                                    |                                                                                                                                                                                                                                                                                                                         |
| 4.1                |                                    |                                                                                                                                                                                                                                                                                                                         |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                         |
| 5.1                |                                    |                                                                                                                                                                                                                                                                                                                         |
| 5.2                | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                     |
| 6.1                |                                    | If <b>M</b> flag is set, set <b>Z</b> based off ( <b>A</b> .L & Operand.L), set <b>N</b> based off (Operand.L & \$80), and set <b>V</b> based off (Operand.L & \$40), otherwise set <b>Z</b> based off ( <b>A</b> & Operand), set <b>N</b> based off (Operand.H & \$80), and set <b>V</b> based off (Operand.H & \$40). |
| 6.2                | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                        |
| +X.1               |                                    |                                                                                                                                                                                                                                                                                                                         |

|                    |                                    |                                                                                                                                                                                                                                 |         |
|--------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AND (3D)           |                                    | Cycles: 4                                                                                                                                                                                                                       | Size: 3 |
| Absolute, X (Read) |                                    |                                                                                                                                                                                                                                 |         |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                           |         |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| 1.1                | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 1.2                |                                    | Store as Pointer.                                                                                                                                                                                                               |         |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 2.2                |                                    | Store as Pointer.H.                                                                                                                                                                                                             |         |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>X</b> ).<br>Pointer = ui16(Tmp). If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                |         |
| 3.2                |                                    |                                                                                                                                                                                                                                 |         |
| 4.1                |                                    |                                                                                                                                                                                                                                 |         |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |         |
| 5.1                | Read Pointer + 1:Bank (With Carry) |                                                                                                                                                                                                                                 |         |
| 5.2                |                                    | Store as Operand.H.                                                                                                                                                                                                             |         |
| 6.1                | Read <b>PC:PB</b>                  | Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 6.2                |                                    | Store as OpCode.                                                                                                                                                                                                                |         |
| +X.1               |                                    |                                                                                                                                                                                                                                 |         |

| ROL (3E)                       |                                        | Cycles: 7                                                                                                                                                                                                                                                                                                                                                                                                                          | Size: 3 |
|--------------------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Absolute, X (Read/Modiy/Write) |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| Cycle                          | R/W                                    | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                              |         |
| -X.2                           | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
| 1.1                            |                                        | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
| 1.2                            | Read <b>PC:PB</b>                      | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                  |         |
| 2.1                            |                                        | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
| 2.2                            | Read <b>PC:PB</b>                      | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| 3.1                            |                                        | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{X})$ .                                                                                                                                                                                                                                                                                                                                                 |         |
|                                |                                        | Pointer = $\text{ui16}(\text{Tmp})$ .                                                                                                                                                                                                                                                                                                                                                                                              |         |
|                                |                                        | Bank = $\mathbf{DB} + (\text{Tmp} \gg 16)$ .                                                                                                                                                                                                                                                                                                                                                                                       |         |
| 3.2                            |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| 4.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| 4.2                            | Read Pointer:Bank                      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                    |         |
| 5.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| 5.2                            | Read Pointer + 1:Bank (With Carry)     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| 6.1                            |                                        | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$80), perform $\text{Operand.L} = (\text{Operand.L} \ll 1) \mid \mathbf{C}$ , and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.H & \$80), perform $\text{Operand} = (\text{Operand} \ll 1) \mid \mathbf{C}$ , and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |         |
| 6.2                            |                                        | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                      |         |
| 7.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| 7.2                            | Write to Pointer + 1:Bank (With Carry) | Write Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
| 8.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| 8.2                            | Write to Pointer:Bank                  | Write Operand.L.                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
| 9.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| 9.2                            | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
| +X.1                           |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |

|                         |                                    |                                                                                                                                                                                                                                 |         |
|-------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| AND (3F)                |                                    | Cycles: 5                                                                                                                                                                                                                       | Size: 4 |
| Absolute Long, X (Read) |                                    |                                                                                                                                                                                                                                 |         |
| Cycle                   | R/W                                | Desc.                                                                                                                                                                                                                           |         |
| -X.2                    | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| 1.1                     |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 1.2                     | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                               |         |
| 2.1                     |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 2.2                     | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                             |         |
| 3.1                     |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 3.2                     | Read <b>PC:PB</b>                  | Stores as Bank.                                                                                                                                                                                                                 |         |
| 4.1                     |                                    | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$ .                                                                                                                                                |         |
|                         |                                    | Pointer = $\text{ui16}(\text{Tmp})$ .                                                                                                                                                                                           |         |
|                         |                                    | Bank += (Tmp >> 16).                                                                                                                                                                                                            |         |
| 4.2                     | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |         |
| 5.1                     |                                    |                                                                                                                                                                                                                                 |         |
| 5.2                     | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                             |         |
| 6.1                     |                                    | Perform <b>A</b> &= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 6.2                     | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| +X.1                    |                                    |                                                                                                                                                                                                                                 |         |



|          |                     |                                                                                                                                                       |
|----------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| RTI (40) | Cycles: 7           | Size: 1                                                                                                                                               |
| Implied  |                     |                                                                                                                                                       |
| Cycle    | R/W                 | Desc.                                                                                                                                                 |
| -X.2     | Read <b>PC:PB</b>   | Store as OpCode.                                                                                                                                      |
| 1.1      |                     | Inc. <b>PC</b> .                                                                                                                                      |
| 1.2      |                     |                                                                                                                                                       |
| 2.1      |                     |                                                                                                                                                       |
| 2.2      |                     |                                                                                                                                                       |
| 3.1      |                     | Inc. <b>S</b> by 1.                                                                                                                                   |
| 3.2      | Read ( <b>S</b> +0) | Store as Operand.L.                                                                                                                                   |
| 4.1      |                     | Set <b>P</b> to Operand.                                                                                                                              |
| 4.2      | Read ( <b>S</b> +1) | Store as Operand.L.                                                                                                                                   |
| 5.1      |                     |                                                                                                                                                       |
| 5.2      | Read ( <b>S</b> +2) | Store as Operand.H.                                                                                                                                   |
| 6.1      |                     |                                                                                                                                                       |
| 6.2      | Read ( <b>S</b> +3) | Store as <b>PB</b> .                                                                                                                                  |
| 7.1      |                     | Inc. <b>S</b> by 3. Set <b>PB</b> to Bank. Set <b>PC</b> to Operand. If <b>X</b> flag is set, set <b>X</b> to <b>X.L</b> and <b>Y</b> to <b>Y.L</b> . |
| 7.2      | Read <b>PC:PB</b>   | Store as OpCode.                                                                                                                                      |
| +X.1     |                     |                                                                                                                                                       |

| EOR (41)                             | Cycles: 6                   | Size: 2                                                                                                                                                                                                                       |
|--------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indexed Indirect (d,x) (Read) |                             |                                                                                                                                                                                                                               |
| Cycle                                | R/W                         | Desc.                                                                                                                                                                                                                         |
| -X.2                                 | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                              |
| 1.1                                  |                             | Inc. <b>PC</b> .                                                                                                                                                                                                              |
| 1.2                                  | Read <b>PC:PB</b>           | Store as Pointer.                                                                                                                                                                                                             |
| 2.1                                  |                             | Inc. <b>PC</b> . Set Pointer to Pointer + <b>X</b> + <b>D</b> . Address = Pointer.                                                                                                                                            |
| 2.2                                  |                             | If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                      |
| 3.1                                  |                             | Address.H = Pointer.H (fixes high byte of address).                                                                                                                                                                           |
| 3.2                                  |                             |                                                                                                                                                                                                                               |
| 4.1                                  |                             |                                                                                                                                                                                                                               |
| 4.2                                  | Read Address                | Store as Pointer.                                                                                                                                                                                                             |
| 5.1                                  |                             |                                                                                                                                                                                                                               |
| 5.2                                  | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                           |
| 6.1                                  |                             |                                                                                                                                                                                                                               |
| 6.2                                  | Read Pointer: <b>DB</b>     | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |
| 7.1                                  |                             |                                                                                                                                                                                                                               |
| 7.2                                  | Read Pointer + 1: <b>DB</b> | Store as Operand.H.                                                                                                                                                                                                           |
| 8.1                                  |                             | Perform <b>A</b> ^= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |
| 8.2                                  | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                              |
| +X.1                                 |                             |                                                                                                                                                                                                                               |

|                    |                   |                   |
|--------------------|-------------------|-------------------|
| WDM (42) Cycles: 2 |                   | Size: 2           |
| Immediate          |                   |                   |
| Cycle              | R/W               | Desc.             |
| -X.2               | Read <b>PC:PB</b> | Store as OpCode.  |
| 1.1                |                   | Inc. <b>PC</b> .  |
| 1.2                | Read <b>PC:PB</b> | Store as Operand. |
| 2.1                |                   | Inc. <b>PC</b> .  |
| 2.2                | Read <b>PC:PB</b> | Store as OpCode.  |
| +X.1               |                   |                   |

|                       |                   |                                                                                                                                                                                                                                 |
|-----------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EOR (43)              | Cycles: 4         | Size: 2                                                                                                                                                                                                                         |
| Stack Relative (Read) |                   |                                                                                                                                                                                                                                 |
| Cycle                 | R/W               | Desc.                                                                                                                                                                                                                           |
| -X.2                  | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                |
| 1.1                   | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 1.2                   |                   | Store as Pointer.                                                                                                                                                                                                               |
| 2.1                   |                   | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                                                                                            |
| 2.2                   |                   |                                                                                                                                                                                                                                 |
| 3.1                   |                   |                                                                                                                                                                                                                                 |
| 3.2                   | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |
| 4.1                   | Read Address + 1  |                                                                                                                                                                                                                                 |
| 4.2                   |                   | Store as Operand.H.                                                                                                                                                                                                             |
| 5.1                   | Read <b>PC:PB</b> | Perform <b>A</b> ^= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |
| 5.2                   |                   | Store as OpCode.                                                                                                                                                                                                                |
| +X.1                  |                   |                                                                                                                                                                                                                                 |

|                       |                    |                                                                                                                                                                                                    |
|-----------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MVP (44)              | Cycles: 7          | Size: 3                                                                                                                                                                                            |
| Block Move (Positive) |                    |                                                                                                                                                                                                    |
| Cycle                 | R/W                | Desc.                                                                                                                                                                                              |
| -X.2                  | Read <b>PC:PB</b>  | Store as OpCode.                                                                                                                                                                                   |
| 1.1                   |                    | Inc. <b>PC</b> .                                                                                                                                                                                   |
| 1.2                   | Read <b>PC:PB</b>  | Store as Operand.                                                                                                                                                                                  |
| 2.1                   |                    | Inc. <b>PC</b> . Set <b>DB</b> to Operand.L.                                                                                                                                                       |
| 2.2                   | Read <b>PC:PB</b>  | Stores as Bank.                                                                                                                                                                                    |
| 3.1                   |                    | Inc. <b>PC</b> .                                                                                                                                                                                   |
| 3.2                   | Read <b>X:Bank</b> | Store as Operand.L.                                                                                                                                                                                |
| 4.1                   |                    |                                                                                                                                                                                                    |
| 4.2                   | Write <b>Y:DB</b>  | Write Operand.L.                                                                                                                                                                                   |
| 5.1                   |                    | Perform <b>A</b> -= 1. If <b>X</b> flag is set, perform <b>X.L</b> -= 1 and <b>Y.L</b> -= 1, otherwise perform <b>X</b> -= 1 and <b>Y</b> -= 1. If <b>A</b> is not \$FFFF, perform <b>PC</b> -= 3. |
| 5.2                   |                    |                                                                                                                                                                                                    |
| 6.1                   |                    |                                                                                                                                                                                                    |
| 6.2                   |                    |                                                                                                                                                                                                    |
| 7.1                   |                    |                                                                                                                                                                                                    |
| 7.2                   | Read <b>PC:PB</b>  | Store as OpCode.                                                                                                                                                                                   |
| +X.1                  |                    |                                                                                                                                                                                                    |

|               |                   |                                                                                                                                                                                                                               |         |
|---------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| EOR (45)      |                   | Cycles: 3                                                                                                                                                                                                                     | Size: 2 |
| Direct (Read) |                   |                                                                                                                                                                                                                               |         |
| Cycle         | R/W               | Desc.                                                                                                                                                                                                                         |         |
| -X.2          | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2           | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                    |         |
| 2.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 2.2           |                   |                                                                                                                                                                                                                               |         |
| 3.1           |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                             |         |
| 3.2           | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 4.1           |                   |                                                                                                                                                                                                                               |         |
| 4.2           | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                           |         |
| 5.1           |                   | Perform <b>A</b> ^= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 5.2           | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1          |                   |                                                                                                                                                                                                                               |         |

| LSR (46)                   | Cycles: 5            | Size: 2                                                                                                                                                                                                                                                                                                                                |
|----------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct (Read/Modify/Write) |                      |                                                                                                                                                                                                                                                                                                                                        |
| Cycle                      | R/W                  | Desc.                                                                                                                                                                                                                                                                                                                                  |
| -X.2                       | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| 1.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 1.2                        | Read <b>PC:PB</b>    | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                             |
| 2.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 2.2                        |                      |                                                                                                                                                                                                                                                                                                                                        |
| 3.1                        |                      | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                      |
| 3.2                        | Read Address         | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                        |
| 4.1                        |                      |                                                                                                                                                                                                                                                                                                                                        |
| 4.2                        | Read Address + 1     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                    |
| 5.1                        |                      | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$01), perform Operand.L >>= 1, and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.L & \$01), perform Operand >>= 1, and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |
| 5.2                        |                      | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                          |
| 6.1                        |                      |                                                                                                                                                                                                                                                                                                                                        |
| 6.2                        | Write to Address + 1 | Write Operand.H.                                                                                                                                                                                                                                                                                                                       |
| 7.1                        |                      |                                                                                                                                                                                                                                                                                                                                        |
| 7.2                        | Write to Address     | Write Operand.L.                                                                                                                                                                                                                                                                                                                       |
| 8.1                        |                      |                                                                                                                                                                                                                                                                                                                                        |
| 8.2                        | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| +X.1                       |                      |                                                                                                                                                                                                                                                                                                                                        |

|                             |                                    |                                                                                                                                                                                                                               |         |
|-----------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| EOR (47)                    |                                    | Cycles: 6                                                                                                                                                                                                                     | Size: 2 |
| Direct Indirect Long (Read) |                                    |                                                                                                                                                                                                                               |         |
| Cycle                       | R/W                                | Desc.                                                                                                                                                                                                                         |         |
| -X.2                        | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1                         |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2                         | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                    |         |
| 2.1                         |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 2.2                         |                                    |                                                                                                                                                                                                                               |         |
| 3.1                         |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                             |         |
| 3.2                         | Read Address                       | Store as Pointer.                                                                                                                                                                                                             |         |
| 4.1                         |                                    |                                                                                                                                                                                                                               |         |
| 4.2                         | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                           |         |
| 5.1                         |                                    |                                                                                                                                                                                                                               |         |
| 5.2                         | Read Address + 2                   | Store as Bank.                                                                                                                                                                                                                |         |
| 6.1                         |                                    |                                                                                                                                                                                                                               |         |
| 6.2                         | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 7.1                         |                                    |                                                                                                                                                                                                                               |         |
| 7.2                         | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                           |         |
| 8.1                         |                                    | Perform <b>A</b> ^= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 8.2                         | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1                        |                                    |                                                                                                                                                                                                                               |         |



|          |                         |                                               |
|----------|-------------------------|-----------------------------------------------|
| PHA (48) | Cycles: 3               | Size: 1                                       |
| Implied  |                         |                                               |
| Cycle    | R/W                     | Desc.                                         |
| -X.2     | Read <b>PC:PB</b>       | Store as OpCode.                              |
| 1.1      |                         | Inc. <b>PC</b> .                              |
| 1.2      |                         | If <b>M</b> flag is set, skip the next cycle. |
| 2.1      |                         |                                               |
| 2.2      | Write to ( <b>S</b> +0) | Push <b>A.H</b> onto stack.                   |
| 3.1      |                         | Dec. <b>S</b> by 1.                           |
| 3.2      | Write to ( <b>S</b> +0) | Push <b>A.L</b> onto stack.                   |
| 4.1      |                         | Dec. <b>S</b> by 1.                           |
| 4.2      | Read <b>PC:PB</b>       | Store as OpCode.                              |
| +X.1     |                         |                                               |

|           |                   |                                                                                                                                                                                                                                                           |
|-----------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EOR (49)  | Cycles: 3         | Size: 2                                                                                                                                                                                                                                                   |
| Immediate |                   |                                                                                                                                                                                                                                                           |
| Cycle     | R/W               | Desc.                                                                                                                                                                                                                                                     |
| -X.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                          |
| 1.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                          |
| 1.2       | Read <b>PC:PB</b> | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                           |
| 2.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                          |
| 2.2       | Read <b>PC:PB</b> | Store as Operand.H.                                                                                                                                                                                                                                       |
| 3.1       |                   | Inc. <b>PC</b> . Perform <b>A ^= Operand</b> . If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L &amp; \$80</b> ) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H &amp; \$80</b> ) and <b>Z</b> based off <b>A</b> . |
| 3.2       | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                          |
| +X.1      |                   |                                                                                                                                                                                                                                                           |

|          |                   |                                                                                                                                                                                                                                                                                                                                                      |
|----------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LSR (4A) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                                                                                                              |
| Implied  |                   |                                                                                                                                                                                                                                                                                                                                                      |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                     |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                     |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                                                                                                             |
| 2.1      |                   | If <b>M</b> flag is set, set <b>C</b> based off ( <b>A.L</b> & \$01), perform <b>A.L</b> >>= 1, and set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>C</b> based off ( <b>A.L</b> & \$01), perform <b>A</b> >>= 1, and set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                                                                                                                     |
| +X.1     |                   |                                                                                                                                                                                                                                                                                                                                                      |

|          |                         |           |                            |
|----------|-------------------------|-----------|----------------------------|
| PHK (4B) |                         | Cycles: 3 | Size: 1                    |
| Implied  |                         |           |                            |
| Cycle    | R/W                     |           | Desc.                      |
| -X.2     | Read <b>PC:PB</b>       |           | Store as OpCode.           |
| 1.1      |                         |           | Inc. <b>PC</b> .           |
| 1.2      |                         |           |                            |
| 2.1      |                         |           |                            |
| 2.2      | Write to ( <b>S</b> +0) |           | Push <b>PB</b> onto stack. |
| 3.1      |                         |           | Dec. <b>S</b> by 1.        |
| 3.2      | Read <b>PC:PB</b>       |           | Store as OpCode.           |
| +X.1     |                         |           |                            |

|          |                   |                           |
|----------|-------------------|---------------------------|
| JMP (4C) | Cycles: 3         | Size: 3                   |
| Absolute |                   |                           |
| Cycle    | R/W               | Desc.                     |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.          |
| 1.1      |                   | Inc. <b>PC</b> .          |
| 1.2      | Read <b>PC:PB</b> | Store as Address.         |
| 2.1      |                   | Inc. <b>PC</b> .          |
| 2.2      | Read <b>PC:PB</b> | Store as Address.L.       |
| 3.1      |                   | Set <b>PC</b> to Address. |
| 3.2      | Read <b>PC:PB</b> | Store as OpCode.          |
| +X.1     |                   |                           |

|                 |                                          |                                                                                                                                                                                                                                 |         |
|-----------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| EOR (4D)        |                                          | Cycles: 4                                                                                                                                                                                                                       | Size: 3 |
| Absolute (Read) |                                          |                                                                                                                                                                                                                                 |         |
| Cycle           | R/W                                      | Desc.                                                                                                                                                                                                                           |         |
| -X.2            | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                |         |
| 1.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 1.2             | Read <b>PC:PB</b>                        | Store as Address.                                                                                                                                                                                                               |         |
| 2.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 2.2             | Read <b>PC:PB</b>                        | Store as Address.L.                                                                                                                                                                                                             |         |
| 3.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 3.2             | Read Address: <b>DB</b>                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |         |
| 4.1             |                                          |                                                                                                                                                                                                                                 |         |
| 4.2             | Read Address + 1: <b>DB</b> (With Carry) | Store as Operand.H.                                                                                                                                                                                                             |         |
| 5.1             |                                          | Perform <b>A</b> ^= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 5.2             | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                |         |
| +X.1            |                                          |                                                                                                                                                                                                                                 |         |

| LSR (4E)                     | Cycles: 6                                    | Size: 3                                                                                                                                                                                                                                                                                                                                |
|------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute (Read/Modify/Write) |                                              |                                                                                                                                                                                                                                                                                                                                        |
| Cycle                        | R/W                                          | Desc.                                                                                                                                                                                                                                                                                                                                  |
| -X.2                         | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| 1.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 1.2                          | Read <b>PC:PB</b>                            | Store as Address.                                                                                                                                                                                                                                                                                                                      |
| 2.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 2.2                          | Read <b>PC:PB</b>                            | Store as Address.L.                                                                                                                                                                                                                                                                                                                    |
| 3.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 3.2                          | Read Address: <b>DB</b>                      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                        |
| 4.1                          |                                              |                                                                                                                                                                                                                                                                                                                                        |
| 4.2                          | Read Address + 1: <b>DB</b> (With Carry)     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                    |
| 5.1                          |                                              | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$01), perform Operand.L >>= 1, and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.L & \$01), perform Operand >>= 1, and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |
| 5.2                          |                                              | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                          |
| 6.1                          |                                              |                                                                                                                                                                                                                                                                                                                                        |
| 6.2                          | Write to Address + 1: <b>DB</b> (With Carry) | Write Operand.H.                                                                                                                                                                                                                                                                                                                       |
| 7.1                          |                                              |                                                                                                                                                                                                                                                                                                                                        |
| 7.2                          | Write to Address: <b>DB</b>                  | Write Operand.L.                                                                                                                                                                                                                                                                                                                       |
| 8.1                          |                                              |                                                                                                                                                                                                                                                                                                                                        |
| 8.2                          | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| +X.1                         |                                              |                                                                                                                                                                                                                                                                                                                                        |

|                      |                                    |                                                                                                                                                                                                                                 |
|----------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EOR (4F)             | Cycles: 5                          | Size: 4                                                                                                                                                                                                                         |
| Absolute Long (Read) |                                    |                                                                                                                                                                                                                                 |
| Cycle                | R/W                                | Desc.                                                                                                                                                                                                                           |
| -X.2                 | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |
| 1.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 1.2                  | Read <b>PC:PB</b>                  | Store as Address.                                                                                                                                                                                                               |
| 2.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 2.2                  | Read <b>PC:PB</b>                  | Store as Address.L.                                                                                                                                                                                                             |
| 3.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 3.2                  | Read <b>PC:PB</b>                  | Stores as Bank.                                                                                                                                                                                                                 |
| 4.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |
| 4.2                  | Read Address:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |
| 5.1                  |                                    |                                                                                                                                                                                                                                 |
| 5.2                  | Read Address + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                             |
| 6.1                  |                                    | Perform <b>A</b> ^= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |
| 6.2                  | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |
| +X.1                 |                                    |                                                                                                                                                                                                                                 |



|          |                   |                                                                                                 |
|----------|-------------------|-------------------------------------------------------------------------------------------------|
| BVC (50) | Cycles: 2         | Size: 2                                                                                         |
| Relative |                   |                                                                                                 |
| Cycle    | R/W               | Desc.                                                                                           |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                |
| 1.1      |                   | Inc. <b>PC</b> . Decide to branch if <b>V</b> is 0.                                             |
| 1.2      | Read <b>PC:PB</b> | Store as Operand. Address = i16(i8(Operand.L)) + <b>PC</b> . If not branching, poll interrupts. |
| 2.1      |                   | Inc. <b>PC</b> . If not branching, end (next half-cycle is 3.2).                                |
| 2.2      |                   | Poll interrupts.                                                                                |
| 3.1      |                   | Set <b>PC</b> to Address.                                                                       |
| 3.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                |
| +X.1     |                   |                                                                                                 |

| EOR (51)                             |                                    | Cycles: 5                                                                                                                                                                                                                     | Size: 2 |
|--------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Direct Indirect Indexed (d),y (Read) |                                    |                                                                                                                                                                                                                               |         |
| Cycle                                | R/W                                | Desc.                                                                                                                                                                                                                         |         |
| -X.2                                 | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| 1.1                                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 1.2                                  | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                    |         |
| 2.1                                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                              |         |
| 2.2                                  |                                    |                                                                                                                                                                                                                               |         |
| 3.1                                  |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                             |         |
| 3.2                                  | Read Address                       | Store as Pointer.                                                                                                                                                                                                             |         |
| 4.1                                  |                                    |                                                                                                                                                                                                                               |         |
| 4.2                                  | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                           |         |
| 5.1                                  |                                    | Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                            |         |
| 5.2                                  |                                    |                                                                                                                                                                                                                               |         |
| 6.1                                  |                                    |                                                                                                                                                                                                                               |         |
| 6.2                                  | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |         |
| 7.1                                  |                                    |                                                                                                                                                                                                                               |         |
| 7.2                                  | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                           |         |
| 8.1                                  |                                    | Perform <b>A</b> ^= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 8.2                                  | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                              |         |
| +X.1                                 |                                    |                                                                                                                                                                                                                               |         |

| EOR (52)               | Cycles: 5                   | Size: 2                                                                                                                                                                                                                       |
|------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indirect (Read) |                             |                                                                                                                                                                                                                               |
| Cycle                  | R/W                         | Desc.                                                                                                                                                                                                                         |
| -X.2                   | Read PC:PB                  | Store as OpCode.                                                                                                                                                                                                              |
| 1.1                    |                             | Inc. <b>PC</b> .                                                                                                                                                                                                              |
| 1.2                    | Read <b>PC:PB</b>           | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                    |
| 2.1                    |                             | Inc. <b>PC</b> .                                                                                                                                                                                                              |
| 2.2                    |                             |                                                                                                                                                                                                                               |
| 3.1                    |                             | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                             |
| 3.2                    | Read Address                | Store as Pointer.                                                                                                                                                                                                             |
| 4.1                    |                             |                                                                                                                                                                                                                               |
| 4.2                    | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                           |
| 5.1                    |                             |                                                                                                                                                                                                                               |
| 5.2                    | Read Pointer: <b>DB</b>     | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |
| 6.1                    |                             |                                                                                                                                                                                                                               |
| 6.2                    | Read Pointer + 1: <b>DB</b> | Store as Operand.H.                                                                                                                                                                                                           |
| 7.1                    |                             | Perform <b>A</b> ^= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |
| 7.2                    | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                              |
| +X.1                   |                             |                                                                                                                                                                                                                               |

|                                        |                                    |                                                                                                                                                                                                                                 |         |
|----------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| EOR (53)                               |                                    | Cycles: 7                                                                                                                                                                                                                       | Size: 2 |
| Stack Relative Indirect Indexed (Read) |                                    |                                                                                                                                                                                                                                 |         |
| Cycle                                  | R/W                                | Desc.                                                                                                                                                                                                                           |         |
| -X.2                                   | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| 1.1                                    | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 1.2                                    |                                    | Store as Pointer.                                                                                                                                                                                                               |         |
| 2.1                                    |                                    | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                                                                                            |         |
| 2.2                                    |                                    |                                                                                                                                                                                                                                 |         |
| 3.1                                    |                                    |                                                                                                                                                                                                                                 |         |
| 3.2                                    | Read Address                       | Store as Pointer.                                                                                                                                                                                                               |         |
| 4.1                                    | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                             |         |
| 4.2                                    |                                    | Perform Tmp = ui32(Pointer + <b>Y</b> ).                                                                                                                                                                                        |         |
| 5.1                                    |                                    | Pointer = ui16(Tmp).                                                                                                                                                                                                            |         |
|                                        |                                    | Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                                 |         |
| 5.2                                    |                                    |                                                                                                                                                                                                                                 |         |
| 6.1                                    | Read Pointer:Bank                  |                                                                                                                                                                                                                                 |         |
| 6.2                                    |                                    | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |         |
| 7.1                                    | Read Pointer + 1:Bank (With Carry) |                                                                                                                                                                                                                                 |         |
| 7.2                                    |                                    | Store as Operand.H.                                                                                                                                                                                                             |         |
| 8.1                                    | Read <b>PC:PB</b>                  | Perform <b>A</b> ^= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 8.2                                    |                                    | Store as OpCode.                                                                                                                                                                                                                |         |
| +X.1                                   |                                    |                                                                                                                                                                                                                                 |         |

|                       |                    |                                                                                                                                                                                                    |
|-----------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MVN (54)              | Cycles: 7          | Size: 3                                                                                                                                                                                            |
| Block Move (Negative) |                    |                                                                                                                                                                                                    |
| Cycle                 | R/W                | Desc.                                                                                                                                                                                              |
| -X.2                  | Read <b>PC:PB</b>  | Store as OpCode.                                                                                                                                                                                   |
| 1.1                   |                    | Inc. <b>PC</b> .                                                                                                                                                                                   |
| 1.2                   | Read <b>PC:PB</b>  | Store as Operand.                                                                                                                                                                                  |
| 2.1                   |                    | Inc. <b>PC</b> . Set <b>DB</b> to Operand.L.                                                                                                                                                       |
| 2.2                   | Read <b>PC:PB</b>  | Stores as Bank.                                                                                                                                                                                    |
| 3.1                   |                    | Inc. <b>PC</b> .                                                                                                                                                                                   |
| 3.2                   | Read <b>X:Bank</b> | Store as Operand.L.                                                                                                                                                                                |
| 4.1                   |                    |                                                                                                                                                                                                    |
| 4.2                   | Write <b>Y:DB</b>  | Write Operand.L.                                                                                                                                                                                   |
| 5.1                   |                    | Perform <b>A</b> -= 1. If <b>X</b> flag is set, perform <b>X.L</b> += 1 and <b>Y.L</b> += 1, otherwise perform <b>X</b> += 1 and <b>Y</b> += 1. If <b>A</b> is not \$FFFF, perform <b>PC</b> -= 3. |
| 5.2                   |                    |                                                                                                                                                                                                    |
| 6.1                   |                    |                                                                                                                                                                                                    |
| 6.2                   |                    |                                                                                                                                                                                                    |
| 7.1                   |                    |                                                                                                                                                                                                    |
| 7.2                   | Read <b>PC:PB</b>  | Store as OpCode.                                                                                                                                                                                   |
| +X.1                  |                    |                                                                                                                                                                                                    |

| EOR (55)         | Cycles: 4         | Size: 2                                                                                                                                                                                                                       |
|------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct, X (Read) |                   |                                                                                                                                                                                                                               |
| Cycle            | R/W               | Desc.                                                                                                                                                                                                                         |
| -X.2             | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                              |
| 1.1              |                   | Inc. <b>PC</b> .                                                                                                                                                                                                              |
| 1.2              | Read <b>PC:PB</b> | Store as Operand.                                                                                                                                                                                                             |
| 2.1              |                   | Inc. <b>PC</b> .                                                                                                                                                                                                              |
| 2.2              |                   |                                                                                                                                                                                                                               |
| 3.1              |                   | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                             |
| 3.2              |                   |                                                                                                                                                                                                                               |
| 4.1              |                   |                                                                                                                                                                                                                               |
| 4.2              | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                               |
| 5.1              |                   |                                                                                                                                                                                                                               |
| 5.2              | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                           |
| 6.1              |                   | Perform <b>A</b> ^= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |
| 6.2              | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                              |
| +X.1             |                   |                                                                                                                                                                                                                               |

| LSR (56)                      | Cycles: 6            | Size: 2                                                                                                                                                                                                                                                                                                                                |
|-------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct, X (Read/Modify/Write) |                      |                                                                                                                                                                                                                                                                                                                                        |
| Cycle                         | R/W                  | Desc.                                                                                                                                                                                                                                                                                                                                  |
| -X.2                          | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| 1.1                           | Read <b>PC:PB</b>    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 1.2                           |                      | Store as Operand.                                                                                                                                                                                                                                                                                                                      |
| 2.1                           |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 2.2                           |                      |                                                                                                                                                                                                                                                                                                                                        |
| 3.1                           |                      | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                                                                                                                                      |
| 3.2                           |                      |                                                                                                                                                                                                                                                                                                                                        |
| 4.1                           |                      |                                                                                                                                                                                                                                                                                                                                        |
| 4.2                           | Read Address         | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                        |
| 5.1                           |                      |                                                                                                                                                                                                                                                                                                                                        |
| 5.2                           | Read Address + 1     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                    |
| 6.1                           |                      | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$01), perform Operand.L >>= 1, and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.L & \$01), perform Operand >>= 1, and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |
| 6.2                           |                      | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                          |
| 7.1                           |                      |                                                                                                                                                                                                                                                                                                                                        |
| 7.2                           | Write to Address + 1 | Write Operand.H.                                                                                                                                                                                                                                                                                                                       |
| 8.1                           |                      |                                                                                                                                                                                                                                                                                                                                        |
| 8.2                           | Write to Address     | Write Operand.L.                                                                                                                                                                                                                                                                                                                       |
| 9.1                           |                      |                                                                                                                                                                                                                                                                                                                                        |
| 9.2                           | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| +X.1                          |                      |                                                                                                                                                                                                                                                                                                                                        |

| EOR (57)                     |                                    | Cycles: 6                                                                                                                                                                                                                                       | Size: 2 |
|------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Direct Indirect Indexed Long |                                    |                                                                                                                                                                                                                                                 |         |
| Cycle                        | R/W                                | Desc.                                                                                                                                                                                                                                           |         |
| -X.2                         | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                |         |
| 1.1                          |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                |         |
| 1.2                          | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                      |         |
| 2.1                          |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                |         |
| 2.2                          |                                    |                                                                                                                                                                                                                                                 |         |
| 3.1                          |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                               |         |
| 3.2                          | Read Address                       | Store as Pointer.                                                                                                                                                                                                                               |         |
| 4.1                          |                                    |                                                                                                                                                                                                                                                 |         |
| 4.2                          | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                             |         |
| 5.1                          |                                    |                                                                                                                                                                                                                                                 |         |
| 5.2                          | Read Address + 2                   | Store as Bank.                                                                                                                                                                                                                                  |         |
| 6.1                          |                                    | Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{Y})$ .                                                                                                                                                                               |         |
|                              |                                    | Pointer = $\text{ui16}(\text{Tmp})$ .                                                                                                                                                                                                           |         |
|                              |                                    | Bank += (Tmp >> 16).                                                                                                                                                                                                                            |         |
| 6.2                          | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                 |         |
| 7.1                          |                                    |                                                                                                                                                                                                                                                 |         |
| 7.2                          | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                             |         |
| 8.1                          |                                    | Perform $\mathbf{A} \wedge = \text{Operand}$ . If <b>M</b> flag is set, set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 8.2                          | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                |         |
| +X.1                         |                                    |                                                                                                                                                                                                                                                 |         |



|          |                   |                      |
|----------|-------------------|----------------------|
| CLI (58) | Cycles: 2         | Size: 1              |
| Implied  |                   |                      |
| Cycle    | R/W               | Desc.                |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.     |
| 1.1      |                   | Inc. <b>PC</b> .     |
| 1.2      | Read <b>PC:PB</b> | Discard.             |
| 2.1      |                   | Set the I flag to 0. |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.     |
| +X.1     |                   |                      |

|                    |                                    |                                                                                                                                                                                                                                 |         |
|--------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| EOR (59)           |                                    | Cycles: 4                                                                                                                                                                                                                       | Size: 3 |
| Absolute, Y (Read) |                                    |                                                                                                                                                                                                                                 |         |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                           |         |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| 1.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 1.2                | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                               |         |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 2.2                | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                             |         |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).             |         |
| 3.2                |                                    |                                                                                                                                                                                                                                 |         |
| 4.1                |                                    |                                                                                                                                                                                                                                 |         |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |         |
| 5.1                |                                    |                                                                                                                                                                                                                                 |         |
| 5.2                | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                             |         |
| 6.1                |                                    | Perform <b>A</b> ^= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 6.2                | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| +X.1               |                                    |                                                                                                                                                                                                                                 |         |

|          |                         |                                               |
|----------|-------------------------|-----------------------------------------------|
| PHY (5A) | Cycles: 3               | Size: 1                                       |
| Implied  |                         |                                               |
| Cycle    | R/W                     | Desc.                                         |
| -X.2     | Read <b>PC:PB</b>       | Store as OpCode.                              |
| 1.1      |                         | Inc. <b>PC</b> .                              |
| 1.2      |                         | If <b>X</b> flag is set, skip the next cycle. |
| 2.1      |                         |                                               |
| 2.2      | Write to ( <b>S</b> +0) | Push <b>Y</b> .H onto stack.                  |
| 3.1      |                         | Dec. <b>S</b> by 1.                           |
| 3.2      | Write to ( <b>S</b> +0) | Push <b>Y</b> .L onto stack.                  |
| 4.1      |                         | Dec. <b>S</b> by 1.                           |
| 4.2      | Read <b>PC:PB</b>       | Store as OpCode.                              |
| +X.1     |                         |                                               |

|          |                   |                                                                                                           |
|----------|-------------------|-----------------------------------------------------------------------------------------------------------|
| TCD (5B) | Cycles: 2         | Size: 1                                                                                                   |
| Implied  |                   |                                                                                                           |
| Cycle    | R/W               | Desc.                                                                                                     |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                          |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                          |
| 1.2      |                   | Discard.                                                                                                  |
| 2.1      |                   | Set <b>D</b> to <b>A</b> . Set <b>N</b> based off ( <b>D</b> .H & \$80) and <b>Z</b> based off <b>D</b> . |
| 2.2      |                   | Store as OpCode.                                                                                          |
| +X.1     |                   |                                                                                                           |

|               |                   |                                                      |
|---------------|-------------------|------------------------------------------------------|
| JML (5C)      | Cycles: 4         | Size: 4                                              |
| Absolute Long |                   |                                                      |
| Cycle         | R/W               | Desc.                                                |
| -X.2          | Read <b>PC:PB</b> | Store as OpCode.                                     |
| 1.1           |                   | Inc. <b>PC</b> .                                     |
| 1.2           | Read <b>PC:PB</b> | Store as Address.                                    |
| 2.1           |                   | Inc. <b>PC</b> .                                     |
| 2.2           | Read <b>PC:PB</b> | Store as Address.H.                                  |
| 3.1           |                   | Inc. <b>PC</b> .                                     |
| 3.2           | Read <b>PC:PB</b> | Stores as Bank.                                      |
| 4.1           |                   | Copy Address to <b>PC</b> . Copy Bank to <b>PB</b> . |
| 4.2           | Read <b>PC:PB</b> | Store as OpCode.                                     |
| +X.1          |                   |                                                      |

|                    |                                    |                                                                                                                                                                                                                                 |         |
|--------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| EOR (5D)           |                                    | Cycles: 4                                                                                                                                                                                                                       | Size: 3 |
| Absolute, X (Read) |                                    |                                                                                                                                                                                                                                 |         |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                           |         |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                |         |
| 1.1                | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 1.2                |                                    | Store as Pointer.                                                                                                                                                                                                               |         |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                |         |
| 2.2                |                                    | Store as Pointer.H.                                                                                                                                                                                                             |         |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>X</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).             |         |
| 3.2                |                                    |                                                                                                                                                                                                                                 |         |
| 4.1                |                                    |                                                                                                                                                                                                                                 |         |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                 |         |
| 5.1                | Read Pointer + 1:Bank (With Carry) |                                                                                                                                                                                                                                 |         |
| 5.2                |                                    | Store as Operand.H.                                                                                                                                                                                                             |         |
| 6.1                | Read <b>PC:PB</b>                  | Perform <b>A</b> ^= Operand. If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 6.2                |                                    | Store as OpCode.                                                                                                                                                                                                                |         |
| +X.1               |                                    |                                                                                                                                                                                                                                 |         |

| LSR (5E)                       | Cycles: 7                              | Size: 3                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute, X (Read/Modiy/Write) |                                        |                                                                                                                                                                                                                                                                                                                                                              |
| Cycle                          | R/W                                    | Desc.                                                                                                                                                                                                                                                                                                                                                        |
| -X.2                           | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                                                                                                                                                                                                             |
| 1.1                            |                                        | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                             |
| 1.2                            | Read <b>PC:PB</b>                      | Store as Pointer.                                                                                                                                                                                                                                                                                                                                            |
| 2.1                            |                                        | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                             |
| 2.2                            | Read <b>PC:PB</b>                      | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                          |
| 3.1                            |                                        | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank = $\text{DB} + (\text{Tmp} \gg 16)$ .                                                                                                                                                                                      |
| 3.2                            |                                        |                                                                                                                                                                                                                                                                                                                                                              |
| 4.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                              |
| 4.2                            | Read Pointer:Bank                      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                              |
| 5.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                              |
| 5.2                            | Read Pointer + 1:Bank (With Carry)     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                          |
| 6.1                            |                                        | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$01), perform $\text{Operand.L} \gg= 1$ , and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.L & \$01), perform $\text{Operand} \gg= 1$ , and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |
| 6.2                            |                                        | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                |
| 7.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                              |
| 7.2                            | Write to Pointer + 1:Bank (With Carry) | Write Operand.H.                                                                                                                                                                                                                                                                                                                                             |
| 8.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                              |
| 8.2                            | Write to Pointer:Bank                  | Write Operand.L.                                                                                                                                                                                                                                                                                                                                             |
| 9.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                              |
| 9.2                            | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                                                                                                                                                                                                             |
| +X.1                           |                                        |                                                                                                                                                                                                                                                                                                                                                              |

|                         |                                    |                                                                                                                                                                                                                                                 |         |
|-------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| EOR (5F)                |                                    | Cycles: 5                                                                                                                                                                                                                                       | Size: 4 |
| Absolute Long, X (Read) |                                    |                                                                                                                                                                                                                                                 |         |
| Cycle                   | R/W                                | Desc.                                                                                                                                                                                                                                           |         |
| -X.2                    | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                |         |
| 1.1                     |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                |         |
| 1.2                     | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                               |         |
| 2.1                     |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                |         |
| 2.2                     | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                                             |         |
| 3.1                     |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                |         |
| 3.2                     | Read <b>PC:PB</b>                  | Stores as Bank.                                                                                                                                                                                                                                 |         |
| 4.1                     |                                    | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$ .                                                                                                                                                                |         |
|                         |                                    | Pointer = $\text{ui16}(\text{Tmp})$ .                                                                                                                                                                                                           |         |
|                         |                                    | Bank += (Tmp >> 16).                                                                                                                                                                                                                            |         |
| 4.2                     | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                 |         |
| 5.1                     |                                    |                                                                                                                                                                                                                                                 |         |
| 5.2                     | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                             |         |
| 6.1                     |                                    | Perform $\text{A} \wedge = \text{Operand}$ . If <b>M</b> flag is set, set <b>N</b> based off ( <b>A</b> .L & \$80) and <b>Z</b> based off <b>A</b> .L, otherwise set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |         |
| 6.2                     | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                |         |
| +X.1                    |                                    |                                                                                                                                                                                                                                                 |         |



|          |                   |                                                     |
|----------|-------------------|-----------------------------------------------------|
| RTS (60) | Cycles: 6         | Size: 1                                             |
| Implied  |                   |                                                     |
| Cycle    | R/W               | Desc.                                               |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                    |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                    |
| 1.2      |                   |                                                     |
| 2.1      |                   |                                                     |
| 2.2      |                   |                                                     |
| 3.1      |                   | Inc. <b>S</b> by 1.                                 |
| 3.2      |                   | Store as Operand.L.                                 |
| 4.1      |                   |                                                     |
| 4.2      |                   | Store as Operand.H.                                 |
| 5.1      |                   |                                                     |
| 5.2      |                   |                                                     |
| 6.1      | Read <b>PC:PB</b> | Inc. <b>S</b> by 1. Set <b>PC</b> to (Operand + 1). |
| 6.2      |                   | Store as OpCode.                                    |
| +X.1     |                   |                                                     |

| ADC (61) Cycles: 6                   |                             | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indexed Indirect (d,x) (Read) |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Cycle                                | R/W                         | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| -X.2                                 | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.1                                  |                             | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.2                                  | Read <b>PC:PB</b>           | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2.1                                  |                             | Inc. <b>PC</b> . Set Pointer to Pointer + <b>X</b> + <b>D</b> . Address = Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2.2                                  |                             | If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3.1                                  |                             | Address.H = Pointer.H (fixes high byte of address).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3.2                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4.1                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4.2                                  | Read Address                | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.1                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.2                                  | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6.1                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6.2                                  | Read Pointer: <b>DB</b>     | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 7.1                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 7.2                                  | Read Pointer + 1: <b>DB</b> | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8.1                                  |                             | <p>If <b>M</b> flag == 0 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) + ui16(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80 != 0.</p> <p><b>C</b> flag = Result &gt; \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = ui16(<b>A.L</b> &amp; 0x0F) + ui16(Operand &amp; 0x0F) + <b>C</b>;</p> <p>if (Lo &gt; 9) Lo += 6.</p> <p>CarryToHi = Lo &gt; \$0F ? 1 : 0.</p> <p>HiSum = ui16(<b>A.L</b> &gt;&gt; 4) + ui16(Operand &gt;&gt; 4) + CarryToHi.</p> <p><b>V</b> flag = (~((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &gt; 9) HiSum += 6.</p> <p><b>C</b> flag = HiSum &gt; \$0F.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) + ui32(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000 != 0.</p> <p><b>C</b> flag = Result &gt; \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Carry = ui16(<b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = ui16((<b>A</b> &gt;&gt; i) &amp; \$0F) + ui16((Operand &gt;&gt; i) &amp; \$0F) + Carry.</p> <p>if (i == 12) <b>V</b> flag = (~(<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08 != 0.</p> <p>if (DigitSum &gt; 9) DigitSum += 6.</p> <p>Carry = (DigitSum &gt; \$0F) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Carry != 0. <b>Z</b> flag = Result != \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 8.2                                  | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| +X.1                                 |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                               |                         |                                                       |
|-------------------------------|-------------------------|-------------------------------------------------------|
| PER (62)                      | Cycles: 6               | Size: 3                                               |
| Program Counter Relative Long |                         |                                                       |
| Cycle                         | R/W                     | Desc.                                                 |
| -X.2                          | Read <b>PC:PB</b>       | Store as OpCode.                                      |
| 1.1                           |                         | Inc. <b>PC</b> .                                      |
| 1.2                           | Read <b>PC:PB</b>       | Store as Operand.                                     |
| 2.1                           |                         | Inc. <b>PC</b> .                                      |
| 2.2                           | Read <b>PC:PB</b>       | Store as Operand.H.                                   |
| 3.1                           |                         | Inc. <b>PC</b> . Set Operand to Operand + <b>PC</b> . |
| 3.2                           |                         |                                                       |
| 4.1                           |                         |                                                       |
| 4.2                           | Write to ( <b>S</b> +0) | Push Operand.H onto stack.                            |
| 5.1                           |                         |                                                       |
| 5.2                           | Write to ( <b>S</b> -1) | Push Operand.L onto stack.                            |
| 6.1                           |                         | Dec. <b>S</b> by 2.                                   |
| 6.2                           | Read <b>PC:PB</b>       | Store as OpCode.                                      |
| +X.1                          |                         |                                                       |

| ADC (63) Cycles: 4    |                   | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stack Relative (Read) |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Cycle                 | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| -X.2                  | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1.1                   | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1.2                   |                   | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2.1                   |                   | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2.2                   |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 3.1                   |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 3.2                   | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4.1                   | Read Address + 1  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4.2                   |                   | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.1                   |                   | <p>If <b>M</b> flag == 0 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A</b>.L) + ui16(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui16(<b>A</b>.L) ^ ui16(Operand)) &amp; (ui16(<b>A</b>.L) ^ ui8(Result)) &amp; \$80) != 0.</p> <p><b>C</b> flag = Result &gt; \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = ui16(<b>A</b>.L &amp; 0x0F) + ui16(Operand &amp; 0x0F) + <b>C</b>;</p> <p>if (Lo &gt; 9) Lo += 6.</p> <p>CarryToHi = Lo &gt; \$0F ? 1 : 0.</p> <p>HiSum = ui16(<b>A</b>.L &gt;&gt; 4) + ui16(Operand &gt;&gt; 4) + CarryToHi.</p> <p><b>V</b> flag = (~((<b>A</b>.L &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A</b>.L &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &gt; 9) HiSum += 6.</p> <p><b>C</b> flag = HiSum &gt; \$0F.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) + ui32(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0).</p> <p><b>C</b> flag = Result &gt; \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Carry = ui16(<b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = ui16((<b>A</b> &gt;&gt; i) &amp; \$0F) + ui16((Operand &gt;&gt; i) &amp; \$0F) + Carry.</p> <p>if (i == 12) <b>V</b> flag = (~(<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.</p> <p>if (DigitSum &gt; 9) DigitSum += 6.</p> <p>Carry = (DigitSum &gt; \$0F) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Carry != 0. <b>Z</b> flag = Result != \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 5.2                   |                   | Read <b>PC:PB</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                       |                   | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| +X.1                  |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|                |                      |                                                                                   |
|----------------|----------------------|-----------------------------------------------------------------------------------|
| STZ (64)       | Cycles: 3            | Size: 2                                                                           |
| Direct (Write) |                      |                                                                                   |
| Cycle          | R/W                  | Desc.                                                                             |
| -X.2           | Read <b>PC:PB</b>    | Store as OpCode.                                                                  |
| 1.1            |                      | Inc. <b>PC</b> .                                                                  |
| 1.2            | Read <b>PC:PB</b>    | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                        |
| 2.1            |                      | Inc. <b>PC</b> .                                                                  |
| 2.2            |                      |                                                                                   |
| 3.1            |                      | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> . |
| 3.2            | Write to Address     | Write 0. If <b>M</b> flag is set, skip the next cycle.                            |
| 4.1            |                      |                                                                                   |
| 4.2            | Write to Address + 1 | Write 0.                                                                          |
| 5.1            |                      |                                                                                   |
| 5.2            | Read <b>PC:PB</b>    | Store as OpCode.                                                                  |
| +X.1           |                      |                                                                                   |

| ADC (65) Cycles: 3 |                   | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct (Read)      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Cycle              | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| -X.2               | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.1                |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.2                | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2.1                |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2.2                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3.1                |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3.2                | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4.1                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4.2                | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.1                |                   | <p>If <b>M</b> flag == 0 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) + ui16(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80 != 0.</p> <p><b>C</b> flag = Result &gt; \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = ui16(<b>A.L</b> &amp; 0x0F) + ui16(Operand &amp; 0x0F) + <b>C</b>;</p> <p>if (Lo &gt; 9) Lo += 6.</p> <p>CarryToHi = Lo &gt; \$0F ? 1 : 0.</p> <p>HiSum = ui16(<b>A.L</b> &gt;&gt; 4) + ui16(Operand &gt;&gt; 4) + CarryToHi.</p> <p><b>V</b> flag = (~((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &gt; 9) HiSum += 6.</p> <p><b>C</b> flag = HiSum &gt; \$0F.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) + ui32(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000 != 0.</p> <p><b>C</b> flag = Result &gt; \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Carry = ui16(<b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = ui16((<b>A</b> &gt;&gt; i) &amp; \$0F) + ui16((Operand &gt;&gt; i) &amp; \$0F) + Carry.</p> <p>if (i == 12) <b>V</b> flag = (~(<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08 != 0.</p> <p>if (DigitSum &gt; 9) DigitSum += 6.</p> <p>Carry = (DigitSum &gt; \$0F) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Carry != 0. <b>Z</b> flag = Result != \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 5.2                | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| +X.1               |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| ROR (66)                   |                      | Cycles: 5                                                                                                                                                                                                                                                                                                                                                                                             | Size: 2 |
|----------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Direct (Read/Modify/Write) |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |         |
| Cycle                      | R/W                  | Desc.                                                                                                                                                                                                                                                                                                                                                                                                 |         |
| -X.2                       | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                      |         |
| 1.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                      |         |
| 1.2                        | Read <b>PC:PB</b>    | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                            |         |
| 2.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                      |         |
| 2.2                        |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |         |
| 3.1                        |                      | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                     |         |
| 3.2                        | Read Address         | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                       |         |
| 4.1                        |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |         |
| 4.2                        | Read Address + 1     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                   |         |
| 5.1                        |                      | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$01), perform Operand.L = (Operand.L >> 1)   ( <b>C</b> << 7), and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.L & \$01), perform Operand = (Operand >> 1)   ( <b>C</b> << 15), and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |         |
| 5.2                        |                      | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                         |         |
| 6.1                        |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |         |
| 6.2                        | Write to Address + 1 | Write Operand.H.                                                                                                                                                                                                                                                                                                                                                                                      |         |
| 7.1                        |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |         |
| 7.2                        | Write to Address     | Write Operand.L.                                                                                                                                                                                                                                                                                                                                                                                      |         |
| 8.1                        |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |         |
| 8.2                        | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                      |         |
| +X.1                       |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |         |

| ADC (67) Cycles: 6          |                                    | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indirect Long (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Cycle                       | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| -X.2                        | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1.1                         |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1.2                         | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2.1                         |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2.2                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 3.1                         |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 3.2                         | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 4.1                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4.2                         | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.1                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.2                         | Read Address + 2                   | Store as Bank.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 6.1                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 6.2                         | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7.1                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7.2                         | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 8.1                         |                                    | <p>If <b>M</b> flag == 0 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) + ui16(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.</p> <p><b>C</b> flag = Result &gt; \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = ui16(<b>A.L</b> &amp; 0x0F) + ui16(Operand &amp; 0x0F) + <b>C</b>;</p> <p>if (Lo &gt; 9) Lo += 6.</p> <p>CarryToHi = Lo &gt; \$0F ? 1 : 0.</p> <p>HiSum = ui16(<b>A.L</b> &gt;&gt; 4) + ui16(Operand &gt;&gt; 4) + CarryToHi.</p> <p><b>V</b> flag = (~((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &gt; 9) HiSum += 6.</p> <p><b>C</b> flag = HiSum &gt; \$0F.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) + ui32(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0).</p> <p><b>C</b> flag = Result &gt; \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Carry = ui16(<b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = ui16((<b>A</b> &gt;&gt; i) &amp; \$0F) + ui16((Operand &gt;&gt; i) &amp; \$0F) + Carry.</p> <p>if (i == 12) <b>V</b> flag = (~(<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.</p> <p>if (DigitSum &gt; 9) DigitSum += 6.</p> <p>Carry = (DigitSum &gt; \$0F) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Carry != 0. <b>Z</b> flag = Result != \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 8.2                         | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| +X.1                        |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



|          |                     |                                                                                                                                                                                                                                                                                  |
|----------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PLA (68) | Cycles: 4           | Size: 1                                                                                                                                                                                                                                                                          |
| Implied  |                     |                                                                                                                                                                                                                                                                                  |
| Cycle    | R/W                 | Desc.                                                                                                                                                                                                                                                                            |
| -X.2     | Read <b>PC:PB</b>   | Store as OpCode.                                                                                                                                                                                                                                                                 |
| 1.1      |                     | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                 |
| 1.2      |                     |                                                                                                                                                                                                                                                                                  |
| 2.1      |                     |                                                                                                                                                                                                                                                                                  |
| 2.2      |                     |                                                                                                                                                                                                                                                                                  |
| 3.1      |                     | Inc. S by 1.                                                                                                                                                                                                                                                                     |
| 3.2      | Read ( <b>S</b> +0) | Store as Operand.L. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                |
| 4.1      |                     |                                                                                                                                                                                                                                                                                  |
| 4.2      | Read ( <b>S</b> +1) | Store as Operand.H.                                                                                                                                                                                                                                                              |
| 5.1      |                     | Inc. <b>S</b> by 1. If <b>M</b> flag is set, set <b>A.L</b> to Operand.L and set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand and set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |
| 5.2      | Read <b>PC:PB</b>   | Store as OpCode.                                                                                                                                                                                                                                                                 |
| +X.1     |                     |                                                                                                                                                                                                                                                                                  |

| ADC (69) Cycles: 2 |                   | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Immediate          |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Cycle              | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| -X.2               | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1.1                |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1.2                | Read <b>PC:PB</b> | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2.1                |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2.2                | Read <b>PC:PB</b> | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3.1                |                   | <p>If <b>M</b> flag == 0 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A</b>.L) + ui16(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui16(<b>A</b>.L) ^ ui16(Operand)) &amp; (ui16(<b>A</b>.L) ^ ui8(Result)) &amp; \$80) != 0.</p> <p><b>C</b> flag = Result &gt; \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = ui16(<b>A</b>.L &amp; 0x0F) + ui16(Operand &amp; 0x0F) + <b>C</b>;</p> <p>if (Lo &gt; 9) Lo += 6.</p> <p>CarryToHi = Lo &gt; \$0F ? 1 : 0.</p> <p>HiSum = ui16(<b>A</b>.L &gt;&gt; 4) + ui16(Operand &gt;&gt; 4) + CarryToHi.</p> <p><b>V</b> flag = (~((<b>A</b>.L &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A</b>.L &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &gt; 9) HiSum += 6.</p> <p><b>C</b> flag = HiSum &gt; \$0F.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) + ui32(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0).</p> <p><b>C</b> flag = Result &gt; \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Carry = ui16(<b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = ui16((<b>A</b> &gt;&gt; i) &amp; \$0F) + ui16((Operand &gt;&gt; i) &amp; \$0F) + Carry.</p> <p>if (i == 12) <b>V</b> flag = (~(<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.</p> <p>if (DigitSum &gt; 9) DigitSum += 6.</p> <p>Carry = (DigitSum &gt; \$0F) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Carry != 0. <b>Z</b> flag = Result != \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 3.2                | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| +X.1               |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|          |                   |                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROR (6A) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Implied  |                   |                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                   |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2.1      |                   | If <b>M</b> flag is set, set <b>C</b> based off ( <b>A.L</b> & \$01), perform <b>A.L</b> = ( <b>A.L</b> >> 1)   ( <b>C</b> << 7), and set <b>N</b> based off ( <b>A.L</b> & \$80) and <b>Z</b> based off <b>A.L</b> , otherwise set <b>C</b> based off ( <b>A.L</b> & \$01), perform <b>A</b> = ( <b>A</b> >> 1)   ( <b>C</b> << 15), and set <b>N</b> based off ( <b>A.H</b> & \$80) and <b>Z</b> based off <b>A</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                        |
| +X.1     |                   |                                                                                                                                                                                                                                                                                                                                                                                                                         |

|          |                     |                                                                            |
|----------|---------------------|----------------------------------------------------------------------------|
| RTL (6B) | Cycles: 6           | Size: 1                                                                    |
| Implied  |                     |                                                                            |
| Cycle    | R/W                 | Desc.                                                                      |
| -X.2     | Read <b>PC:PB</b>   | Store as OpCode.                                                           |
| 1.1      |                     | Inc. <b>PC</b> .                                                           |
| 1.2      |                     |                                                                            |
| 2.1      |                     |                                                                            |
| 2.2      |                     |                                                                            |
| 3.1      |                     |                                                                            |
| 3.2      | Read ( <b>S</b> +1) | Store as Operand.L.                                                        |
| 4.1      |                     |                                                                            |
| 4.2      | Read ( <b>S</b> +2) | Store as Operand.H.                                                        |
| 5.1      |                     |                                                                            |
| 5.2      | Read ( <b>S</b> +3) | Store as <b>PB</b> .                                                       |
| 6.1      |                     | Inc. <b>S</b> by 3. Set <b>PC</b> to (Operand + 1). Set <b>PB</b> to Bank. |
| 6.2      | Read <b>PC:PB</b>   | Store as OpCode.                                                           |
| +X.1     |                     |                                                                            |

|                          |                   |                           |         |
|--------------------------|-------------------|---------------------------|---------|
| JMP (6C)                 |                   | Cycles: 5                 | Size: 3 |
| Absolute Indirect (Jump) |                   |                           |         |
| Cycle                    | R/W               | Desc.                     |         |
| -X.2                     | Read <b>PC:PB</b> | Store as OpCode.          |         |
| 1.1                      |                   | Inc. <b>PC</b> .          |         |
| 1.2                      | Read <b>PC:PB</b> | Store as Pointer.         |         |
| 2.1                      |                   | Inc. <b>PC</b> .          |         |
| 2.2                      | Read <b>PC:PB</b> | Store as Pointer.H.       |         |
| 3.1                      |                   | Inc. <b>PC</b> .          |         |
| 3.2                      | Read Pointer      | Store as Address.         |         |
| 4.1                      |                   |                           |         |
| 4.2                      | Read Pointer + 1  | Store as Address.H.       |         |
| 5.1                      |                   | Set <b>PC</b> to Address. |         |
| 5.2                      | Read <b>PC:PB</b> | Store as OpCode.          |         |
| +X.1                     |                   |                           |         |

| ADC (6D) Cycles: 4 |                                          | Size: 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute (Read)    |                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Cycle              | R/W                                      | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| -X.2               | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1.1                |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1.2                | Read <b>PC:PB</b>                        | Store as Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2.1                |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2.2                | Read <b>PC:PB</b>                        | Store as Address.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3.1                |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3.2                | Read Address: <b>DB</b>                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4.1                |                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4.2                | Read Address + 1: <b>DB</b> (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.1                |                                          | <p>If <b>M</b> flag == 0 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) + ui16(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.</p> <p><b>C</b> flag = Result &gt; \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = ui16(<b>A.L</b> &amp; 0x0F) + ui16(Operand &amp; 0x0F) + <b>C</b>;</p> <p>if (Lo &gt; 9) Lo += 6.</p> <p>CarryToHi = Lo &gt; \$0F ? 1 : 0.</p> <p>HiSum = ui16(<b>A.L</b> &gt;&gt; 4) + ui16(Operand &gt;&gt; 4) + CarryToHi.</p> <p><b>V</b> flag = (~((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &gt; 9) HiSum += 6.</p> <p><b>C</b> flag = HiSum &gt; \$0F.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) + ui32(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0).</p> <p><b>C</b> flag = Result &gt; \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Carry = ui16(<b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = ui16((<b>A</b> &gt;&gt; i) &amp; \$0F) + ui16((Operand &gt;&gt; i) &amp; \$0F) + Carry.</p> <p>if (i == 12) <b>V</b> flag = (~(<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.</p> <p>if (DigitSum &gt; 9) DigitSum += 6.</p> <p>Carry = (DigitSum &gt; \$0F) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Carry != 0. <b>Z</b> flag = Result != \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 5.2                | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| +X.1               |                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| ROR (6E)                     | Cycles: 6                                    | Size: 3                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute (Read/Modify/Write) |                                              |                                                                                                                                                                                                                                                                                                                                                                                                       |
| Cycle                        | R/W                                          | Desc.                                                                                                                                                                                                                                                                                                                                                                                                 |
| -X.2                         | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                      |
| 1.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                      |
| 1.2                          | Read <b>PC:PB</b>                            | Store as Address.                                                                                                                                                                                                                                                                                                                                                                                     |
| 2.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                      |
| 2.2                          | Read <b>PC:PB</b>                            | Store as Address.H.                                                                                                                                                                                                                                                                                                                                                                                   |
| 3.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                      |
| 3.2                          | Read Address: <b>DB</b>                      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                       |
| 4.1                          |                                              |                                                                                                                                                                                                                                                                                                                                                                                                       |
| 4.2                          | Read Address + 1: <b>DB</b> (With Carry)     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.1                          |                                              | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$01), perform Operand.L = (Operand.L >> 1)   ( <b>C</b> << 7), and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.L & \$01), perform Operand = (Operand >> 1)   ( <b>C</b> << 15), and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |
| 5.2                          |                                              | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                         |
| 6.1                          |                                              |                                                                                                                                                                                                                                                                                                                                                                                                       |
| 6.2                          | Write to Address + 1: <b>DB</b> (With Carry) | Write Operand.H.                                                                                                                                                                                                                                                                                                                                                                                      |
| 7.1                          |                                              |                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7.2                          | Write to Address: <b>DB</b>                  | Write Operand.L.                                                                                                                                                                                                                                                                                                                                                                                      |
| 8.1                          |                                              |                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8.2                          | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                      |
| +X.1                         |                                              |                                                                                                                                                                                                                                                                                                                                                                                                       |

| ADC (6F)             |                                    | Cycles: 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Size: 4 |
|----------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Absolute Long (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |
| Cycle                | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |
| -X.2                 | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| 1.1                  | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| 1.2                  |                                    | Store as Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 2.1                  | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| 2.2                  |                                    | Store as Address.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |
| 3.1                  | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| 3.2                  |                                    | Stores as Bank.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |
| 4.1                  | Read Address:Bank                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| 4.2                  |                                    | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |
| 5.1                  | Read Address + 1:Bank (With Carry) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |
| 5.2                  |                                    | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |
| 6.1                  |                                    | If <b>M</b> flag == 0 (8-bit):<br>If <b>D</b> flag == 0:<br>Result = ui16( <b>A</b> .L) + ui16(Operand) + <b>C</b> .<br><b>V</b> flag = (~ui16( <b>A</b> .L) ^ ui16(Operand)) & (ui16( <b>A</b> .L) ^ ui8(Result)) & \$80) != 0.<br><b>C</b> flag = Result > \$FF.<br>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result & \$80) != 0.<br>If <b>D</b> flag == 1:<br>Lo = ui16( <b>A</b> .L & 0x0F) + ui16(Operand & 0x0F) + <b>C</b> ;<br>if (Lo > 9) Lo += 6.<br>CarryToHi = Lo > \$0F ? 1 : 0.<br>HiSum = ui16( <b>A</b> .L >> 4) + ui16(Operand >> 4) + CarryToHi.<br><b>V</b> flag = (~(( <b>A</b> .L >> 4) ^ (Operand >> 4)) & (( <b>A</b> .L >> 4) ^ HiSum) & \$08) != 0.<br>if (HiSum > 9) HiSum += 6.<br><b>C</b> flag = HiSum > \$0F.<br>Result = ui8(((HiSum & 0x0F) << 4)   (Lo & \$0F)).<br><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result & \$80) != 0.<br>If <b>M</b> flag == 0 (16-bit):<br>If <b>D</b> flag == 0:<br>Result = ui32( <b>A</b> ) + ui32(Operand) + <b>C</b> .<br><b>V</b> flag = (~ui32( <b>A</b> ) ^ ui32(Operand)) & (ui32( <b>A</b> ) ^ ui16(Result)) & \$8000) != 0).<br><b>C</b> flag = Result > \$FFFF.<br>Result = ui16(Result).<br><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result & \$8000) != 0.<br>If <b>D</b> flag == 1:<br>Result = ui16(0). Carry = ui16( <b>C</b> ).<br>for i = 0, 4, 8, 12:<br>DigitSum = ui16(( <b>A</b> >> i) & \$0F) + ui16((Operand >> i) & \$0F) + Carry.<br>if (i == 12) <b>V</b> flag = (~(( <b>A</b> >> 12) ^ (Operand >> 12)) & (( <b>A</b> >> 12) ^ DigitSum) & \$08) != 0.<br>if (DigitSum > 9) DigitSum += 6.<br>Carry = (DigitSum > \$0F) ? 1 : 0.<br>Result  = ui16((DigitSum & \$0F) << i).<br><b>C</b> flag = Carry != 0. <b>Z</b> flag = Result != \$0000. <b>N</b> flag = (Result & \$8000) != 0. |         |
| 6.2                  | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| +X.1                 |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |



|          |                   |                                                                                                 |
|----------|-------------------|-------------------------------------------------------------------------------------------------|
| BVS (70) | Cycles: 2         | Size: 2                                                                                         |
| Relative |                   |                                                                                                 |
| Cycle    | R/W               | Desc.                                                                                           |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                |
| 1.1      |                   | Inc. <b>PC</b> . Decide to branch if <b>V</b> is 1.                                             |
| 1.2      | Read <b>PC:PB</b> | Store as Operand. Address = i16(i8(Operand.L)) + <b>PC</b> . If not branching, poll interrupts. |
| 2.1      |                   | Inc. <b>PC</b> . If not branching, end (next half-cycle is 3.2).                                |
| 2.2      |                   | Poll interrupts.                                                                                |
| 3.1      |                   | Set <b>PC</b> to Address.                                                                       |
| 3.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                |
| +X.1     |                   |                                                                                                 |

| ADC (71) Cycles: 5 Size: 2           |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indirect Indexed (d),y (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Cycle                                | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| -X.2                                 | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1.1                                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1.2                                  | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2.1                                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2.2                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 3.1                                  |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3.2                                  | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4.1                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 4.2                                  | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.1                                  |                                    | Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.2                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 6.1                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 6.2                                  | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7.1                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7.2                                  | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 8.1                                  |                                    | <p>If <b>M</b> flag == 0 (8-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = ui16(<b>A.L</b>) + ui16(Operand) + <b>C</b>.<br/>             <b>V</b> flag = (~ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80 != 0.<br/>             <b>C</b> flag = Result &gt; \$FF.<br/>             Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:<br/>             Lo = ui16(<b>A.L</b> &amp; 0x0F) + ui16(Operand &amp; 0x0F) + <b>C</b>;<br/>             if (Lo &gt; 9) Lo += 6.<br/>             CarryToHi = Lo &gt; \$0F ? 1 : 0.<br/>             HiSum = ui16(<b>A.L</b> &gt;&gt; 4) + ui16(Operand &gt;&gt; 4) + CarryToHi.<br/>             <b>V</b> flag = (~((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.<br/>             if (HiSum &gt; 9) HiSum += 6.<br/>             <b>C</b> flag = HiSum &gt; \$0F.<br/>             Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).<br/>             <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = ui32(<b>A</b>) + ui32(Operand) + <b>C</b>.<br/>             <b>V</b> flag = (~ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000 != 0).<br/>             <b>C</b> flag = Result &gt; \$FFFF.<br/>             Result = ui16(Result).<br/>             <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:<br/>             Result = ui16(0). Carry = ui16(<b>C</b>).<br/>             for i = 0, 4, 8, 12:<br/>               DigitSum = ui16((<b>A</b> &gt;&gt; i) &amp; \$0F) + ui16((Operand &gt;&gt; i) &amp; \$0F) + Carry.<br/>               if (i == 12) <b>V</b> flag = (~(<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08 != 0.<br/>               if (DigitSum &gt; 9) DigitSum += 6.<br/>               Carry = (DigitSum &gt; \$0F) ? 1 : 0.<br/>               Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).<br/>             <b>C</b> flag = Carry != 0. <b>Z</b> flag = Result != \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 8.2                                  | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| +X.1                                 |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| ADC (72)               |                             | Cycles: 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Size: 2 |
|------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Direct Indirect (Read) |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| Cycle                  | R/W                         | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |
| -X.2                   | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |
| 1.1                    |                             | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |
| 1.2                    | Read <b>PC:PB</b>           | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| 2.1                    |                             | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |
| 2.2                    |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 3.1                    |                             | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |
| 3.2                    | Read Address                | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |
| 4.1                    |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 4.2                    | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 5.1                    |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 5.2                    | Read Pointer: <b>DB</b>     | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 6.1                    |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 6.2                    | Read Pointer + 1: <b>DB</b> | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 7.1                    |                             | <p>If <b>M</b> flag == 0 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) + ui16(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80 != 0.</p> <p><b>C</b> flag = Result &gt; \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = ui16(<b>A.L</b> &amp; 0x0F) + ui16(Operand &amp; 0x0F) + <b>C</b>;</p> <p>if (Lo &gt; 9) Lo += 6.</p> <p>CarryToHi = Lo &gt; \$0F ? 1 : 0.</p> <p>HiSum = ui16(<b>A.L</b> &gt;&gt; 4) + ui16(Operand &gt;&gt; 4) + CarryToHi.</p> <p><b>V</b> flag = (~((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &gt; 9) HiSum += 6.</p> <p><b>C</b> flag = HiSum &gt; \$0F.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) + ui32(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000 != 0.</p> <p><b>C</b> flag = Result &gt; \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Carry = ui16(<b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = ui16((<b>A</b> &gt;&gt; i) &amp; \$0F) + ui16((Operand &gt;&gt; i) &amp; \$0F) + Carry.</p> <p>if (i == 12) <b>V</b> flag = (~(<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08 != 0.</p> <p>if (DigitSum &gt; 9) DigitSum += 6.</p> <p>Carry = (DigitSum &gt; \$0F) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Carry != 0. <b>Z</b> flag = Result != \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |         |
| 7.2                    | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |
| +X.1                   |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |

| ADC (73) Cycles: 7                     |                                    | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stack Relative Indirect Indexed (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Cycle                                  | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| -X.2                                   | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.1                                    |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.2                                    | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2.1                                    |                                    | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2.2                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3.1                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3.2                                    | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4.1                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4.2                                    | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.1                                    |                                    | Perform Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.2                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6.1                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6.2                                    | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 7.1                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 7.2                                    | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 8.1                                    |                                    | <p>If <b>M</b> flag == 0 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) + ui16(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80 != 0.</p> <p><b>C</b> flag = Result &gt; \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = ui16(<b>A.L</b> &amp; 0x0F) + ui16(Operand &amp; 0x0F) + <b>C</b>;</p> <p>if (Lo &gt; 9) Lo += 6.</p> <p>CarryToHi = Lo &gt; \$0F ? 1 : 0.</p> <p>HiSum = ui16(<b>A.L</b> &gt;&gt; 4) + ui16(Operand &gt;&gt; 4) + CarryToHi.</p> <p><b>V</b> flag = (~((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &gt; 9) HiSum += 6.</p> <p><b>C</b> flag = HiSum &gt; \$0F.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) + ui32(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000 != 0.</p> <p><b>C</b> flag = Result &gt; \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Carry = ui16(<b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = ui16((<b>A</b> &gt;&gt; i) &amp; \$0F) + ui16((Operand &gt;&gt; i) &amp; \$0F) + Carry.</p> <p>if (i == 12) <b>V</b> flag = (~(<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08 != 0.</p> <p>if (DigitSum &gt; 9) DigitSum += 6.</p> <p>Carry = (DigitSum &gt; \$0F) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Carry != 0. <b>Z</b> flag = Result != \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 8.2                                    | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| +X.1                                   |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                   |                      |                                                                                   |
|-------------------|----------------------|-----------------------------------------------------------------------------------|
| STZ (74)          | Cycles: 4            | Size: 2                                                                           |
| Direct, X (Write) |                      |                                                                                   |
| Cycle             | R/W                  | Desc.                                                                             |
| -X.2              | Read <b>PC:PB</b>    | Store as OpCode.                                                                  |
| 1.1               |                      | Inc. <b>PC</b> .                                                                  |
| 1.2               | Read <b>PC:PB</b>    | Store as Operand.                                                                 |
| 2.1               |                      | Inc. <b>PC</b> .                                                                  |
| 2.2               |                      |                                                                                   |
| 3.1               |                      | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles. |
| 3.2               |                      |                                                                                   |
| 4.1               |                      |                                                                                   |
| 4.2               | Write to Address     | Write 0. If <b>M</b> flag is set, skip the next cycle.                            |
| 5.1               |                      |                                                                                   |
| 5.2               | Write to Address + 1 | Write 0.                                                                          |
| 6.1               |                      |                                                                                   |
| 6.2               | Read <b>PC:PB</b>    | Store as OpCode.                                                                  |
| +X.1              |                      |                                                                                   |

| ADC (75) Cycles: 4 |                   | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct, X (Read)   |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Cycle              | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| -X.2               | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1.1                | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1.2                |                   | Store as Operand.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2.1                |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2.2                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 3.1                |                   | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 3.2                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4.1                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4.2                | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.1                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.2                | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6.1                |                   | <p>If <b>M</b> flag == 0 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) + ui16(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.</p> <p><b>C</b> flag = Result &gt; \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = ui16(<b>A.L</b> &amp; 0x0F) + ui16(Operand &amp; 0x0F) + <b>C</b>;</p> <p>if (Lo &gt; 9) Lo += 6.</p> <p>CarryToHi = Lo &gt; \$0F ? 1 : 0.</p> <p>HiSum = ui16(<b>A.L</b> &gt;&gt; 4) + ui16(Operand &gt;&gt; 4) + CarryToHi.</p> <p><b>V</b> flag = (~((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &gt; 9) HiSum += 6.</p> <p><b>C</b> flag = HiSum &gt; \$0F.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) + ui32(Operand) + <b>C</b>.</p> <p><b>V</b> flag = (~ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0).</p> <p><b>C</b> flag = Result &gt; \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Carry = ui16(<b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = ui16((<b>A</b> &gt;&gt; i) &amp; \$0F) + ui16((Operand &gt;&gt; i) &amp; \$0F) + Carry.</p> <p>if (i == 12) <b>V</b> flag = (~(<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.</p> <p>if (DigitSum &gt; 9) DigitSum += 6.</p> <p>Carry = (DigitSum &gt; \$0F) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Carry != 0. <b>Z</b> flag = Result != \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 6.2                | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| +X.1               |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

| ROR (76)                      | Cycles: 6            | Size: 2                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct, X (Read/Modify/Write) |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |
| Cycle                         | R/W                  | Desc.                                                                                                                                                                                                                                                                                                                                                                                                 |
| -X.2                          | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                      |
| 1.1                           |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                      |
| 1.2                           | Read <b>PC:PB</b>    | Store as Operand.                                                                                                                                                                                                                                                                                                                                                                                     |
| 2.1                           |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                      |
| 2.2                           |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3.1                           |                      | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                                                                                                                                                                                                     |
| 3.2                           |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |
| 4.1                           |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |
| 4.2                           | Read Address         | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                       |
| 5.1                           |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.2                           | Read Address + 1     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                   |
| 6.1                           |                      | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$01), perform Operand.L = (Operand.L >> 1)   ( <b>C</b> << 7), and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.L & \$01), perform Operand = (Operand >> 1)   ( <b>C</b> << 15), and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |
| 6.2                           |                      | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                         |
| 7.1                           |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7.2                           | Write to Address + 1 | Write Operand.H.                                                                                                                                                                                                                                                                                                                                                                                      |
| 8.1                           |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8.2                           | Write to Address     | Write Operand.L.                                                                                                                                                                                                                                                                                                                                                                                      |
| 9.1                           |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |
| 9.2                           | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                      |
| +X.1                          |                      |                                                                                                                                                                                                                                                                                                                                                                                                       |

| ADC (77) Cycles: 6                        |                                    | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indirect Indexed Long [d],y (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cycle                                     | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| -X.2                                      | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.1                                       |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.2                                       | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2.1                                       |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2.2                                       |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3.1                                       |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3.2                                       | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4.1                                       |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4.2                                       | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.1                                       |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.2                                       | Read Address + 2                   | Store as Bank.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 6.1                                       |                                    | Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{Y})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank += $(\text{Tmp} \gg 16)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6.2                                       | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 7.1                                       |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 7.2                                       | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 8.1                                       |                                    | <p>If <b>M</b> flag == 0 (8-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = <math>\text{ui16}(\mathbf{A.L}) + \text{ui16}(\text{Operand}) + \mathbf{C}</math>.<br/>             <b>V</b> flag = <math>(\sim(\text{ui16}(\mathbf{A.L}) \wedge \text{ui16}(\text{Operand})) \&amp; (\text{ui16}(\mathbf{A.L}) \wedge \text{ui8}(\text{Result})) \&amp; \\$80) \neq 0</math>.<br/>             <b>C</b> flag = Result &gt; \$FF.<br/>             Result = <math>\text{ui8}(\text{Result})</math>. <b>Z</b> flag = Result == \$00. <b>N</b> flag = <math>(\text{Result} \&amp; \\$80) \neq 0</math>.<br/>           If <b>D</b> flag == 1:<br/>             Lo = <math>\text{ui16}(\mathbf{A.L} \&amp; 0x0F) + \text{ui16}(\text{Operand} \&amp; 0x0F) + \mathbf{C}</math>;<br/>             if (Lo &gt; 9) Lo += 6.<br/>             CarryToHi = Lo &gt; \$0F ? 1 : 0.<br/>             HiSum = <math>\text{ui16}(\mathbf{A.L} \gg 4) + \text{ui16}(\text{Operand} \gg 4) + \text{CarryToHi}</math>.<br/>             <b>V</b> flag = <math>(\sim((\mathbf{A.L} \gg 4) \wedge (\text{Operand} \gg 4)) \&amp; ((\mathbf{A.L} \gg 4) \wedge \text{HiSum}) \&amp; \\$08) \neq 0</math>.<br/>             if (HiSum &gt; 9) HiSum += 6.<br/>             <b>C</b> flag = HiSum &gt; \$0F.<br/>             Result = <math>\text{ui8}(((\text{HiSum} \&amp; 0x0F) &lt;&lt; 4) \mid (\text{Lo} \&amp; \\$0F))</math>.<br/>             <b>Z</b> flag = Result == \$00. <b>N</b> flag = <math>(\text{Result} \&amp; \\$80) \neq 0</math>.<br/>           If <b>M</b> flag == 0 (16-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = <math>\text{ui32}(\mathbf{A}) + \text{ui32}(\text{Operand}) + \mathbf{C}</math>.<br/>             <b>V</b> flag = <math>(\sim(\text{ui32}(\mathbf{A}) \wedge \text{ui32}(\text{Operand})) \&amp; (\text{ui32}(\mathbf{A}) \wedge \text{ui16}(\text{Result})) \&amp; \\$8000) \neq 0</math>.<br/>             <b>C</b> flag = Result &gt; \$FFFF.<br/>             Result = <math>\text{ui16}(\text{Result})</math>.<br/>             <b>Z</b> flag = Result == \$0000. <b>N</b> flag = <math>(\text{Result} \&amp; \\$8000) \neq 0</math>.<br/>           If <b>D</b> flag == 1:<br/>             Result = <math>\text{ui16}(0)</math>. Carry = <math>\text{ui16}(\mathbf{C})</math>.<br/>             for i = 0, 4, 8, 12:<br/>               DigitSum = <math>\text{ui16}((\mathbf{A} \gg i) \&amp; \\$0F) + \text{ui16}((\text{Operand} \gg i) \&amp; \\$0F) + \text{Carry}</math>.<br/>               if (i == 12) <b>V</b> flag = <math>(\sim((\mathbf{A} \gg 12) \wedge (\text{Operand} \gg 12)) \&amp; ((\mathbf{A} \gg 12) \wedge \text{DigitSum}) \&amp; \\$08) \neq 0</math>.<br/>               if (DigitSum &gt; 9) DigitSum += 6.<br/>               Carry = <math>(\text{DigitSum} &gt; \\$0F) ? 1 : 0</math>.<br/>               Result  = <math>\text{ui16}((\text{DigitSum} \&amp; \\$0F) &lt;&lt; i)</math>.<br/>             <b>C</b> flag = Carry != 0. <b>Z</b> flag = Result != \$0000. <b>N</b> flag = <math>(\text{Result} \&amp; \\$8000) \neq 0</math>.</p> |
| 8.2                                       | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| +X.1                                      |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



|          |                   |                      |
|----------|-------------------|----------------------|
| SEI (78) | Cycles: 2         | Size: 1              |
| Implied  |                   |                      |
| Cycle    | R/W               | Desc.                |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.     |
| 1.1      |                   | Inc. <b>PC</b> .     |
| 1.2      | Read <b>PC:PB</b> | Discard.             |
| 2.1      |                   | Set the I flag to 1. |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.     |
| +X.1     |                   |                      |

| ADC (79) Cycles: 4 |                                    | Size: 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute, Y (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1.2                | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2.2                | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3.2                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 4.1                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.1                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.2                | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 6.1                |                                    | <p>If <b>M</b> flag == 0 (8-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = ui16(<b>A</b>.L) + ui16(Operand) + <b>C</b>.<br/>             <b>V</b> flag = (~ui16(<b>A</b>.L) ^ ui16(Operand)) &amp; (ui16(<b>A</b>.L) ^ ui8(Result)) &amp; \$80) != 0.<br/>             <b>C</b> flag = Result &gt; \$FF.<br/>             Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:<br/>             Lo = ui16(<b>A</b>.L &amp; 0x0F) + ui16(Operand &amp; 0x0F) + <b>C</b>;<br/>             if (Lo &gt; 9) Lo += 6.<br/>             CarryToHi = Lo &gt; \$0F ? 1 : 0.<br/>             HiSum = ui16(<b>A</b>.L &gt;&gt; 4) + ui16(Operand &gt;&gt; 4) + CarryToHi.<br/>             <b>V</b> flag = (~((<b>A</b>.L &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A</b>.L &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.<br/>             if (HiSum &gt; 9) HiSum += 6.<br/>             <b>C</b> flag = HiSum &gt; \$0F.<br/>             Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).<br/>             <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = ui32(<b>A</b>) + ui32(Operand) + <b>C</b>.<br/>             <b>V</b> flag = (~ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0).<br/>             <b>C</b> flag = Result &gt; \$FFFF.<br/>             Result = ui16(Result).<br/>             <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:<br/>             Result = ui16(0). Carry = ui16(<b>C</b>).<br/>             for i = 0, 4, 8, 12:<br/>               DigitSum = ui16((<b>A</b> &gt;&gt; i) &amp; \$0F) + ui16((Operand &gt;&gt; i) &amp; \$0F) + Carry.<br/>               if (i == 12) <b>V</b> flag = (~(<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.<br/>               if (DigitSum &gt; 9) DigitSum += 6.<br/>               Carry = (DigitSum &gt; \$0F) ? 1 : 0.<br/>               Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).<br/>             <b>C</b> flag = Carry != 0. <b>Z</b> flag = Result != \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 6.2                | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| +X.1               |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|          |                     |                                                                                                                                                                                                                                                                                 |
|----------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PLY (7A) | Cycles: 4           | Size: 1                                                                                                                                                                                                                                                                         |
| Implied  |                     |                                                                                                                                                                                                                                                                                 |
| Cycle    | R/W                 | Desc.                                                                                                                                                                                                                                                                           |
| -X.2     | Read <b>PC:PB</b>   | Store as OpCode.                                                                                                                                                                                                                                                                |
| 1.1      |                     | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                |
| 1.2      |                     |                                                                                                                                                                                                                                                                                 |
| 2.1      |                     |                                                                                                                                                                                                                                                                                 |
| 2.2      |                     |                                                                                                                                                                                                                                                                                 |
| 3.1      |                     | Inc. <b>S</b> by 1.                                                                                                                                                                                                                                                             |
| 3.2      | Read ( <b>S</b> +0) | Store as Operand.L. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                               |
| 4.1      |                     |                                                                                                                                                                                                                                                                                 |
| 4.2      | Read ( <b>S</b> +1) | Store as Operand.H.                                                                                                                                                                                                                                                             |
| 5.1      |                     | Inc. <b>S</b> by 1. If <b>X</b> flag is set, set <b>Y</b> .L to Operand.L, set <b>N</b> based off ( <b>Y</b> .L & \$80), and <b>Z</b> based off <b>Y</b> .L, otherwise set <b>Y</b> to Operand, set <b>N</b> based off ( <b>Y</b> .H & \$80), and <b>Z</b> based off <b>Y</b> . |
| 5.2      | Read <b>PC:PB</b>   | Store as OpCode.                                                                                                                                                                                                                                                                |
| +X.1     |                     |                                                                                                                                                                                                                                                                                 |

|          |                   |                                                                                                           |
|----------|-------------------|-----------------------------------------------------------------------------------------------------------|
| TDC (7B) | Cycles: 2         | Size: 1                                                                                                   |
| Implied  |                   |                                                                                                           |
| Cycle    | R/W               | Desc.                                                                                                     |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                          |
| 1.1      |                   | Inc. <b>PC</b> .                                                                                          |
| 1.2      | Read <b>PC:PB</b> | Discard.                                                                                                  |
| 2.1      |                   | Set <b>A</b> to <b>D</b> . Set <b>N</b> based off ( <b>A</b> .H & \$80) and <b>Z</b> based off <b>A</b> . |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                          |
| +X.1     |                   |                                                                                                           |

|                                       |                       |                                                  |         |
|---------------------------------------|-----------------------|--------------------------------------------------|---------|
| JMP (7C)                              |                       | Cycles: 6                                        | Size: 3 |
| Absolute Indexed Indirect (a,x) (Jump |                       |                                                  |         |
| Cycle                                 | R/W                   | Desc.                                            |         |
| -X.2                                  | Read <b>PC:PB</b>     | Store as OpCode.                                 |         |
| 1.1                                   |                       | Inc. <b>PC</b> .                                 |         |
| 1.2                                   | Read <b>PC:PB</b>     | Store as Pointer.                                |         |
| 2.1                                   |                       | Inc. <b>PC</b> .                                 |         |
| 2.2                                   | Read <b>PC:PB</b>     | Store as Pointer.H.                              |         |
| 3.1                                   |                       | Perform Pointer += <b>X</b> . Bank = <b>PB</b> . |         |
| 3.2                                   |                       |                                                  |         |
| 4.1                                   |                       |                                                  |         |
| 4.2                                   | Read Pointer:Bank     | Store as Address.L.                              |         |
| 5.1                                   |                       |                                                  |         |
| 5.2                                   | Read Pointer + 1:Bank | Store as Address.H.                              |         |
| 6.1                                   |                       | Set <b>PC</b> to Address.                        |         |
| 6.2                                   | Read <b>PC:PB</b>     | Store as OpCode.                                 |         |
| +X.1                                  |                       |                                                  |         |

| ADC (7D) Cycles: 4 |                                    | Size: 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute, X (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1.2                | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2.2                | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>X</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3.2                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 4.1                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.1                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.2                | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 6.1                |                                    | <p>If <b>M</b> flag == 0 (8-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = ui16(<b>A.L</b>) + ui16(Operand) + <b>C</b>.<br/>             <b>V</b> flag = (~ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.<br/>             <b>C</b> flag = Result &gt; \$FF.<br/>             Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:<br/>             Lo = ui16(<b>A.L</b> &amp; 0x0F) + ui16(Operand &amp; 0x0F) + <b>C</b>;<br/>             if (Lo &gt; 9) Lo += 6.<br/>             CarryToHi = Lo &gt; \$0F ? 1 : 0.<br/>             HiSum = ui16(<b>A.L</b> &gt;&gt; 4) + ui16(Operand &gt;&gt; 4) + CarryToHi.<br/>             <b>V</b> flag = (~((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.<br/>             if (HiSum &gt; 9) HiSum += 6.<br/>             <b>C</b> flag = HiSum &gt; \$0F.<br/>             Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).<br/>             <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = ui32(<b>A</b>) + ui32(Operand) + <b>C</b>.<br/>             <b>V</b> flag = (~ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0).<br/>             <b>C</b> flag = Result &gt; \$FFFF.<br/>             Result = ui16(Result).<br/>             <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:<br/>             Result = ui16(0). Carry = ui16(<b>C</b>).<br/>             for i = 0, 4, 8, 12:<br/>               DigitSum = ui16((<b>A</b> &gt;&gt; i) &amp; \$0F) + ui16((Operand &gt;&gt; i) &amp; \$0F) + Carry.<br/>               if (i == 12) <b>V</b> flag = (~(<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.<br/>               if (DigitSum &gt; 9) DigitSum += 6.<br/>               Carry = (DigitSum &gt; \$0F) ? 1 : 0.<br/>               Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).<br/>             <b>C</b> flag = Carry != 0. <b>Z</b> flag = Result != \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 6.2                | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| +X.1               |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| ROR (7E) Cycles: 7             |                                        | Size: 3                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute, X (Read/Modiy/Write) |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Cycle                          | R/W                                    | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| -X.2                           | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1.1                            |                                        | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1.2                            | Read <b>PC:PB</b>                      | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2.1                            |                                        | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2.2                            | Read <b>PC:PB</b>                      | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3.1                            |                                        | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank = $\text{DB} + (\text{Tmp} \gg 16)$ .                                                                                                                                                                                                                                                                         |
| 3.2                            |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 4.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 4.2                            | Read Pointer:Bank                      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5.2                            | Read Pointer + 1:Bank (With Carry)     | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 6.1                            |                                        | If <b>M</b> flag is set, set <b>C</b> based off (Operand.L & \$01), perform $\text{Operand.L} = (\text{Operand.L} \gg 1) \mid (\text{C} \ll 7)$ , and set <b>N</b> based off (Operand.L & \$80) and <b>Z</b> based off Operand.L, otherwise set <b>C</b> based off (Operand.L & \$01), perform $\text{Operand} = (\text{Operand} \gg 1) \mid (\text{C} \ll 15)$ , and set <b>N</b> based off (Operand.H & \$80) and <b>Z</b> based off Operand. |
| 6.2                            |                                        | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 7.2                            | Write to Pointer + 1:Bank (With Carry) | Write Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 8.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 8.2                            | Write to Pointer:Bank                  | Write Operand.L.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 9.1                            |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 9.2                            | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| +X.1                           |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| ADC (7F) Cycles: 5      |                                    | Size: 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute Long, X (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Cycle                   | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| -X.2                    | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1.1                     | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1.2                     |                                    | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2.1                     | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2.2                     |                                    | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3.1                     | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3.2                     |                                    | Stores as Bank.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4.1                     | Read <b>PC:PB</b>                  | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                         |                                    | $\text{Pointer} = \text{ui16}(\text{Tmp})$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         |                                    | $\text{Bank} += (\text{Tmp} \gg 16)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4.2                     | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5.1                     | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.2                     |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 6.1                     |                                    | <p>If <b>M</b> flag == 0 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = <math>\text{ui16}(\text{A.L}) + \text{ui16}(\text{Operand}) + \text{C}</math>.</p> <p><b>V</b> flag = <math>(\sim(\text{ui16}(\text{A.L}) \wedge \text{ui16}(\text{Operand})) \&amp; (\text{ui16}(\text{A.L}) \wedge \text{ui8}(\text{Result})) \&amp; \\$80) \neq 0</math>.</p> <p><b>C</b> flag = <math>\text{Result} &gt; \\$\text{FF}</math>.</p> <p>Result = <math>\text{ui8}(\text{Result})</math>. <b>Z</b> flag = <math>\text{Result} == \\$00</math>. <b>N</b> flag = <math>(\text{Result} \&amp; \\$80) \neq 0</math>.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = <math>\text{ui16}(\text{A.L} \&amp; 0x0F) + \text{ui16}(\text{Operand} \&amp; 0x0F) + \text{C}</math>;</p> <p>if (Lo &gt; 9) Lo += 6.</p> <p>CarryToHi = <math>\text{Lo} &gt; \\$0F ? 1 : 0</math>.</p> <p>HiSum = <math>\text{ui16}(\text{A.L} \gg 4) + \text{ui16}(\text{Operand} \gg 4) + \text{CarryToHi}</math>.</p> <p><b>V</b> flag = <math>(\sim((\text{A.L} \gg 4) \wedge (\text{Operand} \gg 4)) \&amp; ((\text{A.L} \gg 4) \wedge \text{HiSum}) \&amp; \\$08) \neq 0</math>.</p> <p>if (HiSum &gt; 9) HiSum += 6.</p> <p><b>C</b> flag = <math>\text{HiSum} &gt; \\$0F</math>.</p> <p>Result = <math>\text{ui8}(((\text{HiSum} \&amp; 0x0F) &lt;&lt; 4) \mid (\text{Lo} \&amp; \\$0F))</math>.</p> <p><b>Z</b> flag = <math>\text{Result} == \\$00</math>. <b>N</b> flag = <math>(\text{Result} \&amp; \\$80) \neq 0</math>.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = <math>\text{ui32}(\text{A}) + \text{ui32}(\text{Operand}) + \text{C}</math>.</p> <p><b>V</b> flag = <math>(\sim(\text{ui32}(\text{A}) \wedge \text{ui32}(\text{Operand})) \&amp; (\text{ui32}(\text{A}) \wedge \text{ui16}(\text{Result})) \&amp; \\$8000) \neq 0</math>.</p> <p><b>C</b> flag = <math>\text{Result} &gt; \\$\text{FFFF}</math>.</p> <p>Result = <math>\text{ui16}(\text{Result})</math>.</p> <p><b>Z</b> flag = <math>\text{Result} == \\$0000</math>. <b>N</b> flag = <math>(\text{Result} \&amp; \\$8000) \neq 0</math>.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = <math>\text{ui16}(0)</math>. Carry = <math>\text{ui16}(\text{C})</math>.</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = <math>\text{ui16}((\text{A} \gg i) \&amp; \\$0F) + \text{ui16}((\text{Operand} \gg i) \&amp; \\$0F) + \text{Carry}</math>.</p> <p>if (i == 12) <b>V</b> flag = <math>(\sim((\text{A} \gg 12) \wedge (\text{Operand} \gg 12)) \&amp; ((\text{A} \gg 12) \wedge \text{DigitSum}) \&amp; \\$08) \neq 0</math>.</p> <p>if (DigitSum &gt; 9) DigitSum += 6.</p> <p>Carry = <math>(\text{DigitSum} &gt; \\$0F) ? 1 : 0</math>.</p> <p>Result  = <math>\text{ui16}((\text{DigitSum} \&amp; \\$0F) &lt;&lt; i)</math>.</p> <p><b>C</b> flag = <math>\text{Carry} \neq 0</math>. <b>Z</b> flag = <math>\text{Result} \neq \\$0000</math>. <b>N</b> flag = <math>(\text{Result} \&amp; \\$8000) \neq 0</math>.</p> |
| 6.2                     | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| +X.1                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



|          |                   |                                                                                                 |
|----------|-------------------|-------------------------------------------------------------------------------------------------|
| BRA (80) | Cycles: 3         | Size: 2                                                                                         |
| Relative |                   |                                                                                                 |
| Cycle    | R/W               | Desc.                                                                                           |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                |
| 1.1      |                   | Inc. <b>PC</b> . Decide to branch always.                                                       |
| 1.2      | Read <b>PC:PB</b> | Store as Operand. Address = i16(i8(Operand.L)) + <b>PC</b> . If not branching, poll interrupts. |
| 2.1      |                   | Inc. <b>PC</b> . If not branching, end (next half-cycle is 3.2).                                |
| 2.2      |                   | Poll interrupts.                                                                                |
| 3.1      |                   | Set <b>PC</b> to Address.                                                                       |
| 3.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                |
| +X.1     |                   |                                                                                                 |

| STA (81)                              | Cycles: 6                       | Size: 2                                                                               |
|---------------------------------------|---------------------------------|---------------------------------------------------------------------------------------|
| Direct Indexed Indirect (d,x) (Write) |                                 |                                                                                       |
| Cycle                                 | R/W                             | Desc.                                                                                 |
| -X.2                                  | Read <b>PC:PB</b>               | Store as OpCode.                                                                      |
| 1.1                                   |                                 | Inc. <b>PC</b> .                                                                      |
| 1.2                                   | Read <b>PC:PB</b>               | Store as Pointer.                                                                     |
| 2.1                                   |                                 | Inc. <b>PC</b> . Set Pointer to Pointer + <b>X</b> + <b>D</b> .<br>Address = Pointer. |
| 2.2                                   |                                 | If <b>D.L</b> is 0, skip the next cycle.                                              |
| 3.1                                   |                                 | Address.H = Pointer.H (fixes high byte of address).                                   |
| 3.2                                   |                                 |                                                                                       |
| 4.1                                   |                                 |                                                                                       |
| 4.2                                   | Read Address                    | Store as Pointer.                                                                     |
| 5.1                                   |                                 |                                                                                       |
| 5.2                                   | Read Address + 1                | Store as Pointer.H.                                                                   |
| 6.1                                   |                                 | Set Operand to <b>A</b> .                                                             |
| 6.2                                   | Write to Pointer: <b>DB</b>     | Write Operand.L. If <b>M</b> flag is set, skip the next cycle.                        |
| 7.1                                   |                                 |                                                                                       |
| 7.2                                   | Write to Pointer + 1: <b>DB</b> | Write Operand.H.                                                                      |
| 8.1                                   |                                 |                                                                                       |
| 8.2                                   | Read <b>PC:PB</b>               | Store as OpCode.                                                                      |
| +X.1                                  |                                 |                                                                                       |

|                               |                   |                               |
|-------------------------------|-------------------|-------------------------------|
| BRL (82)                      | Cycles: 4         | Size: 3                       |
| Program Counter Relative Long |                   |                               |
| Cycle                         | R/W               | Desc.                         |
| -X.2                          | Read <b>PC:PB</b> | Store as OpCode.              |
| 1.1                           |                   | Inc. <b>PC</b> .              |
| 1.2                           | Read <b>PC:PB</b> | Store as Operand.             |
| 2.1                           |                   | Inc. <b>PC</b> .              |
| 2.2                           | Read <b>PC:PB</b> | Store as Operand.H.           |
| 3.1                           |                   | Inc. <b>PC</b> .              |
| 3.2                           |                   |                               |
| 4.1                           |                   | Perform <b>PC</b> += Operand. |
| 4.2                           | Read <b>PC:PB</b> | Store as OpCode.              |
| +X.1                          |                   |                               |

|                        |                      |                                                                |
|------------------------|----------------------|----------------------------------------------------------------|
| STA (83)               | Cycles: 4            | Size: 2                                                        |
| Stack Relative (Write) |                      |                                                                |
| Cycle                  | R/W                  | Desc.                                                          |
| -X.2                   | Read <b>PC:PB</b>    | Store as OpCode.                                               |
| 1.1                    | Read <b>PC:PB</b>    | Inc. <b>PC</b> .                                               |
| 1.2                    |                      | Store as Pointer.                                              |
| 2.1                    |                      | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .           |
| 2.2                    |                      |                                                                |
| 3.1                    |                      | Set Operand to <b>A</b> .                                      |
| 3.2                    | Write to Address     | Write Operand.L. If <b>M</b> flag is set, skip the next cycle. |
| 4.1                    | Write to Address + 1 |                                                                |
| 4.2                    |                      | Write Operand.H.                                               |
| 5.1                    | Read <b>PC:PB</b>    |                                                                |
| 5.2                    |                      | Store as OpCode.                                               |
| +X.1                   |                      |                                                                |

|                |                      |                                                                                   |
|----------------|----------------------|-----------------------------------------------------------------------------------|
| STY (84)       | Cycles: 3            | Size: 2                                                                           |
| Direct (Write) |                      |                                                                                   |
| Cycle          | R/W                  | Desc.                                                                             |
| -X.2           | Read <b>PC:PB</b>    | Store as OpCode.                                                                  |
| 1.1            |                      | Inc. <b>PC</b> .                                                                  |
| 1.2            | Read <b>PC:PB</b>    | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                        |
| 2.1            |                      | Inc. <b>PC</b> .                                                                  |
| 2.2            |                      |                                                                                   |
| 3.1            |                      | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> . |
| 3.2            | Write to Address     | Write <b>Y.L</b> . If <b>X</b> flag is set, skip the next cycle.                  |
| 4.1            |                      |                                                                                   |
| 4.2            | Write to Address + 1 | Write <b>Y.H</b> .                                                                |
| 5.1            |                      |                                                                                   |
| 5.2            | Read <b>PC:PB</b>    | Store as OpCode.                                                                  |
| +X.1           |                      |                                                                                   |

|                |                      |                                                                                   |
|----------------|----------------------|-----------------------------------------------------------------------------------|
| STA (85)       | Cycles: 3            | Size: 2                                                                           |
| Direct (Write) |                      |                                                                                   |
| Cycle          | R/W                  | Desc.                                                                             |
| -X.2           | Read <b>PC:PB</b>    | Store as OpCode.                                                                  |
| 1.1            |                      | Inc. <b>PC</b> .                                                                  |
| 1.2            | Read <b>PC:PB</b>    | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                        |
| 2.1            |                      | Inc. <b>PC</b> .                                                                  |
| 2.2            |                      |                                                                                   |
| 3.1            |                      | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> . |
| 3.2            | Write to Address     | Write <b>A.L</b> . If <b>M</b> flag is set, skip the next cycle.                  |
| 4.1            |                      |                                                                                   |
| 4.2            | Write to Address + 1 | Write <b>A.H</b> .                                                                |
| 5.1            |                      |                                                                                   |
| 5.2            | Read <b>PC:PB</b>    | Store as OpCode.                                                                  |
| +X.1           |                      |                                                                                   |

|                |                      |                                                                                   |
|----------------|----------------------|-----------------------------------------------------------------------------------|
| STX (86)       | Cycles: 3            | Size: 2                                                                           |
| Direct (Write) |                      |                                                                                   |
| Cycle          | R/W                  | Desc.                                                                             |
| -X.2           | Read <b>PC:PB</b>    | Store as OpCode.                                                                  |
| 1.1            |                      | Inc. <b>PC</b> .                                                                  |
| 1.2            | Read <b>PC:PB</b>    | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                        |
| 2.1            |                      | Inc. <b>PC</b> .                                                                  |
| 2.2            |                      |                                                                                   |
| 3.1            |                      | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> . |
| 3.2            | Write to Address     | Write <b>X.L</b> . If <b>X</b> flag is set, skip the next cycle.                  |
| 4.1            |                      |                                                                                   |
| 4.2            | Write to Address + 1 | Write <b>X.H</b> .                                                                |
| 5.1            |                      |                                                                                   |
| 5.2            | Read <b>PC:PB</b>    | Store as OpCode.                                                                  |
| +X.1           |                      |                                                                                   |

| STA (87)                     | Cycles: 6                              | Size: 2                                                                           |
|------------------------------|----------------------------------------|-----------------------------------------------------------------------------------|
| Direct Indirect Long (Write) |                                        |                                                                                   |
| Cycle                        | R/W                                    | Desc.                                                                             |
| -X.2                         | Read <b>PC:PB</b>                      | Store as OpCode.                                                                  |
| 1.1                          |                                        | Inc. <b>PC</b> .                                                                  |
| 1.2                          | Read <b>PC:PB</b>                      | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                        |
| 2.1                          |                                        | Inc. <b>PC</b> .                                                                  |
| 2.2                          |                                        |                                                                                   |
| 3.1                          |                                        | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> . |
| 3.2                          | Read Address                           | Store as Pointer.                                                                 |
| 4.1                          |                                        |                                                                                   |
| 4.2                          | Read Address + 1                       | Store as Pointer.H.                                                               |
| 5.1                          |                                        |                                                                                   |
| 5.2                          | Read Address + 2                       | Store as Bank.                                                                    |
| 6.1                          |                                        | Set Operand to <b>A</b> .                                                         |
| 6.2                          | Write to Pointer:Bank                  | Write Operand.L. If <b>M</b> flag is set, skip the next cycle.                    |
| 7.1                          |                                        |                                                                                   |
| 7.2                          | Write to Pointer + 1:Bank (With Carry) | Write Operand.H.                                                                  |
| 8.1                          |                                        |                                                                                   |
| 8.2                          | Read <b>PC:PB</b>                      | Store as OpCode.                                                                  |
| +X.1                         |                                        |                                                                                   |



|          |                   |                                                                                                                                                                                                                                                            |
|----------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEY (88) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                    |
| Implied  |                   |                                                                                                                                                                                                                                                            |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                      |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                           |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                           |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                   |
| 2.1      |                   | If <b>X</b> flag is set, perform <b>Y.L</b> -= 1, set <b>N</b> based off ( <b>Y.L</b> & \$80), and set <b>Z</b> based off <b>Y.L</b> , otherwise perform <b>Y</b> -= 1, set <b>N</b> based off ( <b>Y.H</b> & \$80), and set <b>Z</b> based off <b>Y</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                           |
| +X.1     |                   |                                                                                                                                                                                                                                                            |

|           |                   |                                                                                                                                                      |
|-----------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| BIT (89)  | Cycles: 2         | Size: 2                                                                                                                                              |
| Immediate |                   |                                                                                                                                                      |
| Cycle     | R/W               | Desc.                                                                                                                                                |
| -X.2      | Read PC:PB        | Store as OpCode.                                                                                                                                     |
| 1.1       |                   | Inc. <b>PC</b> .                                                                                                                                     |
| 1.2       | Read <b>PC:PB</b> | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                      |
| 2.1       |                   | Inc. <b>PC</b> .                                                                                                                                     |
| 2.2       | Read <b>PC:PB</b> | Store as Operand.H.                                                                                                                                  |
| 3.1       |                   | Inc. <b>PC</b> . If <b>M</b> flag is set, set <b>Z</b> based off ( <b>A</b> .L & Operand.L), otherwise set <b>Z</b> based off ( <b>A</b> & Operand). |
| 3.2       | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                     |
| +X.1      |                   |                                                                                                                                                      |

|          |                   |                                                                                                                                                                                                                                                              |
|----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TXA (8A) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                      |
| Implied  |                   |                                                                                                                                                                                                                                                              |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                        |
| -X.2     | Read PC:PB        | Store as OpCode.                                                                                                                                                                                                                                             |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                             |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                     |
| 2.1      |                   | If <b>M</b> flag is set, perform <b>A.L = X.L</b> , set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise perform <b>A = X</b> , set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                             |
| +X.1     |                   |                                                                                                                                                                                                                                                              |

|          |                         |           |                            |
|----------|-------------------------|-----------|----------------------------|
| PHB (8B) |                         | Cycles: 3 | Size: 1                    |
| Implied  |                         |           |                            |
| Cycle    | R/W                     |           | Desc.                      |
| -X.2     | Read <b>PC:PB</b>       |           | Store as OpCode.           |
| 1.1      |                         |           | Inc. <b>PC</b> .           |
| 1.2      |                         |           |                            |
| 2.1      |                         |           |                            |
| 2.2      | Write to ( <b>S</b> +0) |           | Push <b>DB</b> onto stack. |
| 3.1      |                         |           | Dec. <b>S</b> by 1.        |
| 3.2      | Read <b>PC:PB</b>       |           | Store as OpCode.           |
| +X.1     |                         |           |                            |

|                  |                                              |                                                                |
|------------------|----------------------------------------------|----------------------------------------------------------------|
| STY (8C)         | Cycles: 4                                    | Size: 3                                                        |
| Absolute (Write) |                                              |                                                                |
| Cycle            | R/W                                          | Desc.                                                          |
| -X.2             | Read <b>PC:PB</b>                            | Store as OpCode.                                               |
| 1.1              |                                              | Inc. <b>PC</b> .                                               |
| 1.2              | Read <b>PC:PB</b>                            | Store as Address.                                              |
| 2.1              |                                              | Inc. <b>PC</b> .                                               |
| 2.2              | Read <b>PC:PB</b>                            | Store as Address.H.                                            |
| 3.1              |                                              | Inc <b>PC</b> . Set Operand to <b>Y</b> .                      |
| 3.2              | Write to Address: <b>DB</b>                  | Write Operand.L. If <b>X</b> flag is set, skip the next cycle. |
| 4.1              |                                              |                                                                |
| 4.2              | Write to Address + 1: <b>DB</b> (With Carry) | Write Operand.H.                                               |
| 5.1              |                                              |                                                                |
| 5.2              | Read <b>PC:PB</b>                            | Store as OpCode.                                               |
| +X.1             |                                              |                                                                |

|                  |                                              |                                                                |
|------------------|----------------------------------------------|----------------------------------------------------------------|
| STA (8D)         | Cycles: 4                                    | Size: 3                                                        |
| Absolute (Write) |                                              |                                                                |
| Cycle            | R/W                                          | Desc.                                                          |
| -X.2             | Read <b>PC:PB</b>                            | Store as OpCode.                                               |
| 1.1              |                                              | Inc. <b>PC</b> .                                               |
| 1.2              | Read <b>PC:PB</b>                            | Store as Address.                                              |
| 2.1              |                                              | Inc. <b>PC</b> .                                               |
| 2.2              | Read <b>PC:PB</b>                            | Store as Address.H.                                            |
| 3.1              |                                              | Inc <b>PC</b> . Set Operand to <b>A</b> .                      |
| 3.2              | Write to Address: <b>DB</b>                  | Write Operand.L. If <b>M</b> flag is set, skip the next cycle. |
| 4.1              |                                              |                                                                |
| 4.2              | Write to Address + 1: <b>DB</b> (With Carry) | Write Operand.H.                                               |
| 5.1              |                                              |                                                                |
| 5.2              | Read <b>PC:PB</b>                            | Store as OpCode.                                               |
| +X.1             |                                              |                                                                |

|                  |                                              |                                                                |
|------------------|----------------------------------------------|----------------------------------------------------------------|
| STX (8E)         | Cycles: 4                                    | Size: 3                                                        |
| Absolute (Write) |                                              |                                                                |
| Cycle            | R/W                                          | Desc.                                                          |
| -X.2             | Read <b>PC:PB</b>                            | Store as OpCode.                                               |
| 1.1              |                                              | Inc. <b>PC</b> .                                               |
| 1.2              | Read <b>PC:PB</b>                            | Store as Address.                                              |
| 2.1              |                                              | Inc. <b>PC</b> .                                               |
| 2.2              | Read <b>PC:PB</b>                            | Store as Address.H.                                            |
| 3.1              |                                              | Inc <b>PC</b> . Set Operand to <b>X</b> .                      |
| 3.2              | Write to Address: <b>DB</b>                  | Write Operand.L. If <b>X</b> flag is set, skip the next cycle. |
| 4.1              |                                              |                                                                |
| 4.2              | Write to Address + 1: <b>DB</b> (With Carry) | Write Operand.H.                                               |
| 5.1              |                                              |                                                                |
| 5.2              | Read <b>PC:PB</b>                            | Store as OpCode.                                               |
| +X.1             |                                              |                                                                |

|                       |                                 |                                                                |
|-----------------------|---------------------------------|----------------------------------------------------------------|
| STA (8F)              | Cycles: 5                       | Size: 4                                                        |
| Absolute Long (Write) |                                 |                                                                |
| Cycle                 | R/W                             | Desc.                                                          |
| -X.2                  | Read <b>PC:PB</b>               | Store as OpCode.                                               |
| 1.1                   |                                 | Inc. <b>PC</b> .                                               |
| 1.2                   | Read <b>PC:PB</b>               | Store as Address.                                              |
| 2.1                   |                                 | Inc. <b>PC</b> .                                               |
| 2.2                   | Read <b>PC:PB</b>               | Store as Address.H.                                            |
| 3.1                   |                                 | Inc. <b>PC</b> .                                               |
| 3.2                   | Read <b>PC:PB</b>               | Stores as Bank.                                                |
| 4.1                   |                                 | Inc <b>PC</b> . Set Operand to <b>A</b> .                      |
| 4.2                   | Write to Address:Bank           | Write Operand.L. If <b>M</b> flag is set, skip the next cycle. |
| 5.1                   |                                 |                                                                |
| 5.2                   | Write to Address + 1:Bank (With | Write Operand.H.                                               |
| 6.1                   |                                 |                                                                |
| 6.2                   | Read <b>PC:PB</b>               | Store as OpCode.                                               |
| +X.1                  |                                 |                                                                |



|          |                   |                                                                                                 |         |
|----------|-------------------|-------------------------------------------------------------------------------------------------|---------|
| BCC (90) |                   | Cycles: 2                                                                                       | Size: 2 |
| Relative |                   |                                                                                                 |         |
| Cycle    | R/W               | Desc.                                                                                           |         |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                |         |
| 1.1      |                   | Inc. <b>PC</b> . Decide to branch if <b>C</b> is 0.                                             |         |
| 1.2      | Read <b>PC:PB</b> | Store as Operand. Address = i16(i8(Operand.L)) + <b>PC</b> . If not branching, poll interrupts. |         |
| 2.1      |                   | Inc. <b>PC</b> . If not branching, end (next half-cycle is 3.2).                                |         |
| 2.2      |                   | Poll interrupts.                                                                                |         |
| 3.1      |                   | Set <b>PC</b> to Address.                                                                       |         |
| 3.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                |         |
| +X.1     |                   |                                                                                                 |         |

| STA (91)                              | Cycles: 6                              | Size: 2                                                                                                                                                |
|---------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indirect Indexed (d),y (Write) |                                        |                                                                                                                                                        |
| Cycle                                 | R/W                                    | Desc.                                                                                                                                                  |
| -X.2                                  | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                       |
| 1.1                                   |                                        | Inc. <b>PC</b> .                                                                                                                                       |
| 1.2                                   | Read <b>PC:PB</b>                      | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                             |
| 2.1                                   |                                        | Inc. <b>PC</b> .                                                                                                                                       |
| 2.2                                   |                                        |                                                                                                                                                        |
| 3.1                                   |                                        | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                      |
| 3.2                                   | Read Address                           | Store as Pointer.                                                                                                                                      |
| 4.1                                   |                                        |                                                                                                                                                        |
| 4.2                                   | Read Address + 1                       | Store as Pointer.H.                                                                                                                                    |
| 5.1                                   |                                        | Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{Y})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank = $\text{DB} + (\text{Tmp} \gg 16)$ . |
| 5.2                                   |                                        |                                                                                                                                                        |
| 6.1                                   |                                        | Set Operand to <b>A</b> .                                                                                                                              |
| 6.2                                   | Write to Pointer:Bank                  | Write Operand.L. If <b>M</b> flag is set, skip the next cycle.                                                                                         |
| 7.1                                   |                                        |                                                                                                                                                        |
| 7.2                                   | Write to Pointer + 1:Bank (With Carry) | Write Operand.H.                                                                                                                                       |
| 8.1                                   |                                        |                                                                                                                                                        |
| 8.2                                   | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                       |
| +X.1                                  |                                        |                                                                                                                                                        |

|                         |                                 |                                                                                   |
|-------------------------|---------------------------------|-----------------------------------------------------------------------------------|
| STA (92)                | Cycles: 5                       | Size: 2                                                                           |
| Direct Indirect (Write) |                                 |                                                                                   |
| Cycle                   | R/W                             | Desc.                                                                             |
| -X.2                    | Read <b>PC:PB</b>               | Store as OpCode.                                                                  |
| 1.1                     |                                 | Inc. <b>PC</b> .                                                                  |
| 1.2                     | Read <b>PC:PB</b>               | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                        |
| 2.1                     |                                 | Inc. <b>PC</b> .                                                                  |
| 2.2                     |                                 |                                                                                   |
| 3.1                     |                                 | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> . |
| 3.2                     | Read Address                    | Store as Pointer.                                                                 |
| 4.1                     |                                 |                                                                                   |
| 4.2                     | Read Address + 1                | Store as Pointer.H.                                                               |
| 5.1                     |                                 | Set Operand to <b>A</b> .                                                         |
| 5.2                     | Write to Pointer: <b>DB</b>     | Write Operand.L. If <b>M</b> flag is set, skip the next cycle.                    |
| 6.1                     |                                 |                                                                                   |
| 6.2                     | Write to Pointer + 1: <b>DB</b> | Write Operand.H.                                                                  |
| 7.1                     |                                 |                                                                                   |
| 7.2                     | Read <b>PC:PB</b>               | Store as OpCode.                                                                  |
| +X.1                    |                                 |                                                                                   |

| STA (93)                                | Cycles: 7                              | Size: 2                                                                                                                                                |
|-----------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stack Relative Indirect Indexed (Write) |                                        |                                                                                                                                                        |
| Cycle                                   | R/W                                    | Desc.                                                                                                                                                  |
| -X.2                                    | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                       |
| 1.1                                     |                                        | Inc. <b>PC</b> .                                                                                                                                       |
| 1.2                                     | Read <b>PC:PB</b>                      | Store as Pointer.                                                                                                                                      |
| 2.1                                     |                                        | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                   |
| 2.2                                     |                                        |                                                                                                                                                        |
| 3.1                                     |                                        |                                                                                                                                                        |
| 3.2                                     | Read Address                           | Store as Pointer.                                                                                                                                      |
| 4.1                                     |                                        |                                                                                                                                                        |
| 4.2                                     | Read Address + 1                       | Store as Pointer.H.                                                                                                                                    |
| 5.1                                     |                                        | Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{Y})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank = $\text{DB} + (\text{Tmp} \gg 16)$ . |
| 5.2                                     |                                        |                                                                                                                                                        |
| 6.1                                     |                                        | Set Operand to <b>A</b> .                                                                                                                              |
| 6.2                                     | Write to Pointer:Bank                  | Write Operand.L. If <b>M</b> flag is set, skip the next cycle.                                                                                         |
| 7.1                                     |                                        |                                                                                                                                                        |
| 7.2                                     | Write to Pointer + 1:Bank (With Carry) | Write Operand.H.                                                                                                                                       |
| 8.1                                     |                                        |                                                                                                                                                        |
| 8.2                                     | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                       |
| +X.1                                    |                                        |                                                                                                                                                        |

|                   |                      |                                                                                   |
|-------------------|----------------------|-----------------------------------------------------------------------------------|
| STY (94)          | Cycles: 4            | Size: 2                                                                           |
| Direct, X (Write) |                      |                                                                                   |
| Cycle             | R/W                  | Desc.                                                                             |
| -X.2              | Read <b>PC:PB</b>    | Store as OpCode.                                                                  |
| 1.1               | Read <b>PC:PB</b>    | Inc. <b>PC</b> .                                                                  |
| 1.2               |                      | Store as Operand.                                                                 |
| 2.1               |                      | Inc. <b>PC</b> .                                                                  |
| 2.2               |                      |                                                                                   |
| 3.1               |                      | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles. |
| 3.2               |                      |                                                                                   |
| 4.1               |                      |                                                                                   |
| 4.2               | Write to Address     | Write <b>Y.L</b> . If <b>X</b> flag is set, skip the next cycle.                  |
| 5.1               | Write to Address + 1 |                                                                                   |
| 5.2               |                      | Write <b>Y.H</b> .                                                                |
| 6.1               | Read <b>PC:PB</b>    |                                                                                   |
| 6.2               |                      | Store as OpCode.                                                                  |
| +X.1              |                      |                                                                                   |

|                   |                      |                                                                                   |
|-------------------|----------------------|-----------------------------------------------------------------------------------|
| STA (95)          | Cycles: 4            | Size: 2                                                                           |
| Direct, X (Write) |                      |                                                                                   |
| Cycle             | R/W                  | Desc.                                                                             |
| -X.2              | Read <b>PC:PB</b>    | Store as OpCode.                                                                  |
| 1.1               |                      | Inc. <b>PC</b> .                                                                  |
| 1.2               | Read <b>PC:PB</b>    | Store as Operand.                                                                 |
| 2.1               |                      | Inc. <b>PC</b> .                                                                  |
| 2.2               |                      |                                                                                   |
| 3.1               |                      | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles. |
| 3.2               |                      |                                                                                   |
| 4.1               |                      |                                                                                   |
| 4.2               | Write to Address     | Write <b>A.L</b> . If <b>M</b> flag is set, skip the next cycle.                  |
| 5.1               |                      |                                                                                   |
| 5.2               | Write to Address + 1 | Write <b>A.H</b> .                                                                |
| 6.1               |                      |                                                                                   |
| 6.2               | Read <b>PC:PB</b>    | Store as OpCode.                                                                  |
| +X.1              |                      |                                                                                   |

|                   |                      |                                                                                   |
|-------------------|----------------------|-----------------------------------------------------------------------------------|
| STX (96)          | Cycles: 4            | Size: 2                                                                           |
| Direct, Y (Write) |                      |                                                                                   |
| Cycle             | R/W                  | Desc.                                                                             |
| -X.2              | Read <b>PC:PB</b>    | Store as OpCode.                                                                  |
| 1.1               | Read <b>PC:PB</b>    | Inc. <b>PC</b> .                                                                  |
| 1.2               |                      | Store as Operand.                                                                 |
| 2.1               |                      | Inc. <b>PC</b> .                                                                  |
| 2.2               |                      |                                                                                   |
| 3.1               |                      | Address = Operand + <b>Y</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles. |
| 3.2               |                      |                                                                                   |
| 4.1               |                      |                                                                                   |
| 4.2               | Write to Address     | Write <b>X.L</b> . If <b>X</b> flag is set, skip the next cycle.                  |
| 5.1               | Write to Address + 1 |                                                                                   |
| 5.2               |                      | Write <b>X.H</b> .                                                                |
| 6.1               | Read <b>PC:PB</b>    |                                                                                   |
| 6.2               |                      | Store as OpCode.                                                                  |
| +X.1              |                      |                                                                                   |

| STA (97)                                   | Cycles: 6                              | Size: 2                                                                                                                                       |
|--------------------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indirect Indexed Long [d],y (Write) |                                        |                                                                                                                                               |
| Cycle                                      | R/W                                    | Desc.                                                                                                                                         |
| -X.2                                       | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                              |
| 1.1                                        |                                        | Inc. <b>PC</b> .                                                                                                                              |
| 1.2                                        | Read <b>PC:PB</b>                      | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                    |
| 2.1                                        |                                        | Inc. <b>PC</b> .                                                                                                                              |
| 2.2                                        |                                        |                                                                                                                                               |
| 3.1                                        |                                        | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                             |
| 3.2                                        | Read Address                           | Store as Pointer.                                                                                                                             |
| 4.1                                        |                                        |                                                                                                                                               |
| 4.2                                        | Read Address + 1                       | Store as Pointer.H.                                                                                                                           |
| 5.1                                        |                                        |                                                                                                                                               |
| 5.2                                        | Read Address + 2                       | Store as Bank.                                                                                                                                |
| 6.1                                        |                                        | Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{Y})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank += $(\text{Tmp} \gg 16)$ . |
| 6.2                                        | Write to Pointer:Bank                  | Write <b>A.L</b> . If <b>M</b> flag is set, skip the next cycle.                                                                              |
| 7.1                                        |                                        |                                                                                                                                               |
| 7.2                                        | Write to Pointer + 1:Bank (With Carry) | Write <b>A.H</b> .                                                                                                                            |
| 8.1                                        |                                        |                                                                                                                                               |
| 8.2                                        | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                              |
| +X.1                                       |                                        |                                                                                                                                               |



|          |                   |                                                                                                                                                                                                                                                              |
|----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TYA (98) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                      |
| Implied  |                   |                                                                                                                                                                                                                                                              |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                        |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                             |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                             |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                     |
| 2.1      |                   | If <b>M</b> flag is set, perform <b>A.L = Y.L</b> , set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise perform <b>A = Y</b> , set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                             |
| +X.1     |                   |                                                                                                                                                                                                                                                              |

|                     |                                        |                                                                                                                                                                             |
|---------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STA (99)            | Cycles: 5                              | Size: 3                                                                                                                                                                     |
| Absolute, Y (Write) |                                        |                                                                                                                                                                             |
| Cycle               | R/W                                    | Desc.                                                                                                                                                                       |
| -X.2                | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                            |
| 1.1                 |                                        | Inc. <b>PC</b> .                                                                                                                                                            |
| 1.2                 | Read <b>PC:PB</b>                      | Store as Pointer.                                                                                                                                                           |
| 2.1                 |                                        | Inc. <b>PC</b> .                                                                                                                                                            |
| 2.2                 | Read <b>PC:PB</b>                      | Store as Pointer.H.                                                                                                                                                         |
| 3.1                 |                                        | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{Y})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank = $\mathbf{DB} + (\text{Tmp} \gg 16)$ . |
| 3.2                 |                                        |                                                                                                                                                                             |
| 4.1                 |                                        | Set Operand to <b>A</b> .                                                                                                                                                   |
| 4.2                 | Write to Pointer:Bank                  | Write Operand.L. If <b>M</b> flag is set, skip the next cycle.                                                                                                              |
| 5.1                 |                                        |                                                                                                                                                                             |
| 5.2                 | Write to Pointer + 1:Bank (With Carry) | Write Operand.H.                                                                                                                                                            |
| 6.1                 |                                        |                                                                                                                                                                             |
| 6.2                 | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                            |
| +X.1                |                                        |                                                                                                                                                                             |

|          |                   |                            |
|----------|-------------------|----------------------------|
| TXS (9A) | Cycles: 2         | Size: 1                    |
| Implied  |                   |                            |
| Cycle    | R/W               | Desc.                      |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.           |
| 1.1      |                   | Inc. <b>PC</b> .           |
| 1.2      | Read <b>PC:PB</b> | Discard.                   |
| 2.1      |                   | Set <b>S</b> to <b>X</b> . |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.           |
| +X.1     |                   |                            |

|          |                   |                                                                                                                                                                                                                                                              |
|----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TXY (9B) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                      |
| Implied  |                   |                                                                                                                                                                                                                                                              |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                        |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                             |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                             |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                     |
| 2.1      |                   | If <b>X</b> flag is set, perform <b>Y.L = X.L</b> , set <b>N</b> based off ( <b>Y.L</b> & \$80), and set <b>Z</b> based off <b>Y.L</b> , otherwise perform <b>Y = X</b> , set <b>N</b> based off ( <b>Y.H</b> & \$80), and set <b>Z</b> based off <b>Y</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                             |
| +X.1     |                   |                                                                                                                                                                                                                                                              |

|                  |                                              |                                                                |
|------------------|----------------------------------------------|----------------------------------------------------------------|
| STZ (9C)         | Cycles: 4                                    | Size: 3                                                        |
| Absolute (Write) |                                              |                                                                |
| Cycle            | R/W                                          | Desc.                                                          |
| -X.2             | Read <b>PC:PB</b>                            | Store as OpCode.                                               |
| 1.1              |                                              | Inc. <b>PC</b> .                                               |
| 1.2              | Read <b>PC:PB</b>                            | Store as Address.                                              |
| 2.1              |                                              | Inc. <b>PC</b> .                                               |
| 2.2              | Read <b>PC:PB</b>                            | Store as Address.H.                                            |
| 3.1              |                                              | Inc <b>PC</b> . Set Operand to 0.                              |
| 3.2              | Write to Address: <b>DB</b>                  | Write Operand.L. If <b>M</b> flag is set, skip the next cycle. |
| 4.1              |                                              |                                                                |
| 4.2              | Write to Address + 1: <b>DB</b> (With Carry) | Write Operand.H.                                               |
| 5.1              |                                              |                                                                |
| 5.2              | Read <b>PC:PB</b>                            | Store as OpCode.                                               |
| +X.1             |                                              |                                                                |

|                     |                                        |                                                                                                                                                                             |
|---------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STA (9D)            | Cycles: 5                              | Size: 3                                                                                                                                                                     |
| Absolute, X (Write) |                                        |                                                                                                                                                                             |
| Cycle               | R/W                                    | Desc.                                                                                                                                                                       |
| -X.2                | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                            |
| 1.1                 |                                        | Inc. <b>PC</b> .                                                                                                                                                            |
| 1.2                 | Read <b>PC:PB</b>                      | Store as Pointer.                                                                                                                                                           |
| 2.1                 |                                        | Inc. <b>PC</b> .                                                                                                                                                            |
| 2.2                 | Read <b>PC:PB</b>                      | Store as Pointer.H.                                                                                                                                                         |
| 3.1                 |                                        | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{X})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank = $\mathbf{DB} + (\text{Tmp} \gg 16)$ . |
| 3.2                 |                                        |                                                                                                                                                                             |
| 4.1                 |                                        | Set Operand to <b>A</b> .                                                                                                                                                   |
| 4.2                 | Write to Pointer:Bank                  | Write Operand.L. If <b>M</b> flag is set, skip the next cycle.                                                                                                              |
| 5.1                 |                                        |                                                                                                                                                                             |
| 5.2                 | Write to Pointer + 1:Bank (With Carry) | Write Operand.H.                                                                                                                                                            |
| 6.1                 |                                        |                                                                                                                                                                             |
| 6.2                 | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                            |
| +X.1                |                                        |                                                                                                                                                                             |

|                     |                                        |                                                                                                                                                                             |
|---------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STZ (9E)            | Cycles: 5                              | Size: 3                                                                                                                                                                     |
| Absolute, X (Write) |                                        |                                                                                                                                                                             |
| Cycle               | R/W                                    | Desc.                                                                                                                                                                       |
| -X.2                | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                            |
| 1.1                 |                                        | Inc. <b>PC</b> .                                                                                                                                                            |
| 1.2                 | Read <b>PC:PB</b>                      | Store as Pointer.                                                                                                                                                           |
| 2.1                 |                                        | Inc. <b>PC</b> .                                                                                                                                                            |
| 2.2                 | Read <b>PC:PB</b>                      | Store as Pointer.H.                                                                                                                                                         |
| 3.1                 |                                        | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{X})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank = $\mathbf{DB} + (\text{Tmp} \gg 16)$ . |
| 3.2                 |                                        |                                                                                                                                                                             |
| 4.1                 |                                        | Set Operand to 0.                                                                                                                                                           |
| 4.2                 | Write to Pointer:Bank                  | Write Operand.L. If <b>M</b> flag is set, skip the next cycle.                                                                                                              |
| 5.1                 |                                        |                                                                                                                                                                             |
| 5.2                 | Write to Pointer + 1:Bank (With Carry) | Write Operand.H.                                                                                                                                                            |
| 6.1                 |                                        |                                                                                                                                                                             |
| 6.2                 | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                            |
| +X.1                |                                        |                                                                                                                                                                             |

|                          |                                        |                                                                                                                                                     |
|--------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| STA (9F)                 | Cycles: 5                              | Size: 4                                                                                                                                             |
| Absolute Long, X (Write) |                                        |                                                                                                                                                     |
| Cycle                    | R/W                                    | Desc.                                                                                                                                               |
| -X.2                     | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                    |
| 1.1                      |                                        | Inc. <b>PC</b> .                                                                                                                                    |
| 1.2                      | Read <b>PC:PB</b>                      | Store as Pointer.                                                                                                                                   |
| 2.1                      |                                        | Inc. <b>PC</b> .                                                                                                                                    |
| 2.2                      | Read <b>PC:PB</b>                      | Store as Pointer.H.                                                                                                                                 |
| 3.1                      |                                        | Inc. <b>PC</b> .                                                                                                                                    |
| 3.2                      | Read <b>PC:PB</b>                      | Stores as Bank.                                                                                                                                     |
| 4.1                      |                                        | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{X})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank += (Tmp >> 16). |
| 4.2                      | Write to Pointer:Bank                  | Write <b>A.L</b> . If <b>M</b> flag is set, skip the next cycle.                                                                                    |
| 5.1                      |                                        |                                                                                                                                                     |
| 5.2                      | Write to Pointer + 1:Bank (With Carry) | Write <b>A.H</b> .                                                                                                                                  |
| 6.1                      |                                        |                                                                                                                                                     |
| 6.2                      | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                    |
| +X.1                     |                                        |                                                                                                                                                     |



|           |                   |                                                                                                                                                                                                                                                                                      |
|-----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDY (A0)  | Cycles: 2         | Size: 2                                                                                                                                                                                                                                                                              |
| Immediate |                   |                                                                                                                                                                                                                                                                                      |
| Cycle     | R/W               | Desc.                                                                                                                                                                                                                                                                                |
| -X.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                     |
| 1.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                     |
| 1.2       | Read <b>PC:PB</b> | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                                      |
| 2.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                     |
| 2.2       | Read <b>PC:PB</b> | Store as Operand.H.                                                                                                                                                                                                                                                                  |
| 3.1       |                   | Inc. <b>PC</b> . If <b>X</b> flag is set, set <b>Y</b> .L to Operand.L, set <b>N</b> based off ( <b>Y</b> .L & \$80), and set <b>Z</b> based off <b>Y</b> .L, otherwise set <b>Y</b> to Operand, set <b>N</b> based off ( <b>Y</b> .H & \$80), and set <b>Z</b> based off <b>Y</b> . |
| 3.2       | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                     |
| +X.1      |                   |                                                                                                                                                                                                                                                                                      |

| LDA (A1)                             | Cycles: 6                   | Size: 2                                                                                                                                                                                                                                                          |
|--------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indexed Indirect (d,x) (Read) |                             |                                                                                                                                                                                                                                                                  |
| Cycle                                | R/W                         | Desc.                                                                                                                                                                                                                                                            |
| -X.2                                 | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1                                  |                             | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2                                  | Read <b>PC:PB</b>           | Store as Pointer.                                                                                                                                                                                                                                                |
| 2.1                                  |                             | Inc. <b>PC</b> . Set Pointer to Pointer + <b>X</b> + <b>D</b> .                                                                                                                                                                                                  |
| 2.2                                  |                             | Address = Pointer.                                                                                                                                                                                                                                               |
|                                      |                             | If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                         |
| 3.1                                  |                             | Address.H = Pointer.H (fixes high byte of address).                                                                                                                                                                                                              |
| 3.2                                  |                             |                                                                                                                                                                                                                                                                  |
| 4.1                                  |                             |                                                                                                                                                                                                                                                                  |
| 4.2                                  | Read Address                | Store as Pointer.                                                                                                                                                                                                                                                |
| 5.1                                  |                             |                                                                                                                                                                                                                                                                  |
| 5.2                                  | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                                                              |
| 6.1                                  |                             |                                                                                                                                                                                                                                                                  |
| 6.2                                  | Read Pointer: <b>DB</b>     | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 7.1                                  |                             |                                                                                                                                                                                                                                                                  |
| 7.2                                  | Read Pointer + 1: <b>DB</b> | Store as Operand.H.                                                                                                                                                                                                                                              |
| 8.1                                  |                             | If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 8.2                                  | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1                                 |                             |                                                                                                                                                                                                                                                                  |

|           |                   |                                                                                                                                                                                                                                                                                   |
|-----------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDX (A2)  | Cycles: 2         | Size: 2                                                                                                                                                                                                                                                                           |
| Immediate |                   |                                                                                                                                                                                                                                                                                   |
| Cycle     | R/W               | Desc.                                                                                                                                                                                                                                                                             |
| -X.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                  |
| 1.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                  |
| 1.2       | Read <b>PC:PB</b> | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                                   |
| 2.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                  |
| 2.2       | Read <b>PC:PB</b> | Store as Operand.H.                                                                                                                                                                                                                                                               |
| 3.1       |                   | Inc. <b>PC</b> . If <b>X</b> flag is set, set <b>X.L</b> to Operand.L, set <b>N</b> based off ( <b>X.L</b> & \$80), and set <b>Z</b> based off <b>X.L</b> , otherwise set <b>X</b> to Operand, set <b>N</b> based off ( <b>X.H</b> & \$80), and set <b>Z</b> based off <b>X</b> . |
| 3.2       | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                  |
| +X.1      |                   |                                                                                                                                                                                                                                                                                   |

|                       |                   |                                                                                                                                                                                                                                                                  |
|-----------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDA (A3)              | Cycles: 4         | Size: 2                                                                                                                                                                                                                                                          |
| Stack Relative (Read) |                   |                                                                                                                                                                                                                                                                  |
| Cycle                 | R/W               | Desc.                                                                                                                                                                                                                                                            |
| -X.2                  | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1                   | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2                   |                   | Store as Pointer.                                                                                                                                                                                                                                                |
| 2.1                   |                   | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                                                                                                                             |
| 2.2                   |                   |                                                                                                                                                                                                                                                                  |
| 3.1                   |                   |                                                                                                                                                                                                                                                                  |
| 3.2                   | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 4.1                   | Read Address + 1  |                                                                                                                                                                                                                                                                  |
| 4.2                   |                   | Store as Operand.H.                                                                                                                                                                                                                                              |
| 5.1                   | Read <b>PC:PB</b> | If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 5.2                   |                   | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1                  |                   |                                                                                                                                                                                                                                                                  |

|               |                   |                                                                                                                                                                                                                                                                  |
|---------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDY (A4)      | Cycles: 3         | Size: 2                                                                                                                                                                                                                                                          |
| Direct (Read) |                   |                                                                                                                                                                                                                                                                  |
| Cycle         | R/W               | Desc.                                                                                                                                                                                                                                                            |
| -X.2          | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2           | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                       |
| 2.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2           |                   |                                                                                                                                                                                                                                                                  |
| 3.1           |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                |
| 3.2           | Read Address      | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 4.1           |                   |                                                                                                                                                                                                                                                                  |
| 4.2           | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                              |
| 5.1           |                   | If <b>X</b> flag is set, set <b>Y.L</b> to Operand.L, set <b>N</b> based off ( <b>Y.L</b> & \$80), and set <b>Z</b> based off <b>Y.L</b> , otherwise set <b>Y</b> to Operand, set <b>N</b> based off ( <b>Y.H</b> & \$80), and set <b>Z</b> based off <b>Y</b> . |
| 5.2           | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1          |                   |                                                                                                                                                                                                                                                                  |

|               |                   |                                                                                                                                                                                                                                                                  |
|---------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDA (A5)      | Cycles: 3         | Size: 2                                                                                                                                                                                                                                                          |
| Direct (Read) |                   |                                                                                                                                                                                                                                                                  |
| Cycle         | R/W               | Desc.                                                                                                                                                                                                                                                            |
| -X.2          | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2           | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                       |
| 2.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2           |                   |                                                                                                                                                                                                                                                                  |
| 3.1           |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                |
| 3.2           | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 4.1           |                   |                                                                                                                                                                                                                                                                  |
| 4.2           | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                              |
| 5.1           |                   | If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 5.2           | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1          |                   |                                                                                                                                                                                                                                                                  |

|               |                   |                                                                                                                                                                                                                                                                  |
|---------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDX (A6)      | Cycles: 3         | Size: 2                                                                                                                                                                                                                                                          |
| Direct (Read) |                   |                                                                                                                                                                                                                                                                  |
| Cycle         | R/W               | Desc.                                                                                                                                                                                                                                                            |
| -X.2          | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2           | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                       |
| 2.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2           |                   |                                                                                                                                                                                                                                                                  |
| 3.1           |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                |
| 3.2           | Read Address      | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 4.1           |                   |                                                                                                                                                                                                                                                                  |
| 4.2           | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                              |
| 5.1           |                   | If <b>X</b> flag is set, set <b>X.L</b> to Operand.L, set <b>N</b> based off ( <b>X.L</b> & \$80), and set <b>Z</b> based off <b>X.L</b> , otherwise set <b>X</b> to Operand, set <b>N</b> based off ( <b>X.H</b> & \$80), and set <b>Z</b> based off <b>X</b> . |
| 5.2           | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1          |                   |                                                                                                                                                                                                                                                                  |

|                             |                                    |                                                                                                                                                                                                                                                                  |
|-----------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDA (A7)                    | Cycles: 6                          | Size: 2                                                                                                                                                                                                                                                          |
| Direct Indirect Long (Read) |                                    |                                                                                                                                                                                                                                                                  |
| Cycle                       | R/W                                | Desc.                                                                                                                                                                                                                                                            |
| -X.2                        | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1                         |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2                         | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                       |
| 2.1                         |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2                         |                                    |                                                                                                                                                                                                                                                                  |
| 3.1                         |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                |
| 3.2                         | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                |
| 4.1                         |                                    |                                                                                                                                                                                                                                                                  |
| 4.2                         | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                              |
| 5.1                         |                                    |                                                                                                                                                                                                                                                                  |
| 5.2                         | Read Address + 2                   | Store as Bank.                                                                                                                                                                                                                                                   |
| 6.1                         |                                    |                                                                                                                                                                                                                                                                  |
| 6.2                         | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 7.1                         |                                    |                                                                                                                                                                                                                                                                  |
| 7.2                         | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                              |
| 8.1                         |                                    | If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 8.2                         | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1                        |                                    |                                                                                                                                                                                                                                                                  |



|          |                   |                                                                                                                                                                                                                                                              |
|----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TAY (A8) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                      |
| Implied  |                   |                                                                                                                                                                                                                                                              |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                        |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                             |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                             |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                     |
| 2.1      |                   | If <b>X</b> flag is set, perform <b>Y.L = A.L</b> , set <b>N</b> based off ( <b>Y.L</b> & \$80), and set <b>Z</b> based off <b>Y.L</b> , otherwise perform <b>Y = A</b> , set <b>N</b> based off ( <b>Y.H</b> & \$80), and set <b>Z</b> based off <b>Y</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                             |
| +X.1     |                   |                                                                                                                                                                                                                                                              |

|           |                   |                                                                                                                                                                                                                                                                                   |
|-----------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDA (A9)  | Cycles: 2         | Size: 2                                                                                                                                                                                                                                                                           |
| Immediate |                   |                                                                                                                                                                                                                                                                                   |
| Cycle     | R/W               | Desc.                                                                                                                                                                                                                                                                             |
| -X.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                  |
| 1.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                  |
| 1.2       | Read <b>PC:PB</b> | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                   |
| 2.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                  |
| 2.2       | Read <b>PC:PB</b> | Store as Operand.H.                                                                                                                                                                                                                                                               |
| 3.1       |                   | Inc. <b>PC</b> . If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 3.2       | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                  |
| +X.1      |                   |                                                                                                                                                                                                                                                                                   |

|          |                   |                                                                                                                                                                                                                                                                            |
|----------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TAX (AA) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                                    |
| Implied  |                   |                                                                                                                                                                                                                                                                            |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                                      |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                           |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                           |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                                   |
| 2.1      |                   | If <b>X</b> flag is set, perform <b>X.L</b> = <b>A.L</b> , set <b>N</b> based off ( <b>X.L</b> & \$80), and set <b>Z</b> based off <b>X.L</b> , otherwise perform <b>X</b> = <b>A</b> , set <b>N</b> based off ( <b>X.H</b> & \$80), and set <b>Z</b> based off <b>X</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                                           |
| +X.1     |                   |                                                                                                                                                                                                                                                                            |

|          |                     |                                                                                                           |
|----------|---------------------|-----------------------------------------------------------------------------------------------------------|
| PLB (AB) | Cycles: 4           | Size: 1                                                                                                   |
| Implied  |                     |                                                                                                           |
| Cycle    | R/W                 | Desc.                                                                                                     |
| -X.2     | Read PC:PB          | Store as OpCode.                                                                                          |
| 1.1      |                     | Inc. <b>PC</b> .                                                                                          |
| 1.2      |                     |                                                                                                           |
| 2.1      |                     |                                                                                                           |
| 2.2      |                     |                                                                                                           |
| 3.1      |                     | Inc. <b>S</b> by 1.                                                                                       |
| 3.2      | Read ( <b>S</b> +0) | Store as Bank.                                                                                            |
| 4.1      |                     | Set <b>DB</b> to Bank, set <b>N</b> based off ( <b>DB</b> & \$80), and set <b>Z</b> based off <b>DB</b> . |
| 4.2      | Read <b>PC:PB</b>   | Store as OpCode.                                                                                          |
| +X.1     |                     |                                                                                                           |

|                 |                                          |                                                                                                                                                                                                                                                                     |
|-----------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDY (AC)        | Cycles: 4                                | Size: 3                                                                                                                                                                                                                                                             |
| Absolute (Read) |                                          |                                                                                                                                                                                                                                                                     |
| Cycle           | R/W                                      | Desc.                                                                                                                                                                                                                                                               |
| -X.2            | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                    |
| 1.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                    |
| 1.2             | Read <b>PC:PB</b>                        | Store as Address.                                                                                                                                                                                                                                                   |
| 2.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                    |
| 2.2             | Read <b>PC:PB</b>                        | Store as Address.H.                                                                                                                                                                                                                                                 |
| 3.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                    |
| 3.2             | Read Address: <b>DB</b>                  | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                     |
| 4.1             |                                          |                                                                                                                                                                                                                                                                     |
| 4.2             | Read Address + 1: <b>DB</b> (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                 |
| 5.1             |                                          | If <b>X</b> flag is set, set <b>Y</b> .L to Operand.L, set <b>N</b> based off ( <b>Y</b> .L & \$80), and set <b>Z</b> based off <b>Y</b> .L, otherwise set <b>Y</b> to Operand, set <b>N</b> based off ( <b>Y</b> .H & \$80), and set <b>Z</b> based off <b>Y</b> . |
| 5.2             | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                    |
| +X.1            |                                          |                                                                                                                                                                                                                                                                     |

|                 |                                          |                                                                                                                                                                                                                                                                  |
|-----------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDA (AD)        | Cycles: 4                                | Size: 3                                                                                                                                                                                                                                                          |
| Absolute (Read) |                                          |                                                                                                                                                                                                                                                                  |
| Cycle           | R/W                                      | Desc.                                                                                                                                                                                                                                                            |
| -X.2            | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2             | Read <b>PC:PB</b>                        | Store as Address.                                                                                                                                                                                                                                                |
| 2.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2             | Read <b>PC:PB</b>                        | Store as Address.H.                                                                                                                                                                                                                                              |
| 3.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 3.2             | Read Address: <b>DB</b>                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 4.1             |                                          |                                                                                                                                                                                                                                                                  |
| 4.2             | Read Address + 1: <b>DB</b> (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                              |
| 5.1             |                                          | If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 5.2             | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1            |                                          |                                                                                                                                                                                                                                                                  |

|                 |                                          |                                                                                                                                                                                                                                                                  |
|-----------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDX (AE)        | Cycles: 4                                | Size: 3                                                                                                                                                                                                                                                          |
| Absolute (Read) |                                          |                                                                                                                                                                                                                                                                  |
| Cycle           | R/W                                      | Desc.                                                                                                                                                                                                                                                            |
| -X.2            | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2             | Read <b>PC:PB</b>                        | Store as Address.                                                                                                                                                                                                                                                |
| 2.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2             | Read <b>PC:PB</b>                        | Store as Address.H.                                                                                                                                                                                                                                              |
| 3.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 3.2             | Read Address: <b>DB</b>                  | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 4.1             |                                          |                                                                                                                                                                                                                                                                  |
| 4.2             | Read Address + 1: <b>DB</b> (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                              |
| 5.1             |                                          | If <b>X</b> flag is set, set <b>X.L</b> to Operand.L, set <b>N</b> based off ( <b>X.L</b> & \$80), and set <b>Z</b> based off <b>X.L</b> , otherwise set <b>X</b> to Operand, set <b>N</b> based off ( <b>X.H</b> & \$80), and set <b>Z</b> based off <b>X</b> . |
| 5.2             | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1            |                                          |                                                                                                                                                                                                                                                                  |

|                      |                                    |                                                                                                                                                                                                                                                                  |
|----------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDA (AF)             | Cycles: 5                          | Size: 4                                                                                                                                                                                                                                                          |
| Absolute Long (Read) |                                    |                                                                                                                                                                                                                                                                  |
| Cycle                | R/W                                | Desc.                                                                                                                                                                                                                                                            |
| -X.2                 | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2                  | Read <b>PC:PB</b>                  | Store as Address.                                                                                                                                                                                                                                                |
| 2.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2                  | Read <b>PC:PB</b>                  | Store as Address.H.                                                                                                                                                                                                                                              |
| 3.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 3.2                  | Read <b>PC:PB</b>                  | Stores as Bank.                                                                                                                                                                                                                                                  |
| 4.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 4.2                  | Read Address:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 5.1                  |                                    |                                                                                                                                                                                                                                                                  |
| 5.2                  | Read Address + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                              |
| 6.1                  |                                    | If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 6.2                  | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1                 |                                    |                                                                                                                                                                                                                                                                  |



|          |                   |                                                                                                 |         |
|----------|-------------------|-------------------------------------------------------------------------------------------------|---------|
| BCS (B0) |                   | Cycles: 2                                                                                       | Size: 2 |
| Relative |                   |                                                                                                 |         |
| Cycle    | R/W               | Desc.                                                                                           |         |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                |         |
| 1.1      |                   | Inc. <b>PC</b> . Decide to branch if <b>C</b> is 1.                                             |         |
| 1.2      | Read <b>PC:PB</b> | Store as Operand. Address = i16(i8(Operand.L)) + <b>PC</b> . If not branching, poll interrupts. |         |
| 2.1      |                   | Inc. <b>PC</b> . If not branching, end (next half-cycle is 3.2).                                |         |
| 2.2      |                   | Poll interrupts.                                                                                |         |
| 3.1      |                   | Set <b>PC</b> to Address.                                                                       |         |
| 3.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                |         |
| +X.1     |                   |                                                                                                 |         |

| LDA (B1)                             | Cycles: 5                          | Size: 2                                                                                                                                                                                                                                                          |
|--------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indirect Indexed (d),y (Read) |                                    |                                                                                                                                                                                                                                                                  |
| Cycle                                | R/W                                | Desc.                                                                                                                                                                                                                                                            |
| -X.2                                 | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1                                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2                                  | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                       |
| 2.1                                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2                                  |                                    |                                                                                                                                                                                                                                                                  |
| 3.1                                  |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                |
| 3.2                                  | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                |
| 4.1                                  |                                    |                                                                                                                                                                                                                                                                  |
| 4.2                                  | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                              |
| 5.1                                  |                                    | Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                                               |
| 5.2                                  |                                    |                                                                                                                                                                                                                                                                  |
| 6.1                                  |                                    |                                                                                                                                                                                                                                                                  |
| 6.2                                  | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 7.1                                  |                                    |                                                                                                                                                                                                                                                                  |
| 7.2                                  | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                              |
| 8.1                                  |                                    | If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 8.2                                  | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1                                 |                                    |                                                                                                                                                                                                                                                                  |

| LDA (B2)               | Cycles: 5                   | Size: 2                                                                                                                                                                                                                                                          |
|------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indirect (Read) |                             |                                                                                                                                                                                                                                                                  |
| Cycle                  | R/W                         | Desc.                                                                                                                                                                                                                                                            |
| -X.2                   | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1                    |                             | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2                    | Read <b>PC:PB</b>           | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                       |
| 2.1                    |                             | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2                    |                             |                                                                                                                                                                                                                                                                  |
| 3.1                    |                             | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                |
| 3.2                    | Read Address                | Store as Pointer.                                                                                                                                                                                                                                                |
| 4.1                    |                             |                                                                                                                                                                                                                                                                  |
| 4.2                    | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                                                              |
| 5.1                    |                             |                                                                                                                                                                                                                                                                  |
| 5.2                    | Read Pointer: <b>DB</b>     | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 6.1                    |                             |                                                                                                                                                                                                                                                                  |
| 6.2                    | Read Pointer + 1: <b>DB</b> | Store as Operand.H.                                                                                                                                                                                                                                              |
| 7.1                    |                             | If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 7.2                    | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1                   |                             |                                                                                                                                                                                                                                                                  |

| LDA (B3)                               | Cycles: 7                          | Size: 2                                                                                                                                                                                                                                                          |
|----------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stack Relative Indirect Indexed (Read) |                                    |                                                                                                                                                                                                                                                                  |
| Cycle                                  | R/W                                | Desc.                                                                                                                                                                                                                                                            |
| -X.2                                   | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1                                    |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2                                    | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                |
| 2.1                                    |                                    | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                                                                                                                             |
| 2.2                                    |                                    |                                                                                                                                                                                                                                                                  |
| 3.1                                    |                                    |                                                                                                                                                                                                                                                                  |
| 3.2                                    | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                |
| 4.1                                    |                                    |                                                                                                                                                                                                                                                                  |
| 4.2                                    | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                              |
| 5.1                                    |                                    | Perform Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                              |
| 5.2                                    |                                    |                                                                                                                                                                                                                                                                  |
| 6.1                                    |                                    |                                                                                                                                                                                                                                                                  |
| 6.2                                    | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 7.1                                    |                                    |                                                                                                                                                                                                                                                                  |
| 7.2                                    | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                              |
| 8.1                                    |                                    | If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 8.2                                    | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1                                   |                                    |                                                                                                                                                                                                                                                                  |

|                  |                   |                                                                                                                                                                                                                                                                  |
|------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDY (B4)         | Cycles: 4         | Size: 2                                                                                                                                                                                                                                                          |
| Direct, X (Read) |                   |                                                                                                                                                                                                                                                                  |
| Cycle            | R/W               | Desc.                                                                                                                                                                                                                                                            |
| -X.2             | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1              | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2              |                   | Store as Operand.                                                                                                                                                                                                                                                |
| 2.1              |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2              |                   |                                                                                                                                                                                                                                                                  |
| 3.1              |                   | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                                                                |
| 3.2              |                   |                                                                                                                                                                                                                                                                  |
| 4.1              |                   |                                                                                                                                                                                                                                                                  |
| 4.2              | Read Address      | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 5.1              | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                              |
| 5.2              |                   |                                                                                                                                                                                                                                                                  |
| 6.1              | Read <b>PC:PB</b> | If <b>X</b> flag is set, set <b>Y.L</b> to Operand.L, set <b>N</b> based off ( <b>Y.L</b> & \$80), and set <b>Z</b> based off <b>Y.L</b> , otherwise set <b>Y</b> to Operand, set <b>N</b> based off ( <b>Y.H</b> & \$80), and set <b>Z</b> based off <b>Y</b> . |
| 6.2              |                   | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1             |                   |                                                                                                                                                                                                                                                                  |

|                  |                   |                                                                                                                                                                                                                                                                  |
|------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDA (B5)         | Cycles: 4         | Size: 2                                                                                                                                                                                                                                                          |
| Direct, X (Read) |                   |                                                                                                                                                                                                                                                                  |
| Cycle            | R/W               | Desc.                                                                                                                                                                                                                                                            |
| -X.2             | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1              |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2              | Read <b>PC:PB</b> | Store as Operand.                                                                                                                                                                                                                                                |
| 2.1              |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2              |                   |                                                                                                                                                                                                                                                                  |
| 3.1              |                   | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                                                                |
| 3.2              |                   |                                                                                                                                                                                                                                                                  |
| 4.1              |                   |                                                                                                                                                                                                                                                                  |
| 4.2              | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 5.1              |                   |                                                                                                                                                                                                                                                                  |
| 5.2              | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                              |
| 6.1              |                   | If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 6.2              | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1             |                   |                                                                                                                                                                                                                                                                  |

|                  |                   |                                                                                                                                                                                                                                                                  |
|------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDX (B6)         | Cycles: 4         | Size: 2                                                                                                                                                                                                                                                          |
| Direct, Y (Read) |                   |                                                                                                                                                                                                                                                                  |
| Cycle            | R/W               | Desc.                                                                                                                                                                                                                                                            |
| -X.2             | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1              | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2              |                   | Store as Operand.                                                                                                                                                                                                                                                |
| 2.1              |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2              |                   |                                                                                                                                                                                                                                                                  |
| 3.1              |                   | Address = Operand + <b>Y</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                                                                |
| 3.2              |                   |                                                                                                                                                                                                                                                                  |
| 4.1              |                   |                                                                                                                                                                                                                                                                  |
| 4.2              | Read Address      | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 5.1              | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                              |
| 5.2              |                   |                                                                                                                                                                                                                                                                  |
| 6.1              | Read <b>PC:PB</b> | If <b>X</b> flag is set, set <b>X.L</b> to Operand.L, set <b>N</b> based off ( <b>X.L</b> & \$80), and set <b>Z</b> based off <b>X.L</b> , otherwise set <b>X</b> to Operand, set <b>N</b> based off ( <b>X.H</b> & \$80), and set <b>Z</b> based off <b>X</b> . |
| 6.2              |                   | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1             |                   |                                                                                                                                                                                                                                                                  |

| LDA (B7)                                  | Cycles: 6                          | Size: 2                                                                                                                                                                                                                                                          |
|-------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indirect Indexed Long [d],y (Read) |                                    |                                                                                                                                                                                                                                                                  |
| Cycle                                     | R/W                                | Desc.                                                                                                                                                                                                                                                            |
| -X.2                                      | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1                                       |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2                                       | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                       |
| 2.1                                       |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2                                       |                                    |                                                                                                                                                                                                                                                                  |
| 3.1                                       |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                |
| 3.2                                       | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                |
| 4.1                                       |                                    |                                                                                                                                                                                                                                                                  |
| 4.2                                       | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                              |
| 5.1                                       |                                    |                                                                                                                                                                                                                                                                  |
| 5.2                                       | Read Address + 2                   | Store as Bank.                                                                                                                                                                                                                                                   |
| 6.1                                       |                                    | Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{Y})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank += (Tmp >> 16).                                                                                                                               |
| 6.2                                       | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 7.1                                       |                                    |                                                                                                                                                                                                                                                                  |
| 7.2                                       | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                              |
| 8.1                                       |                                    | If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 8.2                                       | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1                                      |                                    |                                                                                                                                                                                                                                                                  |



|          |                   |                             |
|----------|-------------------|-----------------------------|
| CLV (B8) | Cycles: 2         | Size: 1                     |
| Implied  |                   |                             |
| Cycle    | R/W               | Desc.                       |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.            |
| 1.1      |                   | Inc. <b>PC</b> .            |
| 1.2      | Read <b>PC:PB</b> | Discard.                    |
| 2.1      |                   | Set the <b>V</b> flag to 0. |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.            |
| +X.1     |                   |                             |

|                    |                                    |                                                                                                                                                                                                                                                                  |
|--------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDA (B9)           | Cycles: 4                          | Size: 3                                                                                                                                                                                                                                                          |
| Absolute, Y (Read) |                                    |                                                                                                                                                                                                                                                                  |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                                                            |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2                | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2                | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                                                              |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                              |
| 3.2                |                                    |                                                                                                                                                                                                                                                                  |
| 4.1                |                                    |                                                                                                                                                                                                                                                                  |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 5.1                |                                    |                                                                                                                                                                                                                                                                  |
| 5.2                | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                              |
| 6.1                |                                    | If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 6.2                | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1               |                                    |                                                                                                                                                                                                                                                                  |

|          |                   |                                                                                                                                                                                                                                                              |
|----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TSX (BA) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                      |
| Implied  |                   |                                                                                                                                                                                                                                                              |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                        |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                             |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                             |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                     |
| 2.1      |                   | If <b>X</b> flag is set, perform <b>X.L = S.L</b> , set <b>N</b> based off ( <b>X.L</b> & \$80), and set <b>Z</b> based off <b>X.L</b> , otherwise perform <b>X = S</b> , set <b>N</b> based off ( <b>X.H</b> & \$80), and set <b>Z</b> based off <b>X</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                             |
| +X.1     |                   |                                                                                                                                                                                                                                                              |

|          |                   |                                                                                                                                                                                                                                                              |
|----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TYX (BB) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                      |
| Implied  |                   |                                                                                                                                                                                                                                                              |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                        |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                             |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                             |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                     |
| 2.1      |                   | If <b>X</b> flag is set, perform <b>X.L = Y.L</b> , set <b>N</b> based off ( <b>X.L</b> & \$80), and set <b>Z</b> based off <b>X.L</b> , otherwise perform <b>X = Y</b> , set <b>N</b> based off ( <b>X.H</b> & \$80), and set <b>Z</b> based off <b>X</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                             |
| +X.1     |                   |                                                                                                                                                                                                                                                              |

| LDY (BC)           | Cycles: 4                          | Size: 3                                                                                                                                                                                                                                                             |
|--------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute, X (Read) |                                    |                                                                                                                                                                                                                                                                     |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                                                               |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                    |
| 1.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                    |
| 1.2                | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                   |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                    |
| 2.2                | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                                                                 |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>X</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                                 |
| 3.2                |                                    |                                                                                                                                                                                                                                                                     |
| 4.1                |                                    |                                                                                                                                                                                                                                                                     |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                     |
| 5.1                |                                    |                                                                                                                                                                                                                                                                     |
| 5.2                | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                 |
| 6.1                |                                    | If <b>X</b> flag is set, set <b>Y</b> .L to Operand.L, set <b>N</b> based off ( <b>Y</b> .L & \$80), and set <b>Z</b> based off <b>Y</b> .L, otherwise set <b>Y</b> to Operand, set <b>N</b> based off ( <b>Y</b> .H & \$80), and set <b>Z</b> based off <b>Y</b> . |
| 6.2                | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                    |
| +X.1               |                                    |                                                                                                                                                                                                                                                                     |

|                    |                                    |                                                                                                                                                                                                                                                                  |
|--------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDA (BD)           | Cycles: 4                          | Size: 3                                                                                                                                                                                                                                                          |
| Absolute, X (Read) |                                    |                                                                                                                                                                                                                                                                  |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                                                            |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2                | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2                | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                                                              |
| 3.1                |                                    | Inc. PC. Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>X</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                                      |
| 3.2                |                                    |                                                                                                                                                                                                                                                                  |
| 4.1                |                                    |                                                                                                                                                                                                                                                                  |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 5.1                |                                    |                                                                                                                                                                                                                                                                  |
| 5.2                | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                              |
| 6.1                |                                    | If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 6.2                | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1               |                                    |                                                                                                                                                                                                                                                                  |

|                    |                                    |                                                                                                                                                                                                                                                                  |
|--------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LDX (BE)           | Cycles: 4                          | Size: 3                                                                                                                                                                                                                                                          |
| Absolute, Y (Read) |                                    |                                                                                                                                                                                                                                                                  |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                                                            |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2                | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2                | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                                                              |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                              |
| 3.2                |                                    |                                                                                                                                                                                                                                                                  |
| 4.1                |                                    |                                                                                                                                                                                                                                                                  |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 5.1                |                                    |                                                                                                                                                                                                                                                                  |
| 5.2                | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                              |
| 6.1                |                                    | If <b>X</b> flag is set, set <b>X.L</b> to Operand.L, set <b>N</b> based off ( <b>X.L</b> & \$80), and set <b>Z</b> based off <b>X.L</b> , otherwise set <b>X</b> to Operand, set <b>N</b> based off ( <b>X.H</b> & \$80), and set <b>Z</b> based off <b>X</b> . |
| 6.2                | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1               |                                    |                                                                                                                                                                                                                                                                  |

| LDA (BF)                | Cycles: 5                          | Size: 4                                                                                                                                                                                                                                                          |
|-------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute Long, X (Read) |                                    |                                                                                                                                                                                                                                                                  |
| Cycle                   | R/W                                | Desc.                                                                                                                                                                                                                                                            |
| -X.2                    | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| 1.1                     |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 1.2                     | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                |
| 2.1                     |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 2.2                     | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                                                              |
| 3.1                     |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                 |
| 3.2                     | Read <b>PC:PB</b>                  | Stores as Bank.                                                                                                                                                                                                                                                  |
| 4.1                     |                                    | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank += (Tmp >> 16).                                                                                                                |
| 4.2                     | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                  |
| 5.1                     |                                    |                                                                                                                                                                                                                                                                  |
| 5.2                     | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                              |
| 6.1                     |                                    | If <b>M</b> flag is set, set <b>A.L</b> to Operand.L, set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> , otherwise set <b>A</b> to Operand, set <b>N</b> based off ( <b>A.H</b> & \$80), and set <b>Z</b> based off <b>A</b> . |
| 6.2                     | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                 |
| +X.1                    |                                    |                                                                                                                                                                                                                                                                  |



|           |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CPY (C0)  | Cycles: 2         | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Immediate |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Cycle     | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| -X.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1.2       | Read <b>PC:PB</b> | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                |
| 2.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2.2       | Read <b>PC:PB</b> | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3.1       |                   | Inc. <b>PC</b> . If <b>X</b> flag is set, set <b>C</b> flag based off ( <b>Y</b> .L >= Operand.L), set <b>Z</b> flag based off ( <b>Y</b> .L == Operand.L), and set <b>N</b> flag based off (( <b>Y</b> .L - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>Y</b> >= Operand), set <b>Z</b> flag based off ( <b>Y</b> == Operand), and set <b>N</b> flag based off (( <b>Y</b> - Operand) & \$8000) != 0. |
| 3.2       | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                               |
| +X.1      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                |

| CMP (C1) Cycles: 6                   |                             | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indexed Indirect (d,x) (Read) |                             |                                                                                                                                                                                                                                                                                                                                                                                                            |
| Cycle                                | R/W                         | Desc.                                                                                                                                                                                                                                                                                                                                                                                                      |
| -X.2                                 | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |
| 1.1                                  |                             | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |
| 1.2                                  | Read <b>PC:PB</b>           | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                          |
| 2.1                                  |                             | Inc. <b>PC</b> . Set Pointer to Pointer + <b>X</b> + <b>D</b> .                                                                                                                                                                                                                                                                                                                                            |
| 2.2                                  |                             | Address = Pointer.                                                                                                                                                                                                                                                                                                                                                                                         |
|                                      |                             | If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                   |
| 3.1                                  |                             | Address.H = Pointer.H (fixes high byte of address).                                                                                                                                                                                                                                                                                                                                                        |
| 3.2                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4.1                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4.2                                  | Read Address                | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                          |
| 5.1                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.2                                  | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                        |
| 6.1                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 6.2                                  | Read Pointer: <b>DB</b>     | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                            |
| 7.1                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7.2                                  | Read Pointer + 1: <b>DB</b> | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                        |
| 8.1                                  |                             | If <b>M</b> flag is set, set <b>C</b> flag based off ( <b>A.L</b> >= Operand.L), set <b>Z</b> flag based off ( <b>A.L</b> == Operand.L), and set <b>N</b> flag based off (( <b>A.L</b> - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>A</b> >= Operand), set <b>Z</b> flag based off ( <b>A</b> == Operand), and set <b>N</b> flag based off (( <b>A</b> - Operand) & \$8000) != 0. |
| 8.2                                  | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |
| +X.1                                 |                             |                                                                                                                                                                                                                                                                                                                                                                                                            |

|           |                   |                                                                                       |
|-----------|-------------------|---------------------------------------------------------------------------------------|
| REP (C2)  | Cycles: 3         | Size: 2                                                                               |
| Immediate |                   |                                                                                       |
| Cycle     | R/W               | Desc.                                                                                 |
| -X.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                      |
| 1.1       | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                      |
| 1.2       |                   | Store as Operand.                                                                     |
| 2.1       |                   | Inc. <b>PC</b> .                                                                      |
| 2.2       |                   |                                                                                       |
| 3.1       |                   | <b>P</b> &= ~Operand.L, set <b>X</b> .H and <b>Y</b> .H to 0 if <b>X</b> flag is set. |
| 3.2       |                   | Store as OpCode.                                                                      |
| +X.1      |                   |                                                                                       |

|                       |                   |                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CMP (C3)              | Cycles: 4         | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                       |
| Stack Relative (Read) |                   |                                                                                                                                                                                                                                                                                                                                                                                                               |
| Cycle                 | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                         |
| -X.2                  | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.1                   |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.2                   | Read <b>PC:PB</b> | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                             |
| 2.1                   |                   | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                                                                                                                                                                                                                                                                          |
| 2.2                   |                   |                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3.1                   |                   |                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3.2                   | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                               |
| 4.1                   |                   |                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4.2                   | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.1                   |                   | If <b>M</b> flag is set, set <b>C</b> flag based off ( <b>A</b> .L >= Operand.L), set <b>Z</b> flag based off ( <b>A</b> .L == Operand.L), and set <b>N</b> flag based off (( <b>A</b> .L - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>A</b> >= Operand), set <b>Z</b> flag based off ( <b>A</b> == Operand), and set <b>N</b> flag based off (( <b>A</b> - Operand) & \$8000) != 0. |
| 5.2                   | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |
| +X.1                  |                   |                                                                                                                                                                                                                                                                                                                                                                                                               |

|               |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CPY (C4)      | Cycles: 3         | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                    |
| Direct (Read) |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |
| Cycle         | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                      |
| -X.2          | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |
| 1.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |
| 1.2           | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                 |
| 2.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |
| 2.2           |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3.1           |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                          |
| 3.2           | Read Address      | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                            |
| 4.1           |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4.2           | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.1           |                   | If <b>X</b> flag is set, set <b>C</b> flag based off ( <b>Y.L</b> >= Operand.L), set <b>Z</b> flag based off ( <b>Y.L</b> == Operand.L), and set <b>N</b> flag based off (( <b>Y.L</b> - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>Y</b> >= Operand), set <b>Z</b> flag based off ( <b>Y</b> == Operand), and set <b>N</b> flag based off (( <b>Y</b> - Operand) & \$8000) != 0. |
| 5.2           | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |
| +X.1          |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |

|               |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CMP (C5)      | Cycles: 3         | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                    |
| Direct (Read) |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |
| Cycle         | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                      |
| -X.2          | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |
| 1.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |
| 1.2           | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                 |
| 2.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |
| 2.2           |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3.1           |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                          |
| 3.2           | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                            |
| 4.1           |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4.2           | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.1           |                   | If <b>M</b> flag is set, set <b>C</b> flag based off ( <b>A.L</b> >= Operand.L), set <b>Z</b> flag based off ( <b>A.L</b> == Operand.L), and set <b>N</b> flag based off (( <b>A.L</b> - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>A</b> >= Operand), set <b>Z</b> flag based off ( <b>A</b> == Operand), and set <b>N</b> flag based off (( <b>A</b> - Operand) & \$8000) != 0. |
| 5.2           | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |
| +X.1          |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |

| DEC (C6)                   | Cycles: 5            | Size: 2                                                                                                                                                                                                                                          |
|----------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct (Read/Modify/Write) |                      |                                                                                                                                                                                                                                                  |
| Cycle                      | R/W                  | Desc.                                                                                                                                                                                                                                            |
| -X.2                       | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                 |
| 1.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                 |
| 1.2                        | Read <b>PC:PB</b>    | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                       |
| 2.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                 |
| 2.2                        |                      |                                                                                                                                                                                                                                                  |
| 3.1                        |                      | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                |
| 3.2                        | Read Address         | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                  |
| 4.1                        |                      |                                                                                                                                                                                                                                                  |
| 4.2                        | Read Address + 1     | Store as Operand.H.                                                                                                                                                                                                                              |
| 5.1                        |                      | If <b>M</b> flag is set, perform Operand.L -= 1, set <b>N</b> based off (Operand.L & \$80), and set <b>Z</b> based off Operand.L, otherwise perform Operand -= 1, set <b>N</b> based off (Operand.H & \$80), and set <b>Z</b> based off Operand. |
| 5.2                        |                      | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                    |
| 6.1                        |                      |                                                                                                                                                                                                                                                  |
| 6.2                        | Write to Address + 1 | Write Operand.H.                                                                                                                                                                                                                                 |
| 7.1                        |                      |                                                                                                                                                                                                                                                  |
| 7.2                        | Write to Address     | Write Operand.L.                                                                                                                                                                                                                                 |
| 8.1                        |                      |                                                                                                                                                                                                                                                  |
| 8.2                        | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                 |
| +X.1                       |                      |                                                                                                                                                                                                                                                  |

|                             |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
|-----------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| CMP (C7)                    |                                    | Cycles: 6                                                                                                                                                                                                                                                                                                                                                                                                  | Size: 2 |
| Direct Indirect Long (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| Cycle                       | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| -X.2                        | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 1.1                         |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 1.2                         | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                 |         |
| 2.1                         |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 2.2                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 3.1                         |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                          |         |
| 3.2                         | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                          |         |
| 4.1                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 4.2                         | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 5.1                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 5.2                         | Read Address + 2                   | Store as Bank.                                                                                                                                                                                                                                                                                                                                                                                             |         |
| 6.1                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 6.2                         | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                            |         |
| 7.1                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 7.2                         | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 8.1                         |                                    | If <b>M</b> flag is set, set <b>C</b> flag based off ( <b>A.L</b> >= Operand.L), set <b>Z</b> flag based off ( <b>A.L</b> == Operand.L), and set <b>N</b> flag based off (( <b>A.L</b> - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>A</b> >= Operand), set <b>Z</b> flag based off ( <b>A</b> == Operand), and set <b>N</b> flag based off (( <b>A</b> - Operand) & \$8000) != 0. |         |
| 8.2                         | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |         |
| +X.1                        |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |



|          |                   |                                                                                                                                                                                                                                                            |
|----------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INY (C8) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                    |
| Implied  |                   |                                                                                                                                                                                                                                                            |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                      |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                           |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                           |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                   |
| 2.1      |                   | If <b>X</b> flag is set, perform <b>Y.L</b> += 1, set <b>N</b> based off ( <b>Y.L</b> & \$80), and set <b>Z</b> based off <b>Y.L</b> , otherwise perform <b>Y</b> += 1, set <b>N</b> based off ( <b>Y.H</b> & \$80), and set <b>Z</b> based off <b>Y</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                           |
| +X.1     |                   |                                                                                                                                                                                                                                                            |

|           |                   |                                                                                                                                                                                                                                                                                                                                                                                                                             |         |
|-----------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| CMP (C9)  |                   | Cycles: 2                                                                                                                                                                                                                                                                                                                                                                                                                   | Size: 2 |
| Immediate |                   |                                                                                                                                                                                                                                                                                                                                                                                                                             |         |
| Cycle     | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                       |         |
| -X.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 1.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 1.2       | Read <b>PC:PB</b> | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                             |         |
| 2.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 2.2       | Read <b>PC:PB</b> | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                         |         |
| 3.1       |                   | Inc. <b>PC</b> . If <b>M</b> flag is set, set <b>C</b> flag based off ( <b>A.L</b> >= Operand.L), set <b>Z</b> flag based off ( <b>A.L</b> == Operand.L), and set <b>N</b> flag based off (( <b>A.L</b> - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>A</b> >= Operand), set <b>Z</b> flag based off ( <b>A</b> == Operand), and set <b>N</b> flag based off (( <b>A</b> - Operand) & \$8000) != 0. |         |
| 3.2       | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| +X.1      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                             |         |

|          |                   |                                                                                                                                                                                                                                                            |
|----------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEX (CA) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                    |
| Implied  |                   |                                                                                                                                                                                                                                                            |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                      |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                           |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                           |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                   |
| 2.1      |                   | If <b>X</b> flag is set, perform <b>X.L</b> -= 1, set <b>N</b> based off ( <b>X.L</b> & \$80), and set <b>Z</b> based off <b>X.L</b> , otherwise perform <b>X</b> -= 1, set <b>N</b> based off ( <b>X.H</b> & \$80), and set <b>Z</b> based off <b>X</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                           |
| +X.1     |                   |                                                                                                                                                                                                                                                            |

|          |                   |                                                 |
|----------|-------------------|-------------------------------------------------|
| WAI (CB) | Cycles: 3         | Size: 1                                         |
| Implied  |                   |                                                 |
| Cycle    | R/W               | Desc.                                           |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                |
| 1.2      |                   | Discard.                                        |
| 2.1      |                   |                                                 |
| 2.2      |                   |                                                 |
| 3.1      |                   |                                                 |
| 3.2      |                   | Enter waiting state. Repeat this cycle forever. |

|                 |                                          |                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CPY (CC)        | Cycles: 4                                | Size: 3                                                                                                                                                                                                                                                                                                                                                                                                       |
| Absolute (Read) |                                          |                                                                                                                                                                                                                                                                                                                                                                                                               |
| Cycle           | R/W                                      | Desc.                                                                                                                                                                                                                                                                                                                                                                                                         |
| -X.2            | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.2             | Read <b>PC:PB</b>                        | Store as Address.                                                                                                                                                                                                                                                                                                                                                                                             |
| 2.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 2.2             | Read <b>PC:PB</b>                        | Store as Address.H.                                                                                                                                                                                                                                                                                                                                                                                           |
| 3.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 3.2             | Read Address: <b>DB</b>                  | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                               |
| 4.1             |                                          |                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4.2             | Read Address + 1: <b>DB</b> (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.1             |                                          | If <b>X</b> flag is set, set <b>C</b> flag based off ( <b>Y</b> .L >= Operand.L), set <b>Z</b> flag based off ( <b>Y</b> .L == Operand.L), and set <b>N</b> flag based off (( <b>Y</b> .L - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>Y</b> >= Operand), set <b>Z</b> flag based off ( <b>Y</b> == Operand), and set <b>N</b> flag based off (( <b>Y</b> - Operand) & \$8000) != 0. |
| 5.2             | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |
| +X.1            |                                          |                                                                                                                                                                                                                                                                                                                                                                                                               |

|                 |                                          |                                                                                                                                                                                                                                                                                                                                                                                                               |         |
|-----------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| CMP (CD)        |                                          | Cycles: 4                                                                                                                                                                                                                                                                                                                                                                                                     | Size: 3 |
| Absolute (Read) |                                          |                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| Cycle           | R/W                                      | Desc.                                                                                                                                                                                                                                                                                                                                                                                                         |         |
| -X.2            | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |         |
| 1.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |         |
| 1.2             | Read <b>PC:PB</b>                        | Store as Address.                                                                                                                                                                                                                                                                                                                                                                                             |         |
| 2.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |         |
| 2.2             | Read <b>PC:PB</b>                        | Store as Address.H.                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 3.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |         |
| 3.2             | Read Address: <b>DB</b>                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                               |         |
| 4.1             |                                          |                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 4.2             | Read Address + 1: <b>DB</b> (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 5.1             |                                          | If <b>M</b> flag is set, set <b>C</b> flag based off ( <b>A</b> .L >= Operand.L), set <b>Z</b> flag based off ( <b>A</b> .L == Operand.L), and set <b>N</b> flag based off (( <b>A</b> .L - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>A</b> >= Operand), set <b>Z</b> flag based off ( <b>A</b> == Operand), and set <b>N</b> flag based off (( <b>A</b> - Operand) & \$8000) != 0. |         |
| 5.2             | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |         |
| +X.1            |                                          |                                                                                                                                                                                                                                                                                                                                                                                                               |         |

| DEC (CE)                     | Cycles: 6                                    | Size: 3                                                                                                                                                                                                                                          |
|------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute (Read/Modify/Write) |                                              |                                                                                                                                                                                                                                                  |
| Cycle                        | R/W                                          | Desc.                                                                                                                                                                                                                                            |
| -X.2                         | Read PC:PB                                   | Store as OpCode.                                                                                                                                                                                                                                 |
| 1.1                          | Read <b>PC:PB</b>                            | Inc. <b>PC</b> .                                                                                                                                                                                                                                 |
| 1.2                          |                                              | Store as Address.                                                                                                                                                                                                                                |
| 2.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                 |
| 2.2                          |                                              | Store as Address.H.                                                                                                                                                                                                                              |
| 3.1                          | Read <b>PC:PB</b>                            | Inc. <b>PC</b> .                                                                                                                                                                                                                                 |
| 3.2                          |                                              | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                  |
| 4.1                          | Read Address + 1: <b>DB</b> (With Carry)     | Store as Operand.H.                                                                                                                                                                                                                              |
| 4.2                          |                                              |                                                                                                                                                                                                                                                  |
| 5.1                          |                                              | If <b>M</b> flag is set, perform Operand.L -= 1, set <b>N</b> based off (Operand.L & \$80), and set <b>Z</b> based off Operand.L, otherwise perform Operand -= 1, set <b>N</b> based off (Operand.H & \$80), and set <b>Z</b> based off Operand. |
| 5.2                          |                                              | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                    |
| 6.1                          | Write to Address + 1: <b>DB</b> (With Carry) | Write Operand.H.                                                                                                                                                                                                                                 |
| 6.2                          |                                              |                                                                                                                                                                                                                                                  |
| 7.1                          | Write to Address: <b>DB</b>                  | Write Operand.L.                                                                                                                                                                                                                                 |
| 7.2                          |                                              |                                                                                                                                                                                                                                                  |
| 8.1                          | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                                                                                                 |
| 8.2                          |                                              |                                                                                                                                                                                                                                                  |
| +X.1                         |                                              |                                                                                                                                                                                                                                                  |

|                      |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CMP (CF)             | Cycles: 5                          | Size: 4                                                                                                                                                                                                                                                                                                                                                                                                       |
| Absolute Long (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
| Cycle                | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                         |
| -X.2                 | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.2                  | Read <b>PC:PB</b>                  | Store as Address.                                                                                                                                                                                                                                                                                                                                                                                             |
| 2.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 2.2                  | Read <b>PC:PB</b>                  | Store as Address.H.                                                                                                                                                                                                                                                                                                                                                                                           |
| 3.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 3.2                  | Read <b>PC:PB</b>                  | Stores as Bank.                                                                                                                                                                                                                                                                                                                                                                                               |
| 4.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 4.2                  | Read Address:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                               |
| 5.1                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.2                  | Read Address + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                           |
| 6.1                  |                                    | If <b>M</b> flag is set, set <b>C</b> flag based off ( <b>A</b> .L >= Operand.L), set <b>Z</b> flag based off ( <b>A</b> .L == Operand.L), and set <b>N</b> flag based off (( <b>A</b> .L - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>A</b> >= Operand), set <b>Z</b> flag based off ( <b>A</b> == Operand), and set <b>N</b> flag based off (( <b>A</b> - Operand) & \$8000) != 0. |
| 6.2                  | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |
| +X.1                 |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |



|          |                   |                                                                                                 |         |
|----------|-------------------|-------------------------------------------------------------------------------------------------|---------|
| BNE (D0) |                   | Cycles: 2                                                                                       | Size: 2 |
| Relative |                   |                                                                                                 |         |
| Cycle    | R/W               | Desc.                                                                                           |         |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                |         |
| 1.1      |                   | Inc. <b>PC</b> . Decide to branch if <b>Z</b> is 0.                                             |         |
| 1.2      | Read <b>PC:PB</b> | Store as Operand. Address = i16(i8(Operand.L)) + <b>PC</b> . If not branching, poll interrupts. |         |
| 2.1      |                   | Inc. <b>PC</b> . If not branching, end (next half-cycle is 3.2).                                |         |
| 2.2      |                   | Poll interrupts.                                                                                |         |
| 3.1      |                   | Set <b>PC</b> to Address.                                                                       |         |
| 3.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                |         |
| +X.1     |                   |                                                                                                 |         |

| CMP (D1)                             |                                    | Cycles: 5                                                                                                                                                                                                                                                                                                                                                                                                  | Size: 2 |
|--------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Direct Indirect Indexed (d),y (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| Cycle                                | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| -X.2                                 | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 1.1                                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 1.2                                  | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                 |         |
| 2.1                                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 2.2                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 3.1                                  |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                          |         |
| 3.2                                  | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                          |         |
| 4.1                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 4.2                                  | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 5.1                                  |                                    | Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the X flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                                                |         |
| 5.2                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 6.1                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 6.2                                  | Read Pointer:Bank                  | Store as Operand. If M flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                   |         |
| 7.1                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 7.2                                  | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 8.1                                  |                                    | If <b>M</b> flag is set, set <b>C</b> flag based off ( <b>A.L</b> >= Operand.L), set <b>Z</b> flag based off ( <b>A.L</b> == Operand.L), and set <b>N</b> flag based off (( <b>A.L</b> - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>A</b> >= Operand), set <b>Z</b> flag based off ( <b>A</b> == Operand), and set <b>N</b> flag based off (( <b>A</b> - Operand) & \$8000) != 0. |         |
| 8.2                                  | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |         |
| +X.1                                 |                                    |                                                                                                                                                                                                                                                                                                                                                                                                            |         |

|                        |                             |                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CMP (D2)               | Cycles: 5                   | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                    |
| Direct Indirect (Read) |                             |                                                                                                                                                                                                                                                                                                                                                                                                            |
| Cycle                  | R/W                         | Desc.                                                                                                                                                                                                                                                                                                                                                                                                      |
| -X.2                   | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |
| 1.1                    |                             | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |
| 1.2                    | Read <b>PC:PB</b>           | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                 |
| 2.1                    |                             | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |
| 2.2                    |                             |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3.1                    |                             | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                          |
| 3.2                    | Read Address                | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                          |
| 4.1                    |                             |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4.2                    | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.1                    |                             |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.2                    | Read Pointer: <b>DB</b>     | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                            |
| 6.1                    |                             |                                                                                                                                                                                                                                                                                                                                                                                                            |
| 6.2                    | Read Pointer + 1: <b>DB</b> | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                        |
| 7.1                    |                             | If <b>M</b> flag is set, set <b>C</b> flag based off ( <b>A.L</b> >= Operand.L), set <b>Z</b> flag based off ( <b>A.L</b> == Operand.L), and set <b>N</b> flag based off (( <b>A.L</b> - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>A</b> >= Operand), set <b>Z</b> flag based off ( <b>A</b> == Operand), and set <b>N</b> flag based off (( <b>A</b> - Operand) & \$8000) != 0. |
| 7.2                    | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |
| +X.1                   |                             |                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                        |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |         |
|----------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| CMP (D3)                               |                                    | Cycles: 7                                                                                                                                                                                                                                                                                                                                                                                                     | Size: 2 |
| Stack Relative Indirect Indexed (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| Cycle                                  | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                         |         |
| -X.2                                   | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |         |
| 1.1                                    | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |         |
| 1.2                                    |                                    | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                             |         |
| 2.1                                    |                                    | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                                                                                                                                                                                                                                                                          |         |
| 2.2                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 3.1                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 3.2                                    | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                             |         |
| 4.1                                    | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 4.2                                    |                                    | Perform Tmp = ui32(Pointer + <b>Y</b> ).                                                                                                                                                                                                                                                                                                                                                                      |         |
| 5.1                                    |                                    | Pointer = ui16(Tmp).                                                                                                                                                                                                                                                                                                                                                                                          |         |
|                                        |                                    | Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                                                                                                                                                                                                               |         |
| 5.2                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 6.1                                    | Read Pointer:Bank                  |                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 6.2                                    |                                    | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                               |         |
| 7.1                                    | Read Pointer + 1:Bank (With Carry) |                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 7.2                                    |                                    | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 8.1                                    | Read <b>PC:PB</b>                  | If <b>M</b> flag is set, set <b>C</b> flag based off ( <b>A</b> .L >= Operand.L), set <b>Z</b> flag based off ( <b>A</b> .L == Operand.L), and set <b>N</b> flag based off (( <b>A</b> .L - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>A</b> >= Operand), set <b>Z</b> flag based off ( <b>A</b> == Operand), and set <b>N</b> flag based off (( <b>A</b> - Operand) & \$8000) != 0. |         |
| 8.2                                    |                                    | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |         |
| +X.1                                   |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |         |

|                      |                                                                                                    |                                                                                   |
|----------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| PEI (D4)             | Cycles: 6                                                                                          | Size: 2                                                                           |
| Direct Page Indirect |                                                                                                    |                                                                                   |
| Cycle                | R/W                                                                                                | Desc.                                                                             |
| -X.2                 | Read <b>PC:PB</b>                                                                                  | Store as OpCode.                                                                  |
| 1.1                  |                                                                                                    | Inc. <b>PC</b> .                                                                  |
| 1.2                  | Read <b>PC:PB</b>                                                                                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                        |
| 2.1                  |                                                                                                    | Inc. <b>PC</b> .                                                                  |
| 2.2                  |                                                                                                    |                                                                                   |
| 3.1                  |                                                                                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> . |
| 3.2                  | Read Address                                                                                       | Store as Operand.                                                                 |
| 4.1                  |                                                                                                    |                                                                                   |
| 4.2                  | If <b>D.L</b> is 0, Read (((Address.L + 1) & \$FF)   (Address.H << 8)), otherwise read Address + 1 | Store as Bank.                                                                    |
| 5.1                  |                                                                                                    |                                                                                   |
| 5.2                  | Write to ( <b>S</b> +0)                                                                            | Push Operand.H onto stack.                                                        |
| 6.1                  |                                                                                                    |                                                                                   |
| 6.2                  | Write to ( <b>S</b> -1)                                                                            | Push Operand.L onto stack.                                                        |
| 7.1                  |                                                                                                    | Dec. <b>S</b> by 2.                                                               |
| 7.2                  | Read <b>PC:PB</b>                                                                                  | Store as OpCode.                                                                  |
| +X.1                 |                                                                                                    |                                                                                   |

|                  |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
|------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| CMP (D5)         |                   | Cycles: 4                                                                                                                                                                                                                                                                                                                                                                                                  | Size: 2 |
| Direct, X (Read) |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| Cycle            | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| -X.2             | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 1.1              | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 1.2              |                   | Store as Operand.                                                                                                                                                                                                                                                                                                                                                                                          |         |
| 2.1              |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 2.2              |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 3.1              |                   | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                                                                                                                                                                                                          |         |
| 3.2              |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 4.1              |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 4.2              | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                            |         |
| 5.1              | Read Address + 1  |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 5.2              |                   | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 6.1              | Read <b>PC:PB</b> | If <b>M</b> flag is set, set <b>C</b> flag based off ( <b>A.L</b> >= Operand.L), set <b>Z</b> flag based off ( <b>A.L</b> == Operand.L), and set <b>N</b> flag based off (( <b>A.L</b> - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>A</b> >= Operand), set <b>Z</b> flag based off ( <b>A</b> == Operand), and set <b>N</b> flag based off (( <b>A</b> - Operand) & \$8000) != 0. |         |
| 6.2              |                   | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |         |
| +X.1             |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |         |

| DEC (D6)                      | Cycles: 6            | Size: 2                                                                                                                                                                                                                                          |
|-------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct, X (Read/Modify/Write) |                      |                                                                                                                                                                                                                                                  |
| Cycle                         | R/W                  | Desc.                                                                                                                                                                                                                                            |
| -X.2                          | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                 |
| 1.1                           |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                 |
| 1.2                           | Read <b>PC:PB</b>    | Store as Operand.                                                                                                                                                                                                                                |
| 2.1                           |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                 |
| 2.2                           |                      |                                                                                                                                                                                                                                                  |
| 3.1                           |                      | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                                                |
| 3.2                           |                      |                                                                                                                                                                                                                                                  |
| 4.1                           |                      |                                                                                                                                                                                                                                                  |
| 4.2                           | Read Address         | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                  |
| 5.1                           |                      |                                                                                                                                                                                                                                                  |
| 5.2                           | Read Address + 1     | Store as Operand.H.                                                                                                                                                                                                                              |
| 6.1                           |                      | If <b>M</b> flag is set, perform Operand.L += 1, set <b>N</b> based off (Operand.L & \$80), and set <b>Z</b> based off Operand.L, otherwise perform Operand += 1, set <b>N</b> based off (Operand.H & \$80), and set <b>Z</b> based off Operand. |
| 6.2                           |                      | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                    |
| 7.1                           |                      |                                                                                                                                                                                                                                                  |
| 7.2                           | Write to Address + 1 | Write Operand.H.                                                                                                                                                                                                                                 |
| 8.1                           |                      |                                                                                                                                                                                                                                                  |
| 8.2                           | Write to Address     | Write Operand.L.                                                                                                                                                                                                                                 |
| 9.1                           |                      |                                                                                                                                                                                                                                                  |
| 9.2                           | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                 |
| +X.1                          |                      |                                                                                                                                                                                                                                                  |

| CMP (D7) Cycles: 6                        |                                    | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indirect Indexed Long [d],y (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Cycle                                     | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| -X.2                                      | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 1.1                                       |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 1.2                                       | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2.1                                       |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2.2                                       |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3.1                                       |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                                                                                          |
| 3.2                                       | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 4.1                                       |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4.2                                       | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.1                                       |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5.2                                       | Read Address + 2                   | Store as Bank.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 6.1                                       |                                    | Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{Y})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank += $(\text{Tmp} \gg 16)$ .                                                                                                                                                                                                                                                                                                                                |
| 6.2                                       | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7.1                                       |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7.2                                       | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 8.1                                       |                                    | If <b>M</b> flag is set, set <b>C</b> flag based off $(\text{A.L} \geq \text{Operand.L})$ , set <b>Z</b> flag based off $(\text{A.L} == \text{Operand.L})$ , and set <b>N</b> flag based off $((\text{A.L} - \text{Operand.L}) \& \$80) \neq 0$ , otherwise set <b>C</b> flag based off $(\text{A} \geq \text{Operand})$ , set <b>Z</b> flag based off $(\text{A} == \text{Operand})$ , and set <b>N</b> flag based off $((\text{A} - \text{Operand}) \& \$8000) \neq 0$ . |
| 8.2                                       | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| +X.1                                      |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |



|          |                   |                             |
|----------|-------------------|-----------------------------|
| CLD (D8) | Cycles: 2         | Size: 1                     |
| Implied  |                   |                             |
| Cycle    | R/W               | Desc.                       |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.            |
| 1.1      |                   | Inc. <b>PC</b> .            |
| 1.2      | Read <b>PC:PB</b> | Discard.                    |
| 2.1      |                   | Set the <b>D</b> flag to 0. |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.            |
| +X.1     |                   |                             |

|                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CMP (D9) Cycles: 4 |                                    | Size: 3                                                                                                                                                                                                                                                                                                                                                                                                       |
| Absolute, Y (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                         |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.2                | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                             |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 2.2                | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                           |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                           |
| 3.2                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4.1                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                               |
| 5.1                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.2                | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                           |
| 6.1                |                                    | If <b>M</b> flag is set, set <b>C</b> flag based off ( <b>A</b> .L >= Operand.L), set <b>Z</b> flag based off ( <b>A</b> .L == Operand.L), and set <b>N</b> flag based off (( <b>A</b> .L - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>A</b> >= Operand), set <b>Z</b> flag based off ( <b>A</b> == Operand), and set <b>N</b> flag based off (( <b>A</b> - Operand) & \$8000) != 0. |
| 6.2                | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |
| +X.1               |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |

|          |                         |                                               |
|----------|-------------------------|-----------------------------------------------|
| PHX (DA) | Cycles: 3               | Size: 1                                       |
| Implied  |                         |                                               |
| Cycle    | R/W                     | Desc.                                         |
| -X.2     | Read <b>PC:PB</b>       | Store as OpCode.                              |
| 1.1      |                         | Inc. <b>PC</b> .                              |
| 1.2      |                         | If <b>X</b> flag is set, skip the next cycle. |
| 2.1      |                         |                                               |
| 2.2      | Write to ( <b>S</b> +0) | Push <b>X.H</b> onto stack.                   |
| 3.1      |                         | Dec. <b>S</b> by 1.                           |
| 3.2      | Write to ( <b>S</b> +0) | Push <b>X.L</b> onto stack.                   |
| 4.1      |                         | Dec. <b>S</b> by 1.                           |
| 4.2      | Read <b>PC:PB</b>       | Store as OpCode.                              |
| +X.1     |                         |                                               |

|          |                   |                  |
|----------|-------------------|------------------|
| STP (DB) | Cycles: 4         | Size: 1          |
| Implied  |                   |                  |
| Cycle    | R/W               | Desc.            |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode. |
| 1.1      |                   | Inc. <b>PC</b> . |
| 1.2      |                   |                  |
| 2.1      |                   |                  |
| 2.2      |                   |                  |
| 3.1      |                   |                  |
| 3.2      |                   |                  |
| 4.1      |                   | Stop processor.  |

|                               |                                                                                                    |                                                  |
|-------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------|
| JML (DC)                      | Cycles: 6                                                                                          | Size: 3                                          |
| Absolute Indirect Long (Jump) |                                                                                                    |                                                  |
| Cycle                         | R/W                                                                                                | Desc.                                            |
| -X.2                          | Read <b>PC:PB</b>                                                                                  | Store as OpCode.                                 |
| 1.1                           |                                                                                                    | Inc. <b>PC</b> .                                 |
| 1.2                           | Read <b>PC:PB</b>                                                                                  | Store as Pointer.                                |
| 2.1                           |                                                                                                    | Inc. <b>PC</b> .                                 |
| 2.2                           | Read <b>PC:PB</b>                                                                                  | Store as Pointer.H.                              |
| 3.1                           |                                                                                                    | Inc. <b>PC</b> .                                 |
| 3.2                           | Read Pointer                                                                                       | Store as Operand.                                |
| 4.1                           |                                                                                                    |                                                  |
| 4.2                           | Read Pointer + 1                                                                                   | Store as Operand.H.                              |
| 5.1                           |                                                                                                    |                                                  |
| 5.2                           | If <b>D.L</b> is 0, Read (((Pointer.L + 2) & \$FF)   (Pointer.H << 8)), otherwise read Pointer + 2 | Store as Bank.                                   |
| 6.1                           |                                                                                                    | Set <b>PC</b> to Operand. Set <b>PB</b> to Bank. |
| 6.2                           | Read <b>PC:PB</b>                                                                                  | Store as OpCode.                                 |
| +X.1                          |                                                                                                    |                                                  |

| CMP (DD)           |                                    | Cycles: 4                                                                                                                                                                                                                                                                                                                                                                                                     | Size: 3 |
|--------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Absolute, X (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                         |         |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |         |
| 1.1                | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |         |
| 1.2                |                                    | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                             |         |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |         |
| 2.2                |                                    | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>X</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                           |         |
| 3.2                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 4.1                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                               |         |
| 5.1                | Read Pointer + 1:Bank (With Carry) |                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| 5.2                |                                    | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 6.1                | Read <b>PC:PB</b>                  | If <b>M</b> flag is set, set <b>C</b> flag based off ( <b>A</b> .L >= Operand.L), set <b>Z</b> flag based off ( <b>A</b> .L == Operand.L), and set <b>N</b> flag based off (( <b>A</b> .L - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>A</b> >= Operand), set <b>Z</b> flag based off ( <b>A</b> == Operand), and set <b>N</b> flag based off (( <b>A</b> - Operand) & \$8000) != 0. |         |
| 6.2                |                                    | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |         |
| +X.1               |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |         |

| DEC (DE)                       | Cycles: 7                              | Size: 3                                                                                                                                                                                                                                                                                                          |
|--------------------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute, X (Read/Modiy/Write) |                                        |                                                                                                                                                                                                                                                                                                                  |
| Cycle                          | R/W                                    | Desc.                                                                                                                                                                                                                                                                                                            |
| -X.2                           | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                                                                                                                                                                 |
| 1.1                            |                                        | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                 |
| 1.2                            | Read <b>PC:PB</b>                      | Store as Pointer.                                                                                                                                                                                                                                                                                                |
| 2.1                            |                                        | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                 |
| 2.2                            | Read <b>PC:PB</b>                      | Store as Pointer.H.                                                                                                                                                                                                                                                                                              |
| 3.1                            |                                        | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{X})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank = $\mathbf{DB} + (\text{Tmp} \gg 16)$ .                                                                                                                                      |
| 3.2                            |                                        |                                                                                                                                                                                                                                                                                                                  |
| 4.1                            |                                        |                                                                                                                                                                                                                                                                                                                  |
| 4.2                            | Read Pointer:Bank                      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                  |
| 5.1                            |                                        |                                                                                                                                                                                                                                                                                                                  |
| 5.2                            | Read Pointer + 1:Bank (With Carry)     | Store as Operand.H.                                                                                                                                                                                                                                                                                              |
| 6.1                            |                                        | If <b>M</b> flag is set, perform $\text{Operand.L} -= 1$ , set <b>N</b> based off ( $\text{Operand.L} \& \$80$ ), and set <b>Z</b> based off $\text{Operand.L}$ , otherwise perform $\text{Operand} -= 1$ , set <b>N</b> based off ( $\text{Operand.H} \& \$80$ ), and set <b>Z</b> based off $\text{Operand}$ . |
| 6.2                            |                                        | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                    |
| 7.1                            |                                        |                                                                                                                                                                                                                                                                                                                  |
| 7.2                            | Write to Pointer + 1:Bank (With Carry) | Write Operand.H.                                                                                                                                                                                                                                                                                                 |
| 8.1                            |                                        |                                                                                                                                                                                                                                                                                                                  |
| 8.2                            | Write to Pointer:Bank                  | Write Operand.L.                                                                                                                                                                                                                                                                                                 |
| 9.1                            |                                        |                                                                                                                                                                                                                                                                                                                  |
| 9.2                            | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                                                                                                                                                                 |
| +X.1                           |                                        |                                                                                                                                                                                                                                                                                                                  |

|                                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CMP (DF)    Cycles: 5                      Size: 4 |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
| Absolute Long, X (Read)                            |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
| Cycle                                              | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                         |
| -X.2                                               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.1                                                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.2                                                | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                             |
| 2.1                                                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 2.2                                                | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                           |
| 3.1                                                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 3.2                                                | Read <b>PC:PB</b>                  | Stores as Bank.                                                                                                                                                                                                                                                                                                                                                                                               |
| 4.1                                                |                                    | Inc. <b>PC</b> . Perform Tmp = ui32(Pointer + <b>X</b> ).<br>Pointer = ui16(Tmp).<br>Bank += (Tmp >> 16).                                                                                                                                                                                                                                                                                                     |
| 4.2                                                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                               |
| 5.1                                                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5.2                                                | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                           |
| 6.1                                                |                                    | If <b>M</b> flag is set, set <b>C</b> flag based off ( <b>A</b> .L >= Operand.L), set <b>Z</b> flag based off ( <b>A</b> .L == Operand.L), and set <b>N</b> flag based off (( <b>A</b> .L - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>A</b> >= Operand), set <b>Z</b> flag based off ( <b>A</b> == Operand), and set <b>N</b> flag based off (( <b>A</b> - Operand) & \$8000) != 0. |
| 6.2                                                | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |
| +X.1                                               |                                    |                                                                                                                                                                                                                                                                                                                                                                                                               |



|           |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CPX (E0)  | Cycles: 2         | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Immediate |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Cycle     | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| -X.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1.2       | Read <b>PC:PB</b> | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                |
| 2.1       |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2.2       | Read <b>PC:PB</b> | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3.1       |                   | Inc. <b>PC</b> . If <b>X</b> flag is set, set <b>C</b> flag based off ( <b>X</b> .L >= Operand.L), set <b>Z</b> flag based off ( <b>X</b> .L == Operand.L), and set <b>N</b> flag based off (( <b>X</b> .L - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>X</b> >= Operand), set <b>Z</b> flag based off ( <b>X</b> == Operand), and set <b>N</b> flag based off (( <b>X</b> - Operand) & \$8000) != 0. |
| 3.2       | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                               |
| +X.1      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                |

| SBC (E1) Cycles: 6                   |                             | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indexed Indirect (d,x) (Read) |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Cycle                                | R/W                         | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| -X.2                                 | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1.1                                  |                             | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1.2                                  | Read <b>PC:PB</b>           | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2.1                                  |                             | Inc. <b>PC</b> . Set Pointer to Pointer + <b>X</b> + <b>D</b> .<br>Address = Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2.2                                  |                             | If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 3.1                                  |                             | Address.H = Pointer.H (fixes high byte of address).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3.2                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4.1                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4.2                                  | Read Address                | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.1                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.2                                  | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 6.1                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6.2                                  | Read Pointer: <b>DB</b>     | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 7.1                                  |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 7.2                                  | Read Pointer + 1: <b>DB</b> | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8.1                                  |                             | <p>If <b>M</b> flag == 1 (8-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = ui16(<b>A.L</b>) - ui16(Operand) - (1 - <b>C</b>).<br/>             <b>V</b> flag = ((ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.<br/>             <b>C</b> flag = Result &lt;= \$FF.<br/>             Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:<br/>             Lo = int16(<b>A.L</b> &amp; 0x0F) - int16(Operand &amp; 0x0F) - (1 - <b>C</b>);<br/>             if (Lo &lt; 0) Lo -= 6.<br/>             BorrowToHi = Lo &lt; 0 ? 1 : 0.<br/>             HiSum = int16(<b>A.L</b> &gt;&gt; 4) - int16(Operand &gt;&gt; 4) - BorrowToHi.<br/>             <b>V</b> flag = (((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.<br/>             if (HiSum &lt; 0) HiSum -= 6.<br/>             <b>C</b> flag = HiSum &gt;= 0.<br/>             Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).<br/>             <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = ui32(<b>A</b>) - ui32(Operand) - (1 - <b>C</b>).<br/>             <b>V</b> flag = ((ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0.<br/>             <b>C</b> flag = Result &lt;= \$FFFF.<br/>             Result = ui16(Result).<br/>             <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:<br/>             Result = ui16(0). Borrow = ui16(1 - <b>C</b>).<br/>             for i = 0, 4, 8, 12:<br/>               DigitSum = int16((<b>A</b> &gt;&gt; i) &amp; \$0F) - int16((Operand &gt;&gt; i) &amp; \$0F) - Borrow.<br/>               if (i == 12) <b>V</b> flag = (((<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.<br/>               if (DigitSum &lt; 0) DigitSum -= 6.<br/>               Borrow = (DigitSum &lt; 0) ? 1 : 0.<br/>               Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).<br/>             <b>C</b> flag = Borrow == 0. <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 8.2                                  | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| +X.1                                 |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|           |                   |                                                                                      |
|-----------|-------------------|--------------------------------------------------------------------------------------|
| SEP (E2)  | Cycles: 3         | Size: 2                                                                              |
| Immediate |                   |                                                                                      |
| Cycle     | R/W               | Desc.                                                                                |
| -X.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                     |
| 1.1       | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                     |
| 1.2       |                   | Store as Operand.                                                                    |
| 2.1       |                   |                                                                                      |
| 2.2       |                   |                                                                                      |
| 3.1       |                   | <b>P</b>  = Operand.L, set <b>X</b> .H and <b>Y</b> .H to 0 if <b>X</b> flag is set. |
| 3.2       |                   | Store as OpCode.                                                                     |
| +X.1      |                   |                                                                                      |

| SBC (E3) Cycles: 4    |                   | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stack Relative (Read) |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cycle                 | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| -X.2                  | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.1                   | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.2                   |                   | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2.1                   |                   | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2.2                   |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3.1                   |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3.2                   | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4.1                   |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4.2                   | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.1                   |                   | <p>If <b>M</b> flag == 1 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) - ui16(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = int16(<b>A.L</b> &amp; 0x0F) - int16(Operand &amp; 0x0F) - (1 - <b>C</b>);</p> <p>if (Lo &lt; 0) Lo -= 6.</p> <p>BorrowToHi = Lo &lt; 0 ? 1 : 0.</p> <p>HiSum = int16(<b>A.L</b> &gt;&gt; 4) - int16(Operand &gt;&gt; 4) - BorrowToHi.</p> <p><b>V</b> flag = (((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &lt; 0) HiSum -= 6.</p> <p><b>C</b> flag = HiSum &gt;= 0.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) - ui32(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Borrow = ui16(1 - <b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = int16((<b>A</b> &gt;&gt; i) &amp; \$0F) - int16((Operand &gt;&gt; i) &amp; \$0F) - Borrow.</p> <p>if (i == 12) <b>V</b> flag = (((<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.</p> <p>if (DigitSum &lt; 0) DigitSum -= 6.</p> <p>Borrow = (DigitSum &lt; 0) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Borrow == 0. <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 5.2                   | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| +X.1                  |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|               |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
|---------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| CPX (E4)      |                   | Cycles: 3                                                                                                                                                                                                                                                                                                                                                                                                  | Size: 2 |
| Direct (Read) |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| Cycle         | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| -X.2          | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 1.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 1.2           | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                 |         |
| 2.1           |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                           |         |
| 2.2           |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 3.1           |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                          |         |
| 3.2           | Read Address      | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                            |         |
| 4.1           |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| 4.2           | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                        |         |
| 5.1           |                   | If <b>X</b> flag is set, set <b>C</b> flag based off ( <b>X.L</b> >= Operand.L), set <b>Z</b> flag based off ( <b>X.L</b> == Operand.L), and set <b>N</b> flag based off (( <b>X.L</b> - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>X</b> >= Operand), set <b>Z</b> flag based off ( <b>X</b> == Operand), and set <b>N</b> flag based off (( <b>X</b> - Operand) & \$8000) != 0. |         |
| 5.2           | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                           |         |
| +X.1          |                   |                                                                                                                                                                                                                                                                                                                                                                                                            |         |

| SBC (E5) Cycles: 3 |                   | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct (Read)      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cycle              | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| -X.2               | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.1                |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.2                | Read <b>PC:PB</b> | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2.1                |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2.2                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3.1                |                   | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3.2                | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4.1                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4.2                | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.1                |                   | <p>If <b>M</b> flag == 1 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) - ui16(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = int16(<b>A.L</b> &amp; 0x0F) - int16(Operand &amp; 0x0F) - (1 - <b>C</b>);</p> <p>if (Lo &lt; 0) Lo -= 6.</p> <p>BorrowToHi = Lo &lt; 0 ? 1 : 0.</p> <p>HiSum = int16(<b>A.L</b> &gt;&gt; 4) - int16(Operand &gt;&gt; 4) - BorrowToHi.</p> <p><b>V</b> flag = (((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &lt; 0) HiSum -= 6.</p> <p><b>C</b> flag = HiSum &gt;= 0.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) - ui32(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Borrow = ui16(1 - <b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = int16((<b>A</b> &gt;&gt; i) &amp; \$0F) - int16((Operand &gt;&gt; i) &amp; \$0F) - Borrow.</p> <p>if (i == 12) <b>V</b> flag = (((<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.</p> <p>if (DigitSum &lt; 0) DigitSum -= 6.</p> <p>Borrow = (DigitSum &lt; 0) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Borrow == 0. <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 5.2                | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| +X.1               |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                            |                      |                                                                                                                                                                                                                                                  |         |
|----------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| INC (E6)                   |                      | Cycles: 5                                                                                                                                                                                                                                        | Size: 2 |
| Direct (Read/Modify/Write) |                      |                                                                                                                                                                                                                                                  |         |
| Cycle                      | R/W                  | Desc.                                                                                                                                                                                                                                            |         |
| -X.2                       | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                 |         |
| 1.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                 |         |
| 1.2                        | Read <b>PC:PB</b>    | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                       |         |
| 2.1                        |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                 |         |
| 2.2                        |                      |                                                                                                                                                                                                                                                  |         |
| 3.1                        |                      | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                |         |
| 3.2                        | Read Address         | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                  |         |
| 4.1                        |                      |                                                                                                                                                                                                                                                  |         |
| 4.2                        | Read Address + 1     | Store as Operand.H.                                                                                                                                                                                                                              |         |
| 5.1                        |                      | If <b>M</b> flag is set, perform Operand.L += 1, set <b>N</b> based off (Operand.L & \$80), and set <b>Z</b> based off Operand.L, otherwise perform Operand += 1, set <b>N</b> based off (Operand.H & \$80), and set <b>Z</b> based off Operand. |         |
| 5.2                        |                      | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                    |         |
| 6.1                        |                      |                                                                                                                                                                                                                                                  |         |
| 6.2                        | Write to Address + 1 | Write Operand.H.                                                                                                                                                                                                                                 |         |
| 7.1                        |                      |                                                                                                                                                                                                                                                  |         |
| 7.2                        | Write to Address     | Write Operand.L.                                                                                                                                                                                                                                 |         |
| 8.1                        |                      |                                                                                                                                                                                                                                                  |         |
| 8.2                        | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                 |         |
| +X.1                       |                      |                                                                                                                                                                                                                                                  |         |

| SBC (E7) Cycles: 6          |                                    | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indirect Long (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cycle                       | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| -X.2                        | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.1                         |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.2                         | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2.1                         |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2.2                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3.1                         |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3.2                         | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4.1                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4.2                         | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.1                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.2                         | Read Address + 2                   | Store as Bank.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 6.1                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 6.2                         | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 7.1                         |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 7.2                         | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 8.1                         |                                    | <p>If <b>M</b> flag == 1 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) - ui16(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = int16(<b>A.L</b> &amp; 0x0F) - int16(Operand &amp; 0x0F) - (1 - <b>C</b>);</p> <p>if (Lo &lt; 0) Lo -= 6.</p> <p>BorrowToHi = Lo &lt; 0 ? 1 : 0.</p> <p>HiSum = int16(<b>A.L</b> &gt;&gt; 4) - int16(Operand &gt;&gt; 4) - BorrowToHi.</p> <p><b>V</b> flag = (((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &lt; 0) HiSum -= 6.</p> <p><b>C</b> flag = HiSum &gt;= 0.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) - ui32(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Borrow = ui16(1 - <b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = int16((<b>A</b> &gt;&gt; i) &amp; \$0F) - int16((Operand &gt;&gt; i) &amp; \$0F) - Borrow.</p> <p>if (i == 12) <b>V</b> flag = (((<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.</p> <p>if (DigitSum &lt; 0) DigitSum -= 6.</p> <p>Borrow = (DigitSum &lt; 0) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Borrow == 0. <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 8.2                         | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| +X.1                        |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



|          |                   |                                                                                                                                                                                                                                                            |
|----------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INX (E8) | Cycles: 2         | Size: 1                                                                                                                                                                                                                                                    |
| Implied  |                   |                                                                                                                                                                                                                                                            |
| Cycle    | R/W               | Desc.                                                                                                                                                                                                                                                      |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                           |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                           |
| 1.2      |                   | Discard.                                                                                                                                                                                                                                                   |
| 2.1      |                   | If <b>X</b> flag is set, perform <b>X.L</b> += 1, set <b>N</b> based off ( <b>X.L</b> & \$80), and set <b>Z</b> based off <b>X.L</b> , otherwise perform <b>X</b> += 1, set <b>N</b> based off ( <b>X.H</b> & \$80), and set <b>Z</b> based off <b>X</b> . |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                                                                                           |
| +X.1     |                   |                                                                                                                                                                                                                                                            |

| SBC (E9) Cycles: 2 |                   | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Immediate          |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cycle              | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| -X.2               | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.1                |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.2                | Read <b>PC:PB</b> | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2.1                |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2.2                | Read <b>PC:PB</b> | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3.1                |                   | <p>If <b>M</b> flag == 1 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) - ui16(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = int16(<b>A.L</b> &amp; 0x0F) - int16(Operand &amp; 0x0F) - (1 - <b>C</b>);</p> <p>if (Lo &lt; 0) Lo -= 6.</p> <p>BorrowToHi = Lo &lt; 0 ? 1 : 0.</p> <p>HiSum = int16(<b>A.L</b> &gt;&gt; 4) - int16(Operand &gt;&gt; 4) - BorrowToHi.</p> <p><b>V</b> flag = (((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &lt; 0) HiSum -= 6.</p> <p><b>C</b> flag = HiSum &gt;= 0.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) - ui32(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Borrow = ui16(1 - <b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = int16((<b>A</b> &gt;&gt; i) &amp; \$0F) - int16((Operand &gt;&gt; i) &amp; \$0F) - Borrow.</p> <p>if (i == 12) <b>V</b> flag = (((<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.</p> <p>if (DigitSum &lt; 0) DigitSum -= 6.</p> <p>Borrow = (DigitSum &lt; 0) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Borrow == 0. <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 3.2                | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| +X.1               |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|          |                   |           |                  |
|----------|-------------------|-----------|------------------|
| NOP (EA) |                   | Cycles: 2 | Size: 1          |
| Implied  |                   |           |                  |
| Cycle    | R/W               |           | Desc.            |
| -X.2     | Read <b>PC:PB</b> |           | Store as OpCode. |
| 1.1      |                   |           | Inc. <b>PC</b> . |
| 1.2      | Read <b>PC:PB</b> |           | Discard.         |
| 2.1      |                   |           |                  |
| 2.2      | Read <b>PC:PB</b> |           | Store as OpCode. |
| +X.1     |                   |           |                  |

|          |                   |                                                                                                                       |
|----------|-------------------|-----------------------------------------------------------------------------------------------------------------------|
| XBA (EB) | Cycles: 3         | Size: 1                                                                                                               |
| Implied  |                   |                                                                                                                       |
| Cycle    | R/W               | Desc.                                                                                                                 |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                      |
| 1.1      |                   | Inc. <b>PC</b> .                                                                                                      |
| 1.2      |                   |                                                                                                                       |
| 2.1      |                   |                                                                                                                       |
| 2.2      |                   |                                                                                                                       |
| 3.1      |                   | Swap <b>A.L</b> and <b>A.H</b> , set <b>N</b> based off ( <b>A.L</b> & \$80), and set <b>Z</b> based off <b>A.L</b> . |
| 3.2      | Read <b>PC:PB</b> | Store as OpCode.                                                                                                      |
| +X.1     |                   |                                                                                                                       |

|                 |                                          |                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CPX (EC)        | Cycles: 4                                | Size: 3                                                                                                                                                                                                                                                                                                                                                                                                       |
| Absolute (Read) |                                          |                                                                                                                                                                                                                                                                                                                                                                                                               |
| Cycle           | R/W                                      | Desc.                                                                                                                                                                                                                                                                                                                                                                                                         |
| -X.2            | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 1.2             | Read <b>PC:PB</b>                        | Store as Address.                                                                                                                                                                                                                                                                                                                                                                                             |
| 2.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 2.2             | Read <b>PC:PB</b>                        | Store as Address.H.                                                                                                                                                                                                                                                                                                                                                                                           |
| 3.1             |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                              |
| 3.2             | Read Address: <b>DB</b>                  | Store as Operand. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                               |
| 4.1             |                                          |                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4.2             | Read Address + 1: <b>DB</b> (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.1             |                                          | If <b>X</b> flag is set, set <b>C</b> flag based off ( <b>X</b> .L >= Operand.L), set <b>Z</b> flag based off ( <b>X</b> .L == Operand.L), and set <b>N</b> flag based off (( <b>X</b> .L - Operand.L) & \$80) != 0, otherwise set <b>C</b> flag based off ( <b>X</b> >= Operand), set <b>Z</b> flag based off ( <b>X</b> == Operand), and set <b>N</b> flag based off (( <b>X</b> - Operand) & \$8000) != 0. |
| 5.2             | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                              |
| +X.1            |                                          |                                                                                                                                                                                                                                                                                                                                                                                                               |

| SBC (ED) Cycles: 4 |                                          | Size: 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute (Read)    |                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cycle              | R/W                                      | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| -X.2               | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.1                |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.2                | Read <b>PC:PB</b>                        | Store as Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2.1                |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2.2                | Read <b>PC:PB</b>                        | Store as Address.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3.1                |                                          | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3.2                | Read Address: <b>DB</b>                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4.1                |                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4.2                | Read Address + 1: <b>DB</b> (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.1                |                                          | <p>If <b>M</b> flag == 1 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) - ui16(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = int16(<b>A.L</b> &amp; 0x0F) - int16(Operand &amp; 0x0F) - (1 - <b>C</b>);</p> <p>if (Lo &lt; 0) Lo -= 6.</p> <p>BorrowToHi = Lo &lt; 0 ? 1 : 0.</p> <p>HiSum = int16(<b>A.L</b> &gt;&gt; 4) - int16(Operand &gt;&gt; 4) - BorrowToHi.</p> <p><b>V</b> flag = (((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &lt; 0) HiSum -= 6.</p> <p><b>C</b> flag = HiSum &gt;= 0.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) - ui32(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Borrow = ui16(1 - <b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = int16((<b>A</b> &gt;&gt; i) &amp; \$0F) - int16((Operand &gt;&gt; i) &amp; \$0F) - Borrow.</p> <p>if (i == 12) <b>V</b> flag = (((<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.</p> <p>if (DigitSum &lt; 0) DigitSum -= 6.</p> <p>Borrow = (DigitSum &lt; 0) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Borrow == 0. <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 5.2                | Read <b>PC:PB</b>                        | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| +X.1               |                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| INC (EE)                     | Cycles: 6                                    | Size: 3                                                                                                                                                                                                                                          |
|------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute (Read/Modify/Write) |                                              |                                                                                                                                                                                                                                                  |
| Cycle                        | R/W                                          | Desc.                                                                                                                                                                                                                                            |
| -X.2                         | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                                                                                                 |
| 1.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                 |
| 1.2                          | Read <b>PC:PB</b>                            | Store as Address.                                                                                                                                                                                                                                |
| 2.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                 |
| 2.2                          | Read <b>PC:PB</b>                            | Store as Address.H.                                                                                                                                                                                                                              |
| 3.1                          |                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                 |
| 3.2                          | Read Address: <b>DB</b>                      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                  |
| 4.1                          |                                              |                                                                                                                                                                                                                                                  |
| 4.2                          | Read Address + 1: <b>DB</b> (With Carry)     | Store as Operand.H.                                                                                                                                                                                                                              |
| 5.1                          |                                              | If <b>M</b> flag is set, perform Operand.L += 1, set <b>N</b> based off (Operand.L & \$80), and set <b>Z</b> based off Operand.L, otherwise perform Operand += 1, set <b>N</b> based off (Operand.H & \$80), and set <b>Z</b> based off Operand. |
| 5.2                          |                                              | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                    |
| 6.1                          |                                              |                                                                                                                                                                                                                                                  |
| 6.2                          | Write to Address + 1: <b>DB</b> (With Carry) | Write Operand.H.                                                                                                                                                                                                                                 |
| 7.1                          |                                              |                                                                                                                                                                                                                                                  |
| 7.2                          | Write to Address: <b>DB</b>                  | Write Operand.L.                                                                                                                                                                                                                                 |
| 8.1                          |                                              |                                                                                                                                                                                                                                                  |
| 8.2                          | Read <b>PC:PB</b>                            | Store as OpCode.                                                                                                                                                                                                                                 |
| +X.1                         |                                              |                                                                                                                                                                                                                                                  |

| SBC (EF)             | Cycles: 5                          | Size: 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute Long (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cycle                | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| -X.2                 | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.2                  | Read <b>PC:PB</b>                  | Store as Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2.2                  | Read <b>PC:PB</b>                  | Store as Address.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 3.2                  | Read <b>PC:PB</b>                  | Stores as Bank.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4.1                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4.2                  | Read Address:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.1                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.2                  | Read Address + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 6.1                  |                                    | <p>If <b>M</b> flag == 1 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) - ui16(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = int16(<b>A.L</b> &amp; 0x0F) - int16(Operand &amp; 0x0F) - (1 - <b>C</b>);</p> <p>if (Lo &lt; 0) Lo -= 6.</p> <p>BorrowToHi = Lo &lt; 0 ? 1 : 0.</p> <p>HiSum = int16(<b>A.L</b> &gt;&gt; 4) - int16(Operand &gt;&gt; 4) - BorrowToHi.</p> <p><b>V</b> flag = (((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &lt; 0) HiSum -= 6.</p> <p><b>C</b> flag = HiSum &gt;= 0.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) - ui32(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Borrow = ui16(1 - <b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = int16((<b>A</b> &gt;&gt; i) &amp; \$0F) - int16((Operand &gt;&gt; i) &amp; \$0F) - Borrow.</p> <p>if (i == 12) <b>V</b> flag = (((<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.</p> <p>if (DigitSum &lt; 0) DigitSum -= 6.</p> <p>Borrow = (DigitSum &lt; 0) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Borrow == 0. <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 6.2                  | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| +X.1                 |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



|          |                   |                                                                                                 |
|----------|-------------------|-------------------------------------------------------------------------------------------------|
| BEQ (F0) | Cycles: 2         | Size: 2                                                                                         |
| Relative |                   |                                                                                                 |
| Cycle    | R/W               | Desc.                                                                                           |
| -X.2     | Read PC:PB        | Store as OpCode.                                                                                |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> . Decide to branch if <b>Z</b> is 1.                                             |
| 1.2      |                   | Store as Operand. Address = i16(i8(Operand.L)) + <b>PC</b> . If not branching, poll interrupts. |
| 2.1      |                   | Inc. <b>PC</b> . If not branching, end (next half-cycle is 3.2).                                |
| 2.2      |                   | Poll interrupts.                                                                                |
| 3.1      |                   | Set <b>PC</b> to Address.                                                                       |
| 3.2      |                   | Store as OpCode.                                                                                |
| +X.1     |                   |                                                                                                 |

| SBC (F1) Cycles: 5                   |                                    | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indirect Indexed (d),y (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Cycle                                | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| -X.2                                 | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1.1                                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1.2                                  | Read <b>PC:PB</b>                  | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2.1                                  |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2.2                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3.1                                  |                                    | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3.2                                  | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 4.1                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4.2                                  | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 5.1                                  |                                    | Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 5.2                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6.1                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6.2                                  | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 7.1                                  |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 7.2                                  | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8.1                                  |                                    | <p>If <b>M</b> flag == 1 (8-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = ui16(<b>A.L</b>) - ui16(Operand) - (1 - <b>C</b>).<br/>             <b>V</b> flag = ((ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.<br/>             <b>C</b> flag = Result &lt;= \$FF.<br/>             Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:<br/>             Lo = int16(<b>A.L</b> &amp; 0x0F) - int16(Operand &amp; 0x0F) - (1 - <b>C</b>);<br/>             if (Lo &lt; 0) Lo -= 6.<br/>             BorrowToHi = Lo &lt; 0 ? 1 : 0.<br/>             HiSum = int16(<b>A.L</b> &gt;&gt; 4) - int16(Operand &gt;&gt; 4) - BorrowToHi.<br/>             <b>V</b> flag = (((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.<br/>             if (HiSum &lt; 0) HiSum -= 6.<br/>             <b>C</b> flag = HiSum &gt;= 0.<br/>             Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).<br/>             <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = ui32(<b>A</b>) - ui32(Operand) - (1 - <b>C</b>).<br/>             <b>V</b> flag = ((ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0.<br/>             <b>C</b> flag = Result &lt;= \$FFFF.<br/>             Result = ui16(Result).<br/>             <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:<br/>             Result = ui16(0). Borrow = ui16(1 - <b>C</b>).<br/>             for i = 0, 4, 8, 12:<br/>               DigitSum = int16((<b>A</b> &gt;&gt; i) &amp; \$0F) - int16((Operand &gt;&gt; i) &amp; \$0F) - Borrow.<br/>               if (i == 12) <b>V</b> flag = (((<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.<br/>               if (DigitSum &lt; 0) DigitSum -= 6.<br/>               Borrow = (DigitSum &lt; 0) ? 1 : 0.<br/>               Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).<br/>             <b>C</b> flag = Borrow == 0. <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 8.2                                  | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| +X.1                                 |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| SBC (F2) Cycles: 5     |                             | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indirect (Read) |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cycle                  | R/W                         | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| -X.2                   | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.1                    |                             | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.2                    | Read <b>PC:PB</b>           | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2.1                    |                             | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2.2                    |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3.1                    |                             | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3.2                    | Read Address                | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4.1                    |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4.2                    | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.1                    |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.2                    | Read Pointer: <b>DB</b>     | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 6.1                    |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 6.2                    | Read Pointer + 1: <b>DB</b> | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 7.1                    |                             | <p>If <b>M</b> flag == 1 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) - ui16(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = int16(<b>A.L</b> &amp; 0x0F) - int16(Operand &amp; 0x0F) - (1 - <b>C</b>);</p> <p>if (Lo &lt; 0) Lo -= 6.</p> <p>BorrowToHi = Lo &lt; 0 ? 1 : 0.</p> <p>HiSum = int16(<b>A.L</b> &gt;&gt; 4) - int16(Operand &gt;&gt; 4) - BorrowToHi.</p> <p><b>V</b> flag = (((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &lt; 0) HiSum -= 6.</p> <p><b>C</b> flag = HiSum &gt;= 0.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) - ui32(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Borrow = ui16(1 - <b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = int16((<b>A</b> &gt;&gt; i) &amp; \$0F) - int16((Operand &gt;&gt; i) &amp; \$0F) - Borrow.</p> <p>if (i == 12) <b>V</b> flag = (((<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.</p> <p>if (DigitSum &lt; 0) DigitSum -= 6.</p> <p>Borrow = (DigitSum &lt; 0) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Borrow == 0. <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 7.2                    | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| +X.1                   |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| SBC (F3) Cycles: 7                     |                                    | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stack Relative Indirect Indexed (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Cycle                                  | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| -X.2                                   | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1.1                                    |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1.2                                    | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 2.1                                    |                                    | Inc. <b>PC</b> . Set Address to Pointer + <b>S</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2.2                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 3.1                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 3.2                                    | Read Address                       | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 4.1                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 4.2                                    | Read Address + 1                   | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 5.1                                    |                                    | Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{Y})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank = $\mathbf{DB} + (\text{Tmp} \gg 16)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.2                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 6.1                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 6.2                                    | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 7.1                                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 7.2                                    | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 8.1                                    |                                    | <p>If <b>M</b> flag == 1 (8-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = <math>\text{ui16}(\mathbf{A.L}) - \text{ui16}(\text{Operand}) - (1 - \mathbf{C})</math>.<br/>             <b>V</b> flag = <math>((\text{ui16}(\mathbf{A.L}) \wedge \text{ui16}(\text{Operand})) \&amp; (\text{ui16}(\mathbf{A.L}) \wedge \text{ui8}(\text{Result})) \&amp; \\$80) \neq 0</math>.<br/>             <b>C</b> flag = Result &lt;= \$FF.<br/>             Result = <math>\text{ui8}(\text{Result})</math>. <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.<br/>           If <b>D</b> flag == 1:<br/>             Lo = <math>\text{int16}(\mathbf{A.L} \&amp; 0x0F) - \text{int16}(\text{Operand} \&amp; 0x0F) - (1 - \mathbf{C})</math>;<br/>             if (Lo &lt; 0) Lo -= 6.<br/>             BorrowToHi = Lo &lt; 0 ? 1 : 0.<br/>             HiSum = <math>\text{int16}(\mathbf{A.L} \gg 4) - \text{int16}(\text{Operand} \gg 4) - \text{BorrowToHi}</math>.<br/>             <b>V</b> flag = <math>((\mathbf{A.L} \gg 4) \wedge (\text{Operand} \gg 4)) \&amp; ((\mathbf{A.L} \gg 4) \wedge \text{HiSum}) \&amp; \\$08 \neq 0</math>.<br/>             if (HiSum &lt; 0) HiSum -= 6.<br/>             <b>C</b> flag = HiSum &gt;= 0.<br/>             Result = <math>\text{ui8}(((\text{HiSum} \&amp; 0x0F) &lt;&lt; 4) \mid (\text{Lo} \&amp; \\$0F))</math>.<br/>             <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = <math>\text{ui32}(\mathbf{A}) - \text{ui32}(\text{Operand}) - (1 - \mathbf{C})</math>.<br/>             <b>V</b> flag = <math>((\text{ui32}(\mathbf{A}) \wedge \text{ui32}(\text{Operand})) \&amp; (\text{ui32}(\mathbf{A}) \wedge \text{ui16}(\text{Result})) \&amp; \\$8000) \neq 0</math>.<br/>             <b>C</b> flag = Result &lt;= \$FFFF.<br/>             Result = <math>\text{ui16}(\text{Result})</math>.<br/>             <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.<br/>           If <b>D</b> flag == 1:<br/>             Result = <math>\text{ui16}(0)</math>. Borrow = <math>\text{ui16}(1 - \mathbf{C})</math>.<br/>             for i = 0, 4, 8, 12:<br/>               DigitSum = <math>\text{int16}((\mathbf{A} \gg i) \&amp; \\$0F) - \text{int16}((\text{Operand} \gg i) \&amp; \\$0F) - \text{Borrow}</math>.<br/>               if (i == 12) <b>V</b> flag = <math>((\mathbf{A} \gg 12) \wedge (\text{Operand} \gg 12)) \&amp; ((\mathbf{A} \gg 12) \wedge \text{DigitSum}) \&amp; \\$08 \neq 0</math>.<br/>               if (DigitSum &lt; 0) DigitSum -= 6.<br/>               Borrow = (DigitSum &lt; 0) ? 1 : 0.<br/>               Result  = <math>\text{ui16}((\text{DigitSum} \&amp; \\$0F) &lt;&lt; i)</math>.<br/>             <b>C</b> flag = Borrow == 0. <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 8.2                                    | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| +X.1                                   |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                 |                         |                            |
|-----------------|-------------------------|----------------------------|
| PEA (F4)        | Cycles: 5               | Size: 3                    |
| Absolute (Push) |                         |                            |
| Cycle           | R/W                     | Desc.                      |
| -X.2            | Read <b>PC:PB</b>       | Store as OpCode.           |
| 1.1             |                         | Inc. <b>PC</b> .           |
| 1.2             | Read <b>PC:PB</b>       | Store as Operand.          |
| 2.1             |                         | Inc. <b>PC</b> .           |
| 2.2             | Read <b>PC:PB</b>       | Store as Operand.H.        |
| 3.1             |                         | Inc. <b>PC</b> .           |
| 3.2             | Write to ( <b>S</b> +0) | Push Operand.H onto stack. |
| 4.1             |                         |                            |
| 4.2             | Write to ( <b>S</b> -1) | Push Operand.L onto stack. |
| 5.1             |                         | Dec. <b>S</b> by 2.        |
| 5.2             | Read <b>PC:PB</b>       | Store as OpCode.           |
| +X.1            |                         |                            |

| SBC (F5) Cycles: 4 |                   | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct, X (Read)   |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cycle              | R/W               | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| -X.2               | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.1                | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 1.2                |                   | Store as Operand.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2.1                |                   | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 2.2                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3.1                |                   | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3.2                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4.1                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4.2                | Read Address      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.1                |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 5.2                | Read Address + 1  | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 6.1                |                   | <p>If <b>M</b> flag == 1 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui16(<b>A.L</b>) - ui16(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FF.</p> <p>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = int16(<b>A.L</b> &amp; 0x0F) - int16(Operand &amp; 0x0F) - (1 - <b>C</b>);</p> <p>if (Lo &lt; 0) Lo -= 6.</p> <p>BorrowToHi = Lo &lt; 0 ? 1 : 0.</p> <p>HiSum = int16(<b>A.L</b> &gt;&gt; 4) - int16(Operand &gt;&gt; 4) - BorrowToHi.</p> <p><b>V</b> flag = (((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.</p> <p>if (HiSum &lt; 0) HiSum -= 6.</p> <p><b>C</b> flag = HiSum &gt;= 0.</p> <p>Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).</p> <p><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = ui32(<b>A</b>) - ui32(Operand) - (1 - <b>C</b>).</p> <p><b>V</b> flag = ((ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0.</p> <p><b>C</b> flag = Result &lt;= \$FFFF.</p> <p>Result = ui16(Result).</p> <p><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = ui16(0). Borrow = ui16(1 - <b>C</b>).</p> <p>for i = 0, 4, 8, 12:</p> <p>DigitSum = int16((<b>A</b> &gt;&gt; i) &amp; \$0F) - int16((Operand &gt;&gt; i) &amp; \$0F) - Borrow.</p> <p>if (i == 12) <b>V</b> flag = (((<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.</p> <p>if (DigitSum &lt; 0) DigitSum -= 6.</p> <p>Borrow = (DigitSum &lt; 0) ? 1 : 0.</p> <p>Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).</p> <p><b>C</b> flag = Borrow == 0. <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 6.2                | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| +X.1               |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| INC (F6)                      | Cycles: 6            | Size: 2                                                                                                                                                                                                                                          |
|-------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct, X (Read/Modify/Write) |                      |                                                                                                                                                                                                                                                  |
| Cycle                         | R/W                  | Desc.                                                                                                                                                                                                                                            |
| -X.2                          | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                 |
| 1.1                           | Read <b>PC:PB</b>    | Inc. <b>PC</b> .                                                                                                                                                                                                                                 |
| 1.2                           |                      | Store as Operand.                                                                                                                                                                                                                                |
| 2.1                           |                      | Inc. <b>PC</b> .                                                                                                                                                                                                                                 |
| 2.2                           |                      |                                                                                                                                                                                                                                                  |
| 3.1                           |                      | Address = Operand + <b>X</b> + <b>D</b> . If <b>D.L</b> is 0, skip 2 half-cycles.                                                                                                                                                                |
| 3.2                           |                      |                                                                                                                                                                                                                                                  |
| 4.1                           |                      |                                                                                                                                                                                                                                                  |
| 4.2                           | Read Address         | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                  |
| 5.1                           | Read Address + 1     | Store as Operand.H.                                                                                                                                                                                                                              |
| 5.2                           |                      |                                                                                                                                                                                                                                                  |
| 6.1                           |                      | If <b>M</b> flag is set, perform Operand.L += 1, set <b>N</b> based off (Operand.L & \$80), and set <b>Z</b> based off Operand.L, otherwise perform Operand += 1, set <b>N</b> based off (Operand.H & \$80), and set <b>Z</b> based off Operand. |
| 6.2                           |                      | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                    |
| 7.1                           | Write to Address + 1 | Write Operand.H.                                                                                                                                                                                                                                 |
| 7.2                           |                      |                                                                                                                                                                                                                                                  |
| 8.1                           | Write to Address     | Write Operand.L.                                                                                                                                                                                                                                 |
| 8.2                           |                      |                                                                                                                                                                                                                                                  |
| 9.1                           | Read <b>PC:PB</b>    | Store as OpCode.                                                                                                                                                                                                                                 |
| 9.2                           |                      |                                                                                                                                                                                                                                                  |
| +X.1                          |                      |                                                                                                                                                                                                                                                  |

| SBC (F7) Cycles: 6                        |                             | Size: 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Direct Indirect Indexed Long [d],y (Read) |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Cycle                                     | R/W                         | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| -X.2                                      | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1.1                                       |                             | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1.2                                       | Read <b>PC:PB</b>           | Store as Operand. If <b>D.L</b> is 0, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2.1                                       |                             | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2.2                                       |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3.1                                       |                             | Inc. <b>PC</b> if previous cycle was skipped. Set Address to Operand + <b>D</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3.2                                       | Read Address                | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 4.1                                       |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 4.2                                       | Read Address + 1            | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5.1                                       |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5.2                                       | Read Address + 2            | Store as Bank.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 6.1                                       |                             | Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{Y})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank += $(\text{Tmp} \gg 16)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 6.2                                       | Read Pointer:Bank           | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 7.1                                       |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 7.2                                       | Read Pointer + 1:Bank (With | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 8.1                                       |                             | <p>If <b>M</b> flag == 1 (8-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = <math>\text{ui16}(\text{A.L}) - \text{ui16}(\text{Operand}) - (1 - \text{C})</math>.<br/> <b>V</b> flag = <math>((\text{ui16}(\text{A.L}) \wedge \text{ui16}(\text{Operand})) \&amp; (\text{ui16}(\text{A.L}) \wedge \text{ui8}(\text{Result})) \&amp; \\$80) \neq 0</math>.<br/> <b>C</b> flag = <math>\text{Result} \leq \\$\text{FF}</math>.<br/> Result = <math>\text{ui8}(\text{Result})</math>. <b>Z</b> flag = <math>\text{Result} == \\$00</math>. <b>N</b> flag = <math>(\text{Result} \&amp; \\$80) \neq 0</math>.</p> <p>If <b>D</b> flag == 1:</p> <p>Lo = <math>\text{int16}(\text{A.L} \&amp; 0x0\text{F}) - \text{int16}(\text{Operand} \&amp; 0x0\text{F}) - (1 - \text{C})</math>;<br/> if <math>(\text{Lo} &lt; 0)</math> Lo -= 6.<br/> BorrowToHi = <math>\text{Lo} &lt; 0 ? 1 : 0</math>.<br/> HiSum = <math>\text{int16}(\text{A.L} \gg 4) - \text{int16}(\text{Operand} \gg 4) - \text{BorrowToHi}</math>.<br/> <b>V</b> flag = <math>((\text{A.L} \gg 4) \wedge (\text{Operand} \gg 4)) \&amp; ((\text{A.L} \gg 4) \wedge \text{HiSum}) \&amp; \\$08) \neq 0</math>.<br/> if <math>(\text{HiSum} &lt; 0)</math> HiSum -= 6.<br/> <b>C</b> flag = <math>\text{HiSum} \geq 0</math>.<br/> Result = <math>\text{ui8}(((\text{HiSum} \&amp; 0x0\text{F}) &lt; 4) \mid (\text{Lo} \&amp; \\$0\text{F}))</math>.<br/> <b>Z</b> flag = <math>\text{Result} == \\$00</math>. <b>N</b> flag = <math>(\text{Result} \&amp; \\$80) \neq 0</math>.</p> <p>If <b>M</b> flag == 0 (16-bit):</p> <p>If <b>D</b> flag == 0:</p> <p>Result = <math>\text{ui32}(\text{A}) - \text{ui32}(\text{Operand}) - (1 - \text{C})</math>.<br/> <b>V</b> flag = <math>((\text{ui32}(\text{A}) \wedge \text{ui32}(\text{Operand})) \&amp; (\text{ui32}(\text{A}) \wedge \text{ui16}(\text{Result})) \&amp; \\$8000) \neq 0</math>.<br/> <b>C</b> flag = <math>\text{Result} \leq \\$\text{FFFF}</math>.<br/> Result = <math>\text{ui16}(\text{Result})</math>.<br/> <b>Z</b> flag = <math>\text{Result} == \\$0000</math>. <b>N</b> flag = <math>(\text{Result} \&amp; \\$8000) \neq 0</math>.</p> <p>If <b>D</b> flag == 1:</p> <p>Result = <math>\text{ui16}(0)</math>. Borrow = <math>\text{ui16}(1 - \text{C})</math>.<br/> for i = 0, 4, 8, 12:<br/> DigitSum = <math>\text{int16}((\text{A} \gg i) \&amp; \\$0\text{F}) - \text{int16}((\text{Operand} \gg i) \&amp; \\$0\text{F}) - \text{Borrow}</math>.<br/> if <math>(i == 12)</math> <b>V</b> flag = <math>((\text{A} \gg 12) \wedge (\text{Operand} \gg 12)) \&amp; ((\text{A} \gg 12) \wedge \text{DigitSum}) \&amp; \\$08) \neq 0</math>.<br/> if <math>(\text{DigitSum} &lt; 0)</math> DigitSum -= 6.<br/> Borrow = <math>(\text{DigitSum} &lt; 0) ? 1 : 0</math>.<br/> Result  = <math>\text{ui16}((\text{DigitSum} \&amp; \\$0\text{F}) &lt; i)</math>.<br/> <b>C</b> flag = <math>\text{Borrow} == 0</math>. <b>Z</b> flag = <math>\text{Result} == \\$0000</math>. <b>N</b> flag = <math>(\text{Result} \&amp; \\$8000) \neq 0</math>.</p> |
| 8.2                                       | Read <b>PC:PB</b>           | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| +X.1                                      |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |



|          |                   |                             |
|----------|-------------------|-----------------------------|
| SED (F8) | Cycles: 2         | Size: 1                     |
| Implied  |                   |                             |
| Cycle    | R/W               | Desc.                       |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.            |
| 1.1      |                   | Inc. <b>PC</b> .            |
| 1.2      | Read <b>PC:PB</b> | Discard.                    |
| 2.1      |                   | Set the <b>D</b> flag to 1. |
| 2.2      | Read <b>PC:PB</b> | Store as OpCode.            |
| +X.1     |                   |                             |

| SBC (F9) Cycles: 4 |                                    | Size: 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute, Y (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1.2                | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2.2                | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>Y</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3.2                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4.1                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.1                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.2                | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 6.1                |                                    | <p>If <b>M</b> flag == 1 (8-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = ui16(<b>A.L</b>) - ui16(Operand) - (1 - <b>C</b>).<br/>             <b>V</b> flag = ((ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.<br/>             <b>C</b> flag = Result &lt;= \$FF.<br/>             Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:<br/>             Lo = int16(<b>A.L</b> &amp; 0x0F) - int16(Operand &amp; 0x0F) - (1 - <b>C</b>);<br/>             if (Lo &lt; 0) Lo -= 6.<br/>             BorrowToHi = Lo &lt; 0 ? 1 : 0.<br/>             HiSum = int16(<b>A.L</b> &gt;&gt; 4) - int16(Operand &gt;&gt; 4) - BorrowToHi.<br/>             <b>V</b> flag = (((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.<br/>             if (HiSum &lt; 0) HiSum -= 6.<br/>             <b>C</b> flag = HiSum &gt;= 0.<br/>             Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).<br/>             <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = ui32(<b>A</b>) - ui32(Operand) - (1 - <b>C</b>).<br/>             <b>V</b> flag = ((ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0.<br/>             <b>C</b> flag = Result &lt;= \$FFFF.<br/>             Result = ui16(Result).<br/>             <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:<br/>             Result = ui16(0). Borrow = ui16(1 - <b>C</b>).<br/>             for i = 0, 4, 8, 12:<br/>               DigitSum = int16((<b>A</b> &gt;&gt; i) &amp; \$0F) - int16((Operand &gt;&gt; i) &amp; \$0F) - Borrow.<br/>               if (i == 12) <b>V</b> flag = (((<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.<br/>               if (DigitSum &lt; 0) DigitSum -= 6.<br/>               Borrow = (DigitSum &lt; 0) ? 1 : 0.<br/>               Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).<br/>             <b>C</b> flag = Borrow == 0. <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 6.2                | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| +X.1               |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|          |                     |                                                                                                                                                                                                                                                                              |
|----------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PLX (FA) | Cycles: 4           | Size: 1                                                                                                                                                                                                                                                                      |
| Implied  |                     |                                                                                                                                                                                                                                                                              |
| Cycle    | R/W                 | Desc.                                                                                                                                                                                                                                                                        |
| -X.2     | Read <b>PC:PB</b>   | Store as OpCode.                                                                                                                                                                                                                                                             |
| 1.1      |                     | Inc. <b>PC</b> .                                                                                                                                                                                                                                                             |
| 1.2      |                     |                                                                                                                                                                                                                                                                              |
| 2.1      |                     |                                                                                                                                                                                                                                                                              |
| 2.2      |                     |                                                                                                                                                                                                                                                                              |
| 3.1      |                     | Inc. <b>S</b> by 1.                                                                                                                                                                                                                                                          |
| 3.2      | Read ( <b>S</b> +0) | Store as Operand.L. If <b>X</b> flag is set, skip the next cycle.                                                                                                                                                                                                            |
| 4.1      |                     |                                                                                                                                                                                                                                                                              |
| 4.2      | Read ( <b>S</b> +1) | Store as Operand.H.                                                                                                                                                                                                                                                          |
| 5.1      |                     | Inc. <b>S</b> by 1. If <b>X</b> flag is set, set <b>X.L</b> to Operand.L, set <b>N</b> based off ( <b>X.L</b> & \$80), and <b>Z</b> based off <b>X.L</b> , otherwise set <b>X</b> to Operand, set <b>N</b> based off ( <b>X.H</b> & \$80), and <b>Z</b> based off <b>X</b> . |
| 5.2      | Read <b>PC:PB</b>   | Store as OpCode.                                                                                                                                                                                                                                                             |
| +X.1     |                     |                                                                                                                                                                                                                                                                              |

|          |                   |                                                                                                                                                                                          |
|----------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| XCE (FB) | Cycles: 2         | Size: 1                                                                                                                                                                                  |
| Implied  |                   |                                                                                                                                                                                          |
| Cycle    | R/W               | Desc.                                                                                                                                                                                    |
| -X.2     | Read <b>PC:PB</b> | Store as OpCode.                                                                                                                                                                         |
| 1.1      | Read <b>PC:PB</b> | Inc. <b>PC</b> .                                                                                                                                                                         |
| 1.2      |                   | Discard.                                                                                                                                                                                 |
| 2.1      |                   | Exchanges the Carry and Emulation flags. If Emulation mode is entered, <b>M</b> and <b>X</b> flags are set to 1, <b>S.H</b> is set to 1, and <b>X.H</b> and <b>Y.H</b> are cleared to 0. |
| 2.2      |                   | Store as OpCode.                                                                                                                                                                         |
| +X.1     |                   |                                                                                                                                                                                          |

|                                        |                         |                               |
|----------------------------------------|-------------------------|-------------------------------|
| JSR (FC)                               | Cycles: 8               | Size: 3                       |
| Absolute Indexed Indirect (a,x) (Jump) |                         |                               |
| Cycle                                  | R/W                     | Desc.                         |
| -X.2                                   | Read <b>PC:PB</b>       | Store as OpCode.              |
| 1.1                                    |                         | Inc. <b>PC</b> .              |
| 1.2                                    | Read <b>PC:PB</b>       | Store as Pointer.             |
| 2.1                                    |                         | Inc. <b>PC</b> .              |
| 2.2                                    | Write to ( <b>S</b> +0) | Push <b>PC</b> .H onto stack. |
| 3.1                                    |                         |                               |
| 3.2                                    | Write to ( <b>S</b> -1) | Push <b>PC</b> .L onto stack. |
| 4.1                                    |                         | Dec. <b>S</b> by 2.           |
| 4.2                                    | Read <b>PC:PB</b>       | Store as Pointer.H.           |
| 5.1                                    |                         | Perform Pointer += <b>X</b> . |
|                                        |                         | Bank = <b>PB</b> .            |
| 5.2                                    |                         |                               |
| 6.1                                    |                         |                               |
| 6.2                                    | Read Pointer:Bank       | Store as Address.L.           |
| 7.1                                    |                         |                               |
| 7.2                                    | Read Pointer + 1:Bank   | Store as Address.H.           |
| 8.1                                    |                         | Set <b>PC</b> to Address.     |
| 8.2                                    | Read <b>PC:PB</b>       | Store as OpCode.              |
| +X.1                                   |                         |                               |

| SBC (FD) Cycles: 4 |                                    | Size: 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute, X (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Cycle              | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| -X.2               | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1.2                | Read <b>PC:PB</b>                  | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2.1                |                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2.2                | Read <b>PC:PB</b>                  | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3.1                |                                    | Inc. <b>PC</b> . Perform Orig = Pointer.<br>Tmp = ui32(Pointer + <b>X</b> ).<br>Pointer = ui16(Tmp).<br>If Orig.H == Pointer.H and the <b>X</b> flag is set, skip 2 half-cycles.<br>Bank = <b>DB</b> + (Tmp >> 16).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3.2                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4.1                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4.2                | Read Pointer:Bank                  | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.1                |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5.2                | Read Pointer + 1:Bank (With Carry) | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 6.1                |                                    | <p>If <b>M</b> flag == 1 (8-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = ui16(<b>A.L</b>) - ui16(Operand) - (1 - <b>C</b>).<br/>             <b>V</b> flag = ((ui16(<b>A.L</b>) ^ ui16(Operand)) &amp; (ui16(<b>A.L</b>) ^ ui8(Result)) &amp; \$80) != 0.<br/>             <b>C</b> flag = Result &lt;= \$FF.<br/>             Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>D</b> flag == 1:<br/>             Lo = int16(<b>A.L</b> &amp; 0x0F) - int16(Operand &amp; 0x0F) - (1 - <b>C</b>);<br/>             if (Lo &lt; 0) Lo -= 6.<br/>             BorrowToHi = Lo &lt; 0 ? 1 : 0.<br/>             HiSum = int16(<b>A.L</b> &gt;&gt; 4) - int16(Operand &gt;&gt; 4) - BorrowToHi.<br/>             <b>V</b> flag = (((<b>A.L</b> &gt;&gt; 4) ^ (Operand &gt;&gt; 4)) &amp; ((<b>A.L</b> &gt;&gt; 4) ^ HiSum) &amp; \$08) != 0.<br/>             if (HiSum &lt; 0) HiSum -= 6.<br/>             <b>C</b> flag = HiSum &gt;= 0.<br/>             Result = ui8(((HiSum &amp; 0x0F) &lt;&lt; 4)   (Lo &amp; \$0F)).<br/>             <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result &amp; \$80) != 0.</p> <p>If <b>M</b> flag == 0 (16-bit):<br/>           If <b>D</b> flag == 0:<br/>             Result = ui32(<b>A</b>) - ui32(Operand) - (1 - <b>C</b>).<br/>             <b>V</b> flag = ((ui32(<b>A</b>) ^ ui32(Operand)) &amp; (ui32(<b>A</b>) ^ ui16(Result)) &amp; \$8000) != 0.<br/>             <b>C</b> flag = Result &lt;= \$FFFF.<br/>             Result = ui16(Result).<br/>             <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> <p>If <b>D</b> flag == 1:<br/>             Result = ui16(0). Borrow = ui16(1 - <b>C</b>).<br/>             for i = 0, 4, 8, 12:<br/>               DigitSum = int16((<b>A</b> &gt;&gt; i) &amp; \$0F) - int16((Operand &gt;&gt; i) &amp; \$0F) - Borrow.<br/>               if (i == 12) <b>V</b> flag = (((<b>A</b> &gt;&gt; 12) ^ (Operand &gt;&gt; 12)) &amp; ((<b>A</b> &gt;&gt; 12) ^ DigitSum) &amp; \$08) != 0.<br/>               if (DigitSum &lt; 0) DigitSum -= 6.<br/>               Borrow = (DigitSum &lt; 0) ? 1 : 0.<br/>               Result  = ui16((DigitSum &amp; \$0F) &lt;&lt; i).<br/>             <b>C</b> flag = Borrow == 0. <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result &amp; \$8000) != 0.</p> |
| 6.2                | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| +X.1               |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| INC (FE)                       | Cycles: 7                              | Size: 3                                                                                                                                                                                                                                                                                                          |
|--------------------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute, X (Read/Modiy/Write) |                                        |                                                                                                                                                                                                                                                                                                                  |
| Cycle                          | R/W                                    | Desc.                                                                                                                                                                                                                                                                                                            |
| -X.2                           | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                                                                                                                                                                 |
| 1.1                            |                                        | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                 |
| 1.2                            | Read <b>PC:PB</b>                      | Store as Pointer.                                                                                                                                                                                                                                                                                                |
| 2.1                            |                                        | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                 |
| 2.2                            | Read <b>PC:PB</b>                      | Store as Pointer.H.                                                                                                                                                                                                                                                                                              |
| 3.1                            |                                        | Inc. <b>PC</b> . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$ .<br>Pointer = $\text{ui16}(\text{Tmp})$ .<br>Bank = $\text{DB} + (\text{Tmp} \gg 16)$ .                                                                                                                                          |
| 3.2                            |                                        |                                                                                                                                                                                                                                                                                                                  |
| 4.1                            |                                        |                                                                                                                                                                                                                                                                                                                  |
| 4.2                            | Read Pointer:Bank                      | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                  |
| 5.1                            |                                        |                                                                                                                                                                                                                                                                                                                  |
| 5.2                            | Read Pointer + 1:Bank (With Carry)     | Store as Operand.H.                                                                                                                                                                                                                                                                                              |
| 6.1                            |                                        | If <b>M</b> flag is set, perform $\text{Operand.L} += 1$ , set <b>N</b> based off ( $\text{Operand.L} \& \$80$ ), and set <b>Z</b> based off $\text{Operand.L}$ , otherwise perform $\text{Operand} += 1$ , set <b>N</b> based off ( $\text{Operand.H} \& \$80$ ), and set <b>Z</b> based off $\text{Operand}$ . |
| 6.2                            |                                        | If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                    |
| 7.1                            |                                        |                                                                                                                                                                                                                                                                                                                  |
| 7.2                            | Write to Pointer + 1:Bank (With Carry) | Write Operand.H.                                                                                                                                                                                                                                                                                                 |
| 8.1                            |                                        |                                                                                                                                                                                                                                                                                                                  |
| 8.2                            | Write to Pointer:Bank                  | Write Operand.L.                                                                                                                                                                                                                                                                                                 |
| 9.1                            |                                        |                                                                                                                                                                                                                                                                                                                  |
| 9.2                            | Read <b>PC:PB</b>                      | Store as OpCode.                                                                                                                                                                                                                                                                                                 |
| +X.1                           |                                        |                                                                                                                                                                                                                                                                                                                  |

| SBC (FF)                |                                    | Cycles: 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Size: 4 |
|-------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Absolute Long, X (Read) |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |
| Cycle                   | R/W                                | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| -X.2                    | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| 1.1                     | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| 1.2                     |                                    | Store as Pointer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
| 2.1                     | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| 2.2                     |                                    | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |
| 3.1                     | Read <b>PC:PB</b>                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| 3.2                     |                                    | Stores as Bank.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |
| 4.1                     | Read <b>PC:PB</b>                  | Inc. <b>PC</b> . Perform Tmp = ui32(Pointer + <b>X</b> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |
|                         |                                    | Pointer = ui16(Tmp).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
|                         | Read Pointer:Bank                  | Bank += (Tmp >> 16).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| 4.2                     |                                    | Store as Operand. If <b>M</b> flag is set, skip the next cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |
| 5.1                     | Read Pointer + 1:Bank (With Carry) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |
| 5.2                     |                                    | Store as Operand.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |
| 6.1                     |                                    | If <b>M</b> flag == 1 (8-bit):<br>If <b>D</b> flag == 0:<br>Result = ui16( <b>A.L</b> ) - ui16(Operand) - (1 - <b>C</b> ).<br><b>V</b> flag = ((ui16( <b>A.L</b> ) ^ ui16(Operand)) & (ui16( <b>A.L</b> ) ^ ui8(Result)) & \$80) != 0.<br><b>C</b> flag = Result <= \$FF.<br>Result = ui8(Result). <b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result & \$80) != 0.<br>If <b>D</b> flag == 1:<br>Lo = int16( <b>A.L</b> & 0x0F) - int16(Operand & 0x0F) - (1 - <b>C</b> );<br>if (Lo < 0) Lo -= 6.<br>BorrowToHi = Lo < 0 ? 1 : 0.<br>HiSum = int16( <b>A.L</b> >> 4) - int16(Operand >> 4) - BorrowToHi.<br><b>V</b> flag = ((( <b>A.L</b> >> 4) ^ (Operand >> 4)) & (( <b>A.L</b> >> 4) ^ HiSum) & \$08) != 0.<br>if (HiSum < 0) HiSum -= 6.<br><b>C</b> flag = HiSum >= 0.<br>Result = ui8(((HiSum & 0x0F) << 4)   (Lo & \$0F)).<br><b>Z</b> flag = Result == \$00. <b>N</b> flag = (Result & \$80) != 0.<br>If <b>M</b> flag == 0 (16-bit):<br>If <b>D</b> flag == 0:<br>Result = ui32( <b>A</b> ) - ui32(Operand) - (1 - <b>C</b> ).<br><b>V</b> flag = ((ui32( <b>A</b> ) ^ ui32(Operand)) & (ui32( <b>A</b> ) ^ ui16(Result)) & \$8000) != 0.<br><b>C</b> flag = Result <= \$FFFF.<br>Result = ui16(Result).<br><b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result & \$8000) != 0.<br>If <b>D</b> flag == 1:<br>Result = ui16(0). Borrow = ui16(1 - <b>C</b> ).<br>for i = 0, 4, 8, 12:<br>DigitSum = int16(( <b>A</b> >> i) & \$0F) - int16((Operand >> i) & \$0F) - Borrow.<br>if (i == 12) <b>V</b> flag = ((( <b>A</b> >> 12) ^ (Operand >> 12)) & (( <b>A</b> >> 12) ^ DigitSum) & \$08) != 0.<br>if (DigitSum < 0) DigitSum -= 6.<br>Borrow = (DigitSum < 0) ? 1 : 0.<br>Result  = ui16((DigitSum & \$0F) << i).<br><b>C</b> flag = Borrow == 0. <b>Z</b> flag = Result == \$0000. <b>N</b> flag = (Result & \$8000) != 0. |         |
| 6.2                     | Read <b>PC:PB</b>                  | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| +X.1                    |                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |